Supplementary Results: Forecast-State Miscalibration in Probabilistic Time-Series Forecasting

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PER-BACKBONE DATASET COMPARISONS

Tables S1–S7 expand the main comparison to every dataset–horizon configuration. Forecast-state probability calibration (FSPC) uses 36 states, validation-fitted residual templates and a mean that blends a training-bias reference with an affine, state-adjusted fit. The blend is selected on validation forecasts using the chronological fitting and selection protocol in the main paper. The w/o Mean Control ablation uses the same source-filtered templates without mean adjustment. The seven backbones cover 83 configurations and 249 independently trained checkpoints. Every correction is paired with the same native forecasts and test coordinates. The training-fitted template variant uses the same affine-plus-state mean and is compared separately in the main paper's temporal component study.

Each cell reports the mean over three independent training runs, with the sample standard deviation in the lower-right subscript (denominator two). Bold identifies the lowest unrounded mean among the six methods in the same row; numerical ties within 10^{-12} share the mark. CRPS and QICE both have downward arrows. CRPS retains original traffic-flow units for DeepSTUQ and model-normalized units for the other backbones; QICE uses percentage points throughout. H denotes the number of forecast steps. NsDiff uses $H = 192$ on ETTm1, ETTm2 and Exchange.

TABLE S1: K^2VAE: all 36 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTh1	96	CRPS ↓	$0.337_{\pm 0.009}$	$0.353_{\pm 0.016}$	$0.346_{\pm 0.003}$	$0.347_{\pm 0.003}$	$0.341_{\pm 0.001}$	$\mathbf{0.337}_{\pm 0.004}$
		QICE ↓	$3.009_{\pm 1.009}$	$3.130_{\pm 0.298}$	$\mathbf{2.178}_{\pm 0.298}$	$2.953_{\pm 0.159}$	$2.809_{\pm 0.159}$	$2.417_{\pm 0.282}$
ETTh1	192	CRPS ↓	$\mathbf{0.372}_{\pm 0.005}$	$0.420_{\pm 0.044}$	$0.414_{\pm 0.024}$	$0.388_{\pm 0.009}$	$0.377_{\pm 0.007}$	$0.372_{\pm 0.006}$
		QICE ↓	$3.175_{\pm 1.022}$	$3.780_{\pm 0.076}$	$\mathbf{1.962}_{\pm 0.296}$	$2.957_{\pm 0.901}$	$2.860_{\pm 0.869}$	$2.866_{\pm 0.457}$
ETTh1	336	CRPS ↓	$\mathbf{0.384}_{\pm 0.005}$	$0.441_{\pm 0.027}$	$0.455_{\pm 0.009}$	$0.418_{\pm 0.015}$	$0.400_{\pm 0.012}$	$0.401_{\pm 0.016}$
		QICE ↓	$3.010_{\pm 0.786}$	$4.359_{\pm 0.170}$	$\mathbf{2.243}_{\pm 0.080}$	$3.745_{\pm 0.196}$	$3.670_{\pm 0.185}$	$3.129_{\pm 0.139}$
ETTh1	720	CRPS ↓	$\mathbf{0.409}_{\pm 0.003}$	$0.459_{\pm 0.067}$	$0.494_{\pm 0.056}$	$0.430_{\pm 0.009}$	$0.416_{\pm 0.007}$	$0.417_{\pm 0.009}$
		QICE ↓	$3.817_{\pm 0.673}$	$4.284_{\pm 0.738}$	$\mathbf{2.514}_{\pm 0.445}$	$4.174_{\pm 0.146}$	$4.057_{\pm 0.109}$	$3.309_{\pm 0.150}$
ETTh2	96	CRPS ↓	$0.300_{\pm 0.007}$	$0.285_{\pm 0.007}$	$\mathbf{0.284}_{\pm 0.006}$	$0.296_{\pm 0.006}$	$0.298_{\pm 0.007}$	$0.294_{\pm 0.005}$
		QICE ↓	$3.939_{\pm 0.175}$	$1.356_{\pm 0.125}$	$\mathbf{1.243}_{\pm 0.190}$	$3.001_{\pm 0.474}$	$3.079_{\pm 0.540}$	$2.844_{\pm 0.143}$
ETTh2	192	CRPS ↓	$0.334_{\pm 0.002}$	$0.315_{\pm 0.004}$	$\mathbf{0.313}_{\pm 0.003}$	$0.330_{\pm 0.002}$	$0.331_{\pm 0.002}$	$0.327_{\pm 0.001}$
		QICE ↓	$4.584_{\pm 0.306}$	$1.381_{\pm 0.316}$	$\mathbf{0.957}_{\pm 0.174}$	$2.705_{\pm 0.263}$	$2.791_{\pm 0.246}$	$3.167_{\pm 0.262}$
ETTh2	336	CRPS ↓	$0.678_{\pm 0.194}$	$0.436_{\pm 0.064}$	$\mathbf{0.434}_{\pm 0.078}$	$0.563_{\pm 0.127}$	$0.567_{\pm 0.129}$	$0.476_{\pm 0.091}$
		QICE ↓	$5.194_{\pm 0.219}$	$2.192_{\pm 0.950}$	$\mathbf{1.259}_{\pm 0.144}$	$3.193_{\pm 0.318}$	$3.246_{\pm 0.348}$	$2.523_{\pm 0.284}$
ETTh2	720	CRPS ↓	$0.502_{\pm 0.135}$	$0.396_{\pm 0.030}$	$\mathbf{0.377}_{\pm 0.018}$	$0.453_{\pm 0.062}$	$0.450_{\pm 0.058}$	$0.400_{\pm 0.029}$
		QICE ↓	$5.422_{\pm 0.460}$	$2.635_{\pm 0.614}$	$\mathbf{1.315}_{\pm 0.350}$	$3.048_{\pm 0.879}$	$3.126_{\pm 0.926}$	$2.887_{\pm 0.797}$
ETTm1	96	CRPS ↓	$0.298_{\pm 0.003}$	$\mathbf{0.280}_{\pm 0.005}$	$0.280_{\pm 0.006}$	$0.291_{\pm 0.004}$	$0.291_{\pm 0.004}$	$0.281_{\pm 0.006}$
		QICE ↓	$3.248_{\pm 0.568}$	$1.217_{\pm 0.064}$	$1.282_{\pm 0.194}$	$1.419_{\pm 0.357}$	$1.344_{\pm 0.421}$	$\mathbf{1.036}_{\pm 0.081}$
ETTm1	192	CRPS ↓	$0.326_{\pm 0.006}$	$\mathbf{0.304}_{\pm 0.005}$	$0.305_{\pm 0.004}$	$0.315_{\pm 0.002}$	$0.314_{\pm 0.001}$	$0.309_{\pm 0.004}$
		QICE ↓	$3.731_{\pm 0.794}$	$1.392_{\pm 0.090}$	$1.304_{\pm 0.197}$	$1.710_{\pm 0.344}$	$1.576_{\pm 0.223}$	$\mathbf{1.139}_{\pm 0.247}$
ETTm1	336	CRPS ↓	$0.344_{\pm 0.011}$	$0.333_{\pm 0.005}$	$\mathbf{0.331}_{\pm 0.008}$	$0.344_{\pm 0.008}$	$0.341_{\pm 0.007}$	$0.333_{\pm 0.006}$
		QICE ↓	$3.242_{\pm 0.494}$	$1.861_{\pm 0.111}$	$1.608_{\pm 0.136}$	$2.146_{\pm 0.144}$	$2.104_{\pm 0.119}$	$\mathbf{1.569}_{\pm 0.056}$
ETTm1	720	CRPS ↓	$0.373_{\pm 0.006}$	$0.382_{\pm 0.009}$	$0.379_{\pm 0.007}$	$0.380_{\pm 0.007}$	$0.372_{\pm 0.006}$	$\mathbf{0.367}_{\pm 0.006}$
		QICE ↓	$3.522_{\pm 0.497}$	$2.982_{\pm 0.122}$	$\mathbf{1.963}_{\pm 0.213}$	$2.900_{\pm 0.129}$	$2.715_{\pm 0.099}$	$2.406_{\pm 0.028}$

Continued

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTm2	96	CRPS ↓	$0.235_{\pm 0.005}$	$\mathbf{0.214}_{\pm 0.004}$	$0.215_{\pm 0.004}$	$0.231_{\pm 0.005}$	$0.232_{\pm 0.005}$	$0.222_{\pm 0.003}$
		QICE ↓	$4.319_{\pm 0.114}$	$1.507_{\pm 0.123}$	$\mathbf{1.213}_{\pm 0.137}$	$2.951_{\pm 0.344}$	$3.022_{\pm 0.359}$	$2.740_{\pm 0.128}$
ETTm2	192	CRPS ↓	$0.276_{\pm 0.006}$	$0.253_{\pm 0.003}$	$\mathbf{0.253}_{\pm 0.003}$	$0.272_{\pm 0.006}$	$0.274_{\pm 0.006}$	$0.260_{\pm 0.003}$
		QICE ↓	$4.334_{\pm 0.245}$	$1.598_{\pm 0.215}$	$\mathbf{1.368}_{\pm 0.086}$	$2.775_{\pm 0.235}$	$2.899_{\pm 0.150}$	$2.759_{\pm 0.149}$
ETTm2	336	CRPS ↓	$0.306_{\pm 0.005}$	$0.280_{\pm 0.003}$	$\mathbf{0.278}_{\pm 0.003}$	$0.298_{\pm 0.005}$	$0.300_{\pm 0.005}$	$0.288_{\pm 0.005}$
		QICE ↓	$4.138_{\pm 0.130}$	$1.690_{\pm 0.554}$	$\mathbf{1.381}_{\pm 0.126}$	$2.912_{\pm 0.202}$	$2.914_{\pm 0.174}$	$2.558_{\pm 0.149}$
ETTm2	720	CRPS ↓	$0.462_{\pm 0.088}$	$0.351_{\pm 0.018}$	$\mathbf{0.344}_{\pm 0.013}$	$0.387_{\pm 0.032}$	$0.392_{\pm 0.034}$	$0.363_{\pm 0.026}$
		QICE ↓	$4.559_{\pm 0.641}$	$2.276_{\pm 0.150}$	$\mathbf{1.158}_{\pm 0.325}$	$2.600_{\pm 0.876}$	$2.623_{\pm 0.896}$	$2.153_{\pm 0.417}$
Electricity	96	CRPS ↓	$0.209_{\pm 0.002}$	$\mathbf{0.204}_{\pm 0.001}$	$0.205_{\pm 0.002}$	$0.207_{\pm 0.002}$	$0.208_{\pm 0.002}$	$0.206_{\pm 0.001}$
		QICE ↓	$1.267_{\pm 0.110}$	$0.408_{\pm 0.154}$	$0.442_{\pm 0.149}$	$\mathbf{0.365}_{\pm 0.140}$	$0.554_{\pm 0.058}$	$0.653_{\pm 0.087}$
Electricity	192	CRPS ↓	$0.219_{\pm 0.005}$	$\mathbf{0.214}_{\pm 0.005}$	$0.215_{\pm 0.007}$	$0.216_{\pm 0.005}$	$0.217_{\pm 0.005}$	$0.216_{\pm 0.006}$
		QICE ↓	$1.363_{\pm 0.153}$	$\mathbf{0.685}_{\pm 0.503}$	$0.873_{\pm 0.626}$	$0.695_{\pm 0.549}$	$0.834_{\pm 0.528}$	$0.943_{\pm 0.631}$
Electricity	336	CRPS ↓	$0.240_{\pm 0.006}$	$\mathbf{0.231}_{\pm 0.004}$	$0.233_{\pm 0.004}$	$0.234_{\pm 0.004}$	$0.235_{\pm 0.004}$	$0.235_{\pm 0.005}$
		QICE ↓	$1.569_{\pm 0.238}$	$0.632_{\pm 0.125}$	$0.684_{\pm 0.392}$	$\mathbf{0.626}_{\pm 0.097}$	$0.756_{\pm 0.084}$	$0.826_{\pm 0.124}$
Electricity	720	CRPS ↓	$0.283_{\pm 0.004}$	$\mathbf{0.266}_{\pm 0.003}$	$0.272_{\pm 0.004}$	$0.272_{\pm 0.003}$	$0.273_{\pm 0.003}$	$0.276_{\pm 0.004}$
		QICE ↓	$1.771_{\pm 0.252}$	$1.012_{\pm 0.235}$	$1.368_{\pm 0.265}$	$\mathbf{0.649}_{\pm 0.257}$	$0.768_{\pm 0.287}$	$1.557_{\pm 0.318}$
Exchange	96	CRPS ↓	$0.271_{\pm 0.010}$	$\mathbf{0.242}_{\pm 0.029}$	$0.270_{\pm 0.008}$	$0.292_{\pm 0.007}$	$0.290_{\pm 0.005}$	$0.289_{\pm 0.005}$
		QICE ↓	$5.156_{\pm 0.599}$	$\mathbf{4.064}_{\pm 0.295}$	$6.247_{\pm 0.148}$	$6.476_{\pm 0.332}$	$6.325_{\pm 0.327}$	$6.215_{\pm 0.284}$
Exchange	192	CRPS ↓	$0.343_{\pm 0.014}$	$\mathbf{0.287}_{\pm 0.013}$	$0.378_{\pm 0.013}$	$0.409_{\pm 0.016}$	$0.408_{\pm 0.014}$	$0.390_{\pm 0.006}$
		QICE ↓	$5.097_{\pm 0.404}$	$\mathbf{4.119}_{\pm 0.335}$	$7.867_{\pm 0.107}$	$7.925_{\pm 0.170}$	$7.757_{\pm 0.198}$	$7.323_{\pm 0.225}$
Exchange	336	CRPS ↓	$0.462_{\pm 0.026}$	$\mathbf{0.388}_{\pm 0.018}$	$0.542_{\pm 0.011}$	$0.565_{\pm 0.028}$	$0.565_{\pm 0.030}$	$0.538_{\pm 0.013}$
		QICE ↓	$5.938_{\pm 1.010}$	$\mathbf{4.106}_{\pm 0.892}$	$8.986_{\pm 0.271}$	$9.187_{\pm 0.224}$	$9.086_{\pm 0.341}$	$8.175_{\pm 0.556}$
Exchange	720	CRPS ↓	$0.690_{\pm 0.058}$	$\mathbf{0.580}_{\pm 0.020}$	$1.171_{\pm 0.206}$	$1.624_{\pm 0.090}$	$1.586_{\pm 0.083}$	$1.479_{\pm 0.071}$
		QICE ↓	$6.643_{\pm 1.832}$	$\mathbf{6.488}_{\pm 0.715}$	$12.054_{\pm 2.888}$	$14.813_{\pm 0.743}$	$14.632_{\pm 0.661}$	$14.413_{\pm 1.240}$
Weather	96	CRPS ↓	$0.204_{\pm 0.004}$	$0.172_{\pm 0.002}$	$\mathbf{0.171}_{\pm 0.003}$	$0.175_{\pm 0.002}$	$0.177_{\pm 0.002}$	$0.175_{\pm 0.003}$
		QICE ↓	$6.176_{\pm 0.143}$	$1.167_{\pm 0.083}$	$0.846_{\pm 0.148}$	$\mathbf{0.797}_{\pm 0.243}$	$0.999_{\pm 0.187}$	$0.957_{\pm 0.213}$
Weather	192	CRPS ↓	$0.233_{\pm 0.004}$	$0.204_{\pm 0.002}$	$\mathbf{0.203}_{\pm 0.001}$	$0.210_{\pm 0.002}$	$0.211_{\pm 0.002}$	$0.207_{\pm 0.002}$
		QICE ↓	$5.236_{\pm 0.153}$	$1.268_{\pm 0.106}$	$0.944_{\pm 0.098}$	$\mathbf{0.507}_{\pm 0.033}$	$0.581_{\pm 0.045}$	$0.698_{\pm 0.188}$
Weather	336	CRPS ↓	$0.266_{\pm 0.006}$	$\mathbf{0.239}_{\pm 0.005}$	$0.239_{\pm 0.005}$	$0.248_{\pm 0.006}$	$0.249_{\pm 0.006}$	$0.243_{\pm 0.005}$
		QICE ↓	$4.654_{\pm 0.427}$	$1.601_{\pm 0.005}$	$1.295_{\pm 0.075}$	$0.937_{\pm 0.103}$	$\mathbf{0.904}_{\pm 0.155}$	$1.001_{\pm 0.118}$
Weather	720	CRPS ↓	$0.309_{\pm 0.005}$	$\mathbf{0.281}_{\pm 0.001}$	$0.282_{\pm 0.001}$	$0.292_{\pm 0.001}$	$0.294_{\pm 0.001}$	$0.286_{\pm 0.001}$
		QICE ↓	$4.314_{\pm 0.327}$	$1.816_{\pm 0.146}$	$1.534_{\pm 0.121}$	$\mathbf{0.998}_{\pm 0.124}$	$1.000_{\pm 0.097}$	$1.241_{\pm 0.073}$
Traffic	96	CRPS ↓	$0.248_{\pm 0.001}$	$\mathbf{0.238}_{\pm 0.001}$	$0.240_{\pm 0.001}$	$0.244_{\pm 0.001}$	$0.245_{\pm 0.001}$	$0.245_{\pm 0.001}$
		QICE ↓	$2.598_{\pm 0.240}$	$1.401_{\pm 0.467}$	$1.264_{\pm 0.409}$	$\mathbf{1.186}_{\pm 0.545}$	$1.268_{\pm 0.538}$	$1.263_{\pm 0.519}$
Traffic	192	CRPS ↓	$0.256_{\pm 0.002}$	$\mathbf{0.246}_{\pm 0.001}$	$0.247_{\pm 0.001}$	$0.251_{\pm 0.001}$	$0.252_{\pm 0.001}$	$0.253_{\pm 0.001}$
		QICE ↓	$2.673_{\pm 0.113}$	$1.242_{\pm 0.300}$	$\mathbf{1.132}_{\pm 0.174}$	$1.151_{\pm 0.185}$	$1.236_{\pm 0.142}$	$1.272_{\pm 0.167}$
Traffic	336	CRPS ↓	$0.276_{\pm 0.001}$	$\mathbf{0.264}_{\pm 0.001}$	$0.266_{\pm 0.001}$	$0.269_{\pm 0.002}$	$0.271_{\pm 0.002}$	$0.271_{\pm 0.002}$
		QICE ↓	$2.842_{\pm 0.420}$	$1.352_{\pm 0.093}$	$\mathbf{1.274}_{\pm 0.114}$	$1.282_{\pm 0.197}$	$1.389_{\pm 0.152}$	$1.396_{\pm 0.142}$
Traffic	720	CRPS ↓	$0.282_{\pm 0.005}$	$\mathbf{0.273}_{\pm 0.010}$	$0.274_{\pm 0.008}$	$0.276_{\pm 0.005}$	$0.277_{\pm 0.005}$	$0.277_{\pm 0.005}$
		QICE ↓	$2.695_{\pm 0.452}$	$1.827_{\pm 0.274}$	$\mathbf{1.726}_{\pm 0.120}$	$1.786_{\pm 0.062}$	$1.866_{\pm 0.058}$	$1.841_{\pm 0.085}$
ILI	24	CRPS ↓	$\mathbf{0.345}_{\pm 0.016}$	$0.348_{\pm 0.006}$	$0.463_{\pm 0.021}$	$0.463_{\pm 0.021}$	$0.460_{\pm 0.020}$	$0.460_{\pm 0.020}$
		QICE ↓	$8.865_{\pm 0.225}$	$\mathbf{6.972}_{\pm 0.694}$	$8.504_{\pm 1.087}$	$8.504_{\pm 1.087}$	$9.405_{\pm 0.598}$	$9.405_{\pm 0.598}$
ILI	36	CRPS ↓	$0.841_{\pm 0.056}$	$0.834_{\pm 0.055}$	$\mathbf{0.795}_{\pm 0.056}$	$\mathbf{0.795}_{\pm 0.056}$	$0.828_{\pm 0.052}$	$0.828_{\pm 0.052}$
		QICE ↓	$9.262_{\pm 0.238}$	$7.259_{\pm 2.083}$	$\mathbf{6.130}_{\pm 1.540}$	$\mathbf{6.130}_{\pm 1.540}$	$7.397_{\pm 1.096}$	$7.397_{\pm 1.096}$
ILI	48	CRPS ↓	$0.838_{\pm 0.062}$	$\mathbf{0.819}_{\pm 0.059}$	$0.820_{\pm 0.079}$	$0.820_{\pm 0.079}$	$0.845_{\pm 0.069}$	$0.845_{\pm 0.069}$
		QICE ↓	$10.335_{\pm 0.715}$	$\mathbf{7.772}_{\pm 0.534}$	$7.950_{\pm 1.473}$	$7.950_{\pm 1.473}$	$8.018_{\pm 1.508}$	$8.018_{\pm 1.508}$
ILI	60	CRPS ↓	$0.802_{\pm 0.059}$	$0.796_{\pm 0.052}$	$\mathbf{0.762}_{\pm 0.066}$	$\mathbf{0.762}_{\pm 0.066}$	$0.790_{\pm 0.066}$	$0.790_{\pm 0.066}$
		QICE ↓	$10.873_{\pm 1.133}$	$8.421_{\pm 0.657}$	$\mathbf{7.516}_{\pm 1.372}$	$\mathbf{7.516}_{\pm 1.372}$	$7.976_{\pm 1.339}$	$7.976_{\pm 1.339}$

TABLE S2: PPM: all 7 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTh1	192	CRPS ↓	$0.337_{\pm 0.000}$	$\mathbf{0.326}_{\pm 0.001}$	$0.349_{\pm 0.001}$	$0.347_{\pm 0.000}$	$0.346_{\pm 0.000}$	$0.340_{\pm 0.000}$
		QICE ↓	$1.813_{\pm 0.011}$	$1.621_{\pm 0.013}$	$2.154_{\pm 0.064}$	$1.275_{\pm 0.065}$	$\mathbf{1.261}_{\pm 0.021}$	$2.321_{\pm 0.042}$
ETTh2	192	CRPS ↓	$0.305_{\pm 0.001}$	$0.302_{\pm 0.001}$	$0.308_{\pm 0.000}$	$\mathbf{0.302}_{\pm 0.001}$	$0.303_{\pm 0.001}$	$0.303_{\pm 0.001}$
		QICE ↓	$3.521_{\pm 0.114}$	$1.476_{\pm 0.089}$	$2.264_{\pm 0.033}$	$\mathbf{1.135}_{\pm 0.034}$	$1.287_{\pm 0.032}$	$1.604_{\pm 0.161}$
ETTm1	192	CRPS ↓	$0.314_{\pm 0.001}$	$\mathbf{0.299}_{\pm 0.000}$	$0.303_{\pm 0.001}$	$0.323_{\pm 0.001}$	$0.323_{\pm 0.001}$	$0.304_{\pm 0.001}$
		QICE ↓	$2.176_{\pm 0.023}$	$0.846_{\pm 0.192}$	$\mathbf{0.824}_{\pm 0.240}$	$1.196_{\pm 0.017}$	$1.149_{\pm 0.083}$	$1.105_{\pm 0.025}$
ETTm2	192	CRPS ↓	$0.236_{\pm 0.000}$	$\mathbf{0.228}_{\pm 0.001}$	$0.238_{\pm 0.000}$	$0.239_{\pm 0.000}$	$0.240_{\pm 0.000}$	$0.237_{\pm 0.000}$
		QICE ↓	$1.971_{\pm 0.119}$	$\mathbf{1.181}_{\pm 0.058}$	$2.154_{\pm 0.215}$	$1.941_{\pm 0.020}$	$1.860_{\pm 0.074}$	$1.431_{\pm 0.031}$
Electricity	192	CRPS ↓	$0.206_{\pm 0.000}$	$\mathbf{0.201}_{\pm 0.000}$	$0.201_{\pm 0.000}$	$0.205_{\pm 0.000}$	$0.206_{\pm 0.000}$	$0.205_{\pm 0.000}$
		QICE ↓	$3.929_{\pm 0.064}$	$1.294_{\pm 0.111}$	$1.222_{\pm 0.088}$	$\mathbf{0.957}_{\pm 0.094}$	$1.110_{\pm 0.117}$	$1.724_{\pm 0.086}$
Weather	192	CRPS ↓	$0.215_{\pm 0.001}$	$\mathbf{0.201}_{\pm 0.001}$	$0.220_{\pm 0.002}$	$0.213_{\pm 0.001}$	$0.215_{\pm 0.001}$	$0.215_{\pm 0.000}$
		QICE ↓	$4.454_{\pm 0.024}$	$2.690_{\pm 0.417}$	$\mathbf{1.486}_{\pm 0.189}$	$2.997_{\pm 0.204}$	$3.602_{\pm 0.020}$	$2.549_{\pm 0.088}$
Traffic	192	CRPS ↓	$0.255_{\pm 0.001}$	$0.253_{\pm 0.000}$	$\mathbf{0.253}_{\pm 0.000}$	$0.255_{\pm 0.001}$	$0.254_{\pm 0.001}$	$0.258_{\pm 0.000}$
		QICE ↓	$3.073_{\pm 0.244}$	$\mathbf{0.486}_{\pm 0.040}$	$0.606_{\pm 0.062}$	$0.575_{\pm 0.058}$	$0.876_{\pm 0.108}$	$0.791_{\pm 0.169}$

TABLE S3: TMDM: all 9 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTh1	192	CRPS ↓	$0.453_{\pm 0.039}$	$\mathbf{0.441}_{\pm 0.021}$	$0.449_{\pm 0.025}$	$0.456_{\pm 0.026}$	$0.457_{\pm 0.028}$	$0.455_{\pm 0.027}$
		QICE ↓	$3.272_{\pm 1.558}$	$\mathbf{2.211}_{\pm 1.036}$	$2.800_{\pm 0.646}$	$3.119_{\pm 0.786}$	$3.142_{\pm 0.818}$	$3.176_{\pm 1.121}$
ETTh2	192	CRPS ↓	$0.401_{\pm 0.022}$	$0.385_{\pm 0.026}$	$0.392_{\pm 0.026}$	$\mathbf{0.381}_{\pm 0.024}$	$0.387_{\pm 0.024}$	$0.393_{\pm 0.021}$
		QICE ↓	$4.488_{\pm 0.828}$	$1.866_{\pm 0.517}$	$1.979_{\pm 0.480}$	$\mathbf{1.314}_{\pm 0.388}$	$1.471_{\pm 0.385}$	$2.115_{\pm 0.206}$
ETTm1	192	CRPS ↓	$0.347_{\pm 0.017}$	$0.338_{\pm 0.016}$	$0.346_{\pm 0.017}$	$0.340_{\pm 0.019}$	$0.340_{\pm 0.018}$	$\mathbf{0.335}_{\pm 0.017}$
		QICE ↓	$2.615_{\pm 0.547}$	$0.850_{\pm 0.284}$	$\mathbf{0.457}_{\pm 0.082}$	$0.950_{\pm 0.010}$	$0.721_{\pm 0.032}$	$0.562_{\pm 0.074}$
ETTm2	192	CRPS ↓	$0.298_{\pm 0.032}$	$0.291_{\pm 0.030}$	$\mathbf{0.281}_{\pm 0.031}$	$0.299_{\pm 0.034}$	$0.301_{\pm 0.035}$	$0.283_{\pm 0.023}$
		QICE ↓	$2.180_{\pm 0.751}$	$2.205_{\pm 0.388}$	$\mathbf{1.837}_{\pm 0.808}$	$2.529_{\pm 0.765}$	$2.523_{\pm 0.792}$	$1.961_{\pm 0.562}$
Electricity	192	CRPS ↓	$0.442_{\pm 0.002}$	$\mathbf{0.248}_{\pm 0.001}$	$0.251_{\pm 0.002}$	$0.252_{\pm 0.002}$	$0.263_{\pm 0.002}$	$0.264_{\pm 0.002}$
		QICE ↓	$10.416_{\pm 0.025}$	$\mathbf{1.199}_{\pm 0.196}$	$1.457_{\pm 0.268}$	$1.242_{\pm 0.177}$	$1.580_{\pm 0.231}$	$1.367_{\pm 0.166}$
Exchange	192	CRPS ↓	$0.392_{\pm 0.047}$	$\mathbf{0.374}_{\pm 0.063}$	$0.419_{\pm 0.068}$	$0.417_{\pm 0.069}$	$0.418_{\pm 0.067}$	$0.418_{\pm 0.065}$
		QICE ↓	$6.437_{\pm 1.957}$	$\mathbf{4.190}_{\pm 0.669}$	$5.192_{\pm 0.696}$	$5.039_{\pm 0.776}$	$5.101_{\pm 0.569}$	$5.140_{\pm 0.523}$
Traffic	192	CRPS ↓	$0.558_{\pm 0.003}$	$\mathbf{0.309}_{\pm 0.006}$	$0.311_{\pm 0.007}$	$0.332_{\pm 0.009}$	$0.345_{\pm 0.009}$	$0.345_{\pm 0.009}$
		QICE ↓	$10.604_{\pm 0.091}$	$1.650_{\pm 0.279}$	$1.594_{\pm 0.274}$	$1.616_{\pm 0.216}$	$1.637_{\pm 0.223}$	$\mathbf{1.526}_{\pm 0.207}$
ILI	36	CRPS ↓	$\mathbf{0.810}_{\pm 0.100}$	$0.820_{\pm 0.107}$	$0.916_{\pm 0.065}$	$0.916_{\pm 0.065}$	$0.914_{\pm 0.068}$	$0.914_{\pm 0.068}$
		QICE ↓	$\mathbf{5.201}_{\pm 0.869}$	$6.026_{\pm 1.178}$	$8.032_{\pm 0.604}$	$8.032_{\pm 0.604}$	$7.561_{\pm 0.406}$	$7.561_{\pm 0.406}$
SolarEnergy	192	CRPS ↓	$0.341_{\pm 0.004}$	$0.191_{\pm 0.003}$	$\mathbf{0.190}_{\pm 0.001}$	$0.239_{\pm 0.003}$	$0.246_{\pm 0.003}$	$0.244_{\pm 0.003}$
		QICE ↓	$9.262_{\pm 0.122}$	$0.991_{\pm 0.275}$	$1.162_{\pm 0.025}$	$0.827_{\pm 0.066}$	$1.057_{\pm 0.066}$	$\mathbf{0.770}_{\pm 0.048}$

TABLE S4: Diffusion-TS: all 9 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTh1	192	CRPS ↓	$0.659_{\pm 0.068}$	$\mathbf{0.586}_{\pm 0.033}$	$0.622_{\pm 0.070}$	$0.660_{\pm 0.115}$	$0.662_{\pm 0.112}$	$0.665_{\pm 0.102}$
		QICE ↓	$2.816_{\pm 1.683}$	$2.540_{\pm 1.074}$	$3.034_{\pm 0.822}$	$2.626_{\pm 1.437}$	$2.530_{\pm 1.480}$	$\mathbf{2.455}_{\pm 1.258}$
ETTh2	192	CRPS ↓	$1.147_{\pm 0.124}$	$0.794_{\pm 0.147}$	$\mathbf{0.771}_{\pm 0.043}$	$0.943_{\pm 0.120}$	$0.952_{\pm 0.116}$	$0.782_{\pm 0.095}$
		QICE ↓	$7.742_{\pm 0.110}$	$3.926_{\pm 2.965}$	$\mathbf{2.098}_{\pm 0.471}$	$4.529_{\pm 1.321}$	$4.430_{\pm 1.347}$	$2.641_{\pm 1.069}$
ETTm1	192	CRPS ↓	$0.873_{\pm 0.056}$	$\mathbf{0.565}_{\pm 0.035}$	$0.578_{\pm 0.033}$	$0.845_{\pm 0.099}$	$0.811_{\pm 0.044}$	$0.807_{\pm 0.042}$
		QICE ↓	$13.310_{\pm 0.288}$	$\mathbf{2.456}_{\pm 0.686}$	$2.489_{\pm 1.089}$	$3.054_{\pm 0.302}$	$11.605_{\pm 0.554}$	$10.696_{\pm 0.658}$
ETTm2	192	CRPS ↓	$0.940_{\pm 0.288}$	$\mathbf{0.616}_{\pm 0.132}$	$0.617_{\pm 0.132}$	$0.733_{\pm 0.214}$	$0.904_{\pm 0.270}$	$0.891_{\pm 0.260}$
		QICE ↓	$15.118_{\pm 0.276}$	$4.394_{\pm 1.292}$	$\mathbf{4.141}_{\pm 0.853}$	$6.359_{\pm 0.944}$	$14.276_{\pm 0.482}$	$14.342_{\pm 0.495}$
Electricity	192	CRPS ↓	$0.453_{\pm 0.010}$	$\mathbf{0.395}_{\pm 0.005}$	$0.402_{\pm 0.004}$	$0.409_{\pm 0.002}$	$0.429_{\pm 0.004}$	$0.431_{\pm 0.006}$
		QICE ↓	$10.318_{\pm 0.294}$	$\mathbf{2.745}_{\pm 0.262}$	$3.178_{\pm 0.164}$	$3.212_{\pm 0.308}$	$6.398_{\pm 0.094}$	$6.612_{\pm 0.291}$
Exchange	192	CRPS ↓	$0.998_{\pm 0.402}$	$\mathbf{0.691}_{\pm 0.103}$	$0.720_{\pm 0.132}$	$0.711_{\pm 0.132}$	$0.814_{\pm 0.291}$	$0.814_{\pm 0.291}$
		QICE ↓	$9.985_{\pm 2.311}$	$4.461_{\pm 0.577}$	$4.376_{\pm 0.514}$	$\mathbf{4.363}_{\pm 0.402}$	$6.747_{\pm 3.524}$	$6.747_{\pm 3.524}$
Traffic	192	CRPS ↓	$0.432_{\pm 0.024}$	$0.334_{\pm 0.013}$	$\mathbf{0.334}_{\pm 0.013}$	$0.349_{\pm 0.015}$	$0.416_{\pm 0.024}$	$0.416_{\pm 0.024}$
		QICE ↓	$14.337_{\pm 0.493}$	$1.256_{\pm 0.338}$	$1.105_{\pm 0.309}$	$\mathbf{0.864}_{\pm 0.291}$	$12.732_{\pm 0.950}$	$12.732_{\pm 0.950}$
ILI	36	CRPS ↓	$\mathbf{1.025}_{\pm 0.049}$	$1.115_{\pm 0.038}$	$1.095_{\pm 0.041}$	$1.095_{\pm 0.041}$	$1.092_{\pm 0.043}$	$1.092_{\pm 0.043}$
		QICE ↓	$\mathbf{4.561}_{\pm 1.212}$	$8.312_{\pm 1.332}$	$7.471_{\pm 1.311}$	$7.471_{\pm 1.311}$	$7.291_{\pm 1.319}$	$7.291_{\pm 1.319}$
SolarEnergy	192	CRPS ↓	$0.365_{\pm 0.016}$	$0.237_{\pm 0.007}$	$\mathbf{0.232}_{\pm 0.001}$	$0.329_{\pm 0.027}$	$0.331_{\pm 0.015}$	$0.300_{\pm 0.010}$
		QICE ↓	$8.069_{\pm 2.068}$	$4.167_{\pm 1.194}$	$\mathbf{2.611}_{\pm 0.689}$	$4.450_{\pm 0.426}$	$4.598_{\pm 1.107}$	$4.140_{\pm 1.186}$

TABLE S5: TimeDiff: all 9 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTh1	192	CRPS ↓	$0.474_{\pm 0.008}$	$\mathbf{0.398}_{\pm 0.002}$	$0.408_{\pm 0.009}$	$0.403_{\pm 0.011}$	$0.435_{\pm 0.013}$	$0.436_{\pm 0.015}$
		QICE ↓	$13.863_{\pm 0.515}$	$2.332_{\pm 0.544}$	$\mathbf{2.004}_{\pm 0.478}$	$3.117_{\pm 0.366}$	$11.496_{\pm 1.093}$	$11.206_{\pm 1.179}$
ETTh2	192	CRPS ↓	$0.491_{\pm 0.098}$	$0.415_{\pm 0.078}$	$\mathbf{0.391}_{\pm 0.078}$	$0.392_{\pm 0.082}$	$0.455_{\pm 0.097}$	$0.456_{\pm 0.092}$
		QICE ↓	$13.457_{\pm 0.655}$	$4.288_{\pm 1.217}$	$\mathbf{3.131}_{\pm 1.445}$	$3.188_{\pm 2.404}$	$10.771_{\pm 1.658}$	$10.809_{\pm 1.636}$
ETTm1	192	CRPS ↓	$0.490_{\pm 0.008}$	$\mathbf{0.387}_{\pm 0.004}$	$0.390_{\pm 0.006}$	$0.390_{\pm 0.006}$	$0.471_{\pm 0.013}$	$0.469_{\pm 0.014}$
		QICE ↓	$14.646_{\pm 0.337}$	$1.500_{\pm 0.283}$	$\mathbf{1.244}_{\pm 0.097}$	$1.675_{\pm 0.157}$	$13.131_{\pm 0.935}$	$13.117_{\pm 0.959}$
ETTm2	192	CRPS ↓	$0.315_{\pm 0.005}$	$\mathbf{0.258}_{\pm 0.005}$	$0.259_{\pm 0.005}$	$0.261_{\pm 0.005}$	$0.299_{\pm 0.007}$	$0.298_{\pm 0.007}$
		QICE ↓	$12.446_{\pm 0.847}$	$1.573_{\pm 0.331}$	$\mathbf{1.563}_{\pm 0.324}$	$1.755_{\pm 0.145}$	$10.299_{\pm 1.435}$	$9.978_{\pm 1.352}$
Electricity	192	CRPS ↓	$0.749_{\pm 0.011}$	$\mathbf{0.525}_{\pm 0.006}$	$0.526_{\pm 0.006}$	$0.531_{\pm 0.009}$	$0.734_{\pm 0.011}$	$0.734_{\pm 0.011}$
		QICE ↓	$15.524_{\pm 0.024}$	$0.773_{\pm 0.118}$	$\mathbf{0.708}_{\pm 0.159}$	$1.433_{\pm 0.160}$	$15.047_{\pm 0.031}$	$15.047_{\pm 0.031}$
Exchange	192	CRPS ↓	$0.443_{\pm 0.010}$	$\mathbf{0.365}_{\pm 0.024}$	$0.410_{\pm 0.007}$	$0.399_{\pm 0.023}$	$0.434_{\pm 0.010}$	$0.433_{\pm 0.010}$
		QICE ↓	$14.438_{\pm 0.680}$	$\mathbf{5.771}_{\pm 1.500}$	$6.569_{\pm 0.355}$	$6.613_{\pm 0.394}$	$12.889_{\pm 1.031}$	$12.889_{\pm 1.031}$
Traffic	192	CRPS ↓	$0.778_{\pm 0.001}$	$\mathbf{0.565}_{\pm 0.001}$	$0.565_{\pm 0.001}$	$0.572_{\pm 0.001}$	$0.763_{\pm 0.001}$	$0.763_{\pm 0.001}$
		QICE ↓	$15.478_{\pm 0.009}$	$0.791_{\pm 0.046}$	$0.799_{\pm 0.033}$	$\mathbf{0.486}_{\pm 0.058}$	$14.959_{\pm 0.018}$	$14.959_{\pm 0.018}$
ILI	36	CRPS ↓	$1.123_{\pm 0.018}$	$\mathbf{0.881}_{\pm 0.105}$	$0.944_{\pm 0.051}$	$0.944_{\pm 0.051}$	$1.041_{\pm 0.023}$	$1.041_{\pm 0.023}$
		QICE ↓	$13.460_{\pm 1.395}$	$\mathbf{7.328}_{\pm 0.816}$	$9.280_{\pm 1.398}$	$9.280_{\pm 1.398}$	$11.323_{\pm 1.461}$	$11.323_{\pm 1.461}$
SolarEnergy	192	CRPS ↓	$0.700_{\pm 0.002}$	$\mathbf{0.396}_{\pm 0.002}$	$0.406_{\pm 0.001}$	$0.487_{\pm 0.006}$	$0.681_{\pm 0.003}$	$0.681_{\pm 0.003}$
		QICE ↓	$15.123_{\pm 0.262}$	$7.018_{\pm 0.617}$	$\mathbf{1.745}_{\pm 0.053}$	$3.554_{\pm 0.158}$	$14.374_{\pm 0.293}$	$14.374_{\pm 0.293}$

TABLE S6: NsDiff: all 9 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
ETTh1	192	CRPS ↓	$0.371_{\pm 0.010}$	$0.389_{\pm 0.008}$	$0.399_{\pm 0.016}$	$\mathbf{0.371}_{\pm 0.008}$	$0.372_{\pm 0.009}$	$0.381_{\pm 0.009}$
		QICE ↓	$2.454_{\pm 0.074}$	$1.510_{\pm 0.508}$	$2.198_{\pm 0.146}$	$1.832_{\pm 0.162}$	$1.765_{\pm 0.330}$	$\mathbf{1.492}_{\pm 0.172}$
ETTh2	192	CRPS ↓	$0.385_{\pm 0.026}$	$0.372_{\pm 0.022}$	$0.376_{\pm 0.026}$	$\mathbf{0.371}_{\pm 0.024}$	$0.375_{\pm 0.025}$	$0.376_{\pm 0.030}$
		QICE ↓	$3.597_{\pm 1.031}$	$2.403_{\pm 0.774}$	$\mathbf{2.110}_{\pm 1.121}$	$2.476_{\pm 1.293}$	$2.453_{\pm 1.234}$	$2.414_{\pm 0.726}$
ETTm1	192	CRPS ↓	$0.343_{\pm 0.018}$	$0.342_{\pm 0.012}$	$0.338_{\pm 0.013}$	$0.339_{\pm 0.014}$	$0.340_{\pm 0.014}$	$\mathbf{0.336}_{\pm 0.011}$
		QICE ↓	$1.927_{\pm 0.105}$	$1.425_{\pm 0.155}$	$1.155_{\pm 0.092}$	$1.071_{\pm 0.187}$	$1.121_{\pm 0.099}$	$\mathbf{0.700}_{\pm 0.017}$
ETTm2	192	CRPS ↓	$0.275_{\pm 0.006}$	$\mathbf{0.264}_{\pm 0.008}$	$0.265_{\pm 0.004}$	$0.276_{\pm 0.004}$	$0.278_{\pm 0.004}$	$0.268_{\pm 0.005}$
		QICE ↓	$3.181_{\pm 0.203}$	$\mathbf{1.750}_{\pm 0.155}$	$1.964_{\pm 0.474}$	$2.584_{\pm 0.603}$	$2.551_{\pm 0.624}$	$2.045_{\pm 0.634}$
Electricity	192	CRPS ↓	$0.277_{\pm 0.002}$	$\mathbf{0.235}_{\pm 0.004}$	$0.237_{\pm 0.004}$	$0.237_{\pm 0.003}$	$0.240_{\pm 0.003}$	$0.241_{\pm 0.003}$
		QICE ↓	$6.542_{\pm 0.086}$	$\mathbf{1.337}_{\pm 0.205}$	$1.490_{\pm 0.226}$	$1.470_{\pm 0.166}$	$1.508_{\pm 0.203}$	$1.669_{\pm 0.184}$
Exchange	192	CRPS ↓	$0.352_{\pm 0.052}$	$\mathbf{0.325}_{\pm 0.039}$	$0.398_{\pm 0.050}$	$0.426_{\pm 0.080}$	$0.404_{\pm 0.056}$	$0.404_{\pm 0.056}$
		QICE ↓	$7.556_{\pm 1.304}$	$\mathbf{4.360}_{\pm 1.136}$	$6.877_{\pm 1.523}$	$6.702_{\pm 2.502}$	$7.142_{\pm 1.849}$	$7.142_{\pm 1.849}$
Traffic	192	CRPS ↓	$0.374_{\pm 0.005}$	$0.276_{\pm 0.009}$	$\mathbf{0.276}_{\pm 0.009}$	$0.292_{\pm 0.009}$	$0.297_{\pm 0.009}$	$0.298_{\pm 0.009}$
		QICE ↓	$8.681_{\pm 0.185}$	$1.960_{\pm 0.422}$	$1.838_{\pm 0.326}$	$\mathbf{1.733}_{\pm 0.427}$	$1.790_{\pm 0.336}$	$1.789_{\pm 0.322}$
ILI	36	CRPS ↓	$\mathbf{0.851}_{\pm 0.184}$	$0.908_{\pm 0.226}$	$0.947_{\pm 0.194}$	$0.947_{\pm 0.194}$	$0.946_{\pm 0.195}$	$0.946_{\pm 0.195}$
		QICE ↓	$\mathbf{6.016}_{\pm 0.936}$	$8.534_{\pm 1.345}$	$9.042_{\pm 1.054}$	$9.042_{\pm 1.054}$	$9.074_{\pm 1.282}$	$9.074_{\pm 1.282}$
SolarEnergy	192	CRPS ↓	$0.288_{\pm 0.015}$	$0.188_{\pm 0.008}$	$\mathbf{0.187}_{\pm 0.008}$	$0.240_{\pm 0.008}$	$0.244_{\pm 0.008}$	$0.237_{\pm 0.008}$
		QICE ↓	$7.461_{\pm 0.130}$	$1.551_{\pm 0.564}$	$1.523_{\pm 0.315}$	$1.347_{\pm 0.252}$	$1.400_{\pm 0.130}$	$\mathbf{1.110}_{\pm 0.340}$

TABLE S7: DeepSTUQ: all 4 complete-test configurations. FSPC uses source-filtered centered templates and the affine-plus-state mean with validation-selected blend strength. The ablation without mean control omits centering and mean adjustment. Subscripts: three-run sample SD; bold: lowest unrounded row mean. All metrics are minimized; QICE is in percentage points.

Dataset	H	Metric	Native	FSPC	w/o Mean Control	Global DC	Global PIT	Cond. PIT
PEMS03	12	CRPS ↓	$11.181_{\pm 0.055}$	$11.093_{\pm 0.044}$	$\mathbf{11.092}_{\pm 0.044}$	$11.157_{\pm 0.048}$	$11.169_{\pm 0.048}$	$11.123_{\pm 0.043}$
		QICE ↓	$1.338_{\pm 0.040}$	$0.909_{\pm 0.070}$	$0.855_{\pm 0.063}$	$\mathbf{0.844}_{\pm 0.063}$	$0.967_{\pm 0.058}$	$0.973_{\pm 0.068}$
PEMS04	12	CRPS ↓	$14.166_{\pm 0.037}$	$14.015_{\pm 0.025}$	$\mathbf{14.013}_{\pm 0.023}$	$14.055_{\pm 0.029}$	$14.063_{\pm 0.029}$	$14.047_{\pm 0.027}$
		QICE ↓	$1.989_{\pm 0.075}$	$0.373_{\pm 0.042}$	$\mathbf{0.362}_{\pm 0.043}$	$0.362_{\pm 0.055}$	$0.512_{\pm 0.101}$	$0.495_{\pm 0.058}$
PEMS07	12	CRPS ↓	$15.235_{\pm 0.032}$	$\mathbf{15.031}_{\pm 0.023}$	$15.033_{\pm 0.012}$	$15.074_{\pm 0.020}$	$15.080_{\pm 0.019}$	$15.079_{\pm 0.016}$
		QICE ↓	$2.448_{\pm 0.047}$	$0.255_{\pm 0.093}$	$\mathbf{0.241}_{\pm 0.153}$	$0.281_{\pm 0.144}$	$0.445_{\pm 0.112}$	$0.262_{\pm 0.144}$
PEMS08	12	CRPS ↓	$11.865_{\pm 0.053}$	$\mathbf{11.617}_{\pm 0.043}$	$11.619_{\pm 0.044}$	$11.633_{\pm 0.038}$	$11.632_{\pm 0.038}$	$11.619_{\pm 0.037}$
		QICE ↓	$2.829_{\pm 0.062}$	$\mathbf{0.220}_{\pm 0.156}$	$0.242_{\pm 0.119}$	$0.245_{\pm 0.119}$	$0.388_{\pm 0.057}$	$0.256_{\pm 0.105}$

DETAILED FULL-METHOD FSPC RESULTS

Tables S8–S21 report all twelve metrics for the full FSPC method across all 83 configurations. Each backbone has a point/distribution table and a coverage/event table. Point metrics use the ensemble mean, except Median MAE, which uses the ensemble median. State and pooled errors use the common training-defined diagnostic partition; undefined coverage cells remain marked. RMSE is the square root of MSE within each run, before averaging across runs. Each value is the mean over three training runs with its sample standard deviation.

TABLE S8: K^2VAE: full FSPC point/distribution results on all 36 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
ETTh1	96	$0.439_{\pm 0.010}$	$0.456_{\pm 0.011}$	$0.663_{\pm 0.007}$	$0.472_{\pm 0.024}$	$0.353_{\pm 0.016}$	$3.130_{\pm 0.298}$
ETTh1	192	$0.543_{\pm 0.008}$	$0.530_{\pm 0.005}$	$0.737_{\pm 0.005}$	$0.554_{\pm 0.070}$	$0.420_{\pm 0.044}$	$3.780_{\pm 0.076}$
ETTh1	336	$0.636_{\pm 0.012}$	$0.583_{\pm 0.008}$	$0.797_{\pm 0.007}$	$0.580_{\pm 0.058}$	$0.441_{\pm 0.027}$	$4.359_{\pm 0.170}$
ETTh1	720	$0.530_{\pm 0.011}$	$0.514_{\pm 0.012}$	$0.728_{\pm 0.008}$	$0.625_{\pm 0.103}$	$0.459_{\pm 0.067}$	$4.284_{\pm 0.738}$
ETTh2	96	$0.317_{\pm 0.014}$	$0.360_{\pm 0.008}$	$0.563_{\pm 0.013}$	$0.364_{\pm 0.007}$	$0.285_{\pm 0.007}$	$1.356_{\pm 0.125}$
ETTh2	192	$0.375_{\pm 0.011}$	$0.398_{\pm 0.006}$	$0.612_{\pm 0.009}$	$0.410_{\pm 0.010}$	$0.315_{\pm 0.004}$	$1.381_{\pm 0.316}$
ETTh2	336	$0.909_{\pm 0.411}$	$0.532_{\pm 0.061}$	$0.934_{\pm 0.234}$	$0.542_{\pm 0.063}$	$0.436_{\pm 0.064}$	$2.192_{\pm 0.950}$
ETTh2	720	$0.490_{\pm 0.041}$	$0.500_{\pm 0.027}$	$0.700_{\pm 0.029}$	$0.512_{\pm 0.033}$	$0.396_{\pm 0.030}$	$2.635_{\pm 0.614}$
ETTm1	96	$0.336_{\pm 0.010}$	$0.366_{\pm 0.004}$	$0.580_{\pm 0.009}$	$0.373_{\pm 0.005}$	$0.280_{\pm 0.005}$	$1.217_{\pm 0.064}$
ETTm1	192	$0.387_{\pm 0.014}$	$0.404_{\pm 0.003}$	$0.622_{\pm 0.011}$	$0.408_{\pm 0.007}$	$0.304_{\pm 0.005}$	$1.392_{\pm 0.090}$
ETTm1	336	$0.441_{\pm 0.018}$	$0.435_{\pm 0.013}$	$0.664_{\pm 0.014}$	$0.449_{\pm 0.007}$	$0.333_{\pm 0.005}$	$1.861_{\pm 0.111}$
ETTm1	720	$0.510_{\pm 0.018}$	$0.494_{\pm 0.012}$	$0.714_{\pm 0.012}$	$0.512_{\pm 0.018}$	$0.382_{\pm 0.009}$	$2.982_{\pm 0.122}$
ETTm2	96	$0.186_{\pm 0.004}$	$0.272_{\pm 0.006}$	$0.432_{\pm 0.005}$	$0.274_{\pm 0.008}$	$0.214_{\pm 0.004}$	$1.507_{\pm 0.123}$
ETTm2	192	$0.259_{\pm 0.005}$	$0.317_{\pm 0.004}$	$0.509_{\pm 0.005}$	$0.321_{\pm 0.005}$	$0.253_{\pm 0.003}$	$1.598_{\pm 0.215}$
ETTm2	336	$0.311_{\pm 0.007}$	$0.352_{\pm 0.008}$	$0.558_{\pm 0.006}$	$0.353_{\pm 0.004}$	$0.280_{\pm 0.003}$	$1.690_{\pm 0.554}$
ETTm2	720	$0.651_{\pm 0.320}$	$0.457_{\pm 0.025}$	$0.792_{\pm 0.191}$	$0.456_{\pm 0.024}$	$0.351_{\pm 0.018}$	$2.276_{\pm 0.150}$
Electricity	96	$0.170_{\pm 0.001}$	$0.273_{\pm 0.001}$	$0.412_{\pm 0.001}$	$0.274_{\pm 0.002}$	$0.204_{\pm 0.001}$	$0.408_{\pm 0.154}$
Electricity	192	$0.184_{\pm 0.008}$	$0.285_{\pm 0.008}$	$0.428_{\pm 0.009}$	$0.285_{\pm 0.007}$	$0.214_{\pm 0.005}$	$0.685_{\pm 0.503}$
Electricity	336	$0.235_{\pm 0.021}$	$0.302_{\pm 0.004}$	$0.485_{\pm 0.022}$	$0.305_{\pm 0.003}$	$0.231_{\pm 0.004}$	$0.632_{\pm 0.125}$
Electricity	720	$0.264_{\pm 0.003}$	$0.344_{\pm 0.005}$	$0.514_{\pm 0.003}$	$0.347_{\pm 0.004}$	$0.266_{\pm 0.003}$	$1.012_{\pm 0.235}$
Exchange	96	$0.144_{\pm 0.014}$	$0.269_{\pm 0.014}$	$0.379_{\pm 0.019}$	$0.322_{\pm 0.058}$	$0.242_{\pm 0.029}$	$4.064_{\pm 0.295}$
Exchange	192	$0.277_{\pm 0.018}$	$0.367_{\pm 0.010}$	$0.526_{\pm 0.017}$	$0.370_{\pm 0.013}$	$0.287_{\pm 0.013}$	$4.119_{\pm 0.335}$
Exchange	336	$0.500_{\pm 0.033}$	$0.506_{\pm 0.015}$	$0.707_{\pm 0.023}$	$0.498_{\pm 0.013}$	$0.388_{\pm 0.018}$	$4.106_{\pm 0.892}$
Exchange	720	$1.062_{\pm 0.125}$	$0.623_{\pm 0.022}$	$1.029_{\pm 0.061}$	$0.628_{\pm 0.021}$	$0.580_{\pm 0.020}$	$6.488_{\pm 0.715}$
Weather	96	$0.171_{\pm 0.006}$	$0.233_{\pm 0.004}$	$0.414_{\pm 0.007}$	$0.222_{\pm 0.004}$	$0.172_{\pm 0.002}$	$1.167_{\pm 0.083}$
Weather	192	$0.217_{\pm 0.003}$	$0.281_{\pm 0.003}$	$0.466_{\pm 0.003}$	$0.264_{\pm 0.003}$	$0.204_{\pm 0.002}$	$1.268_{\pm 0.106}$
Weather	336	$0.280_{\pm 0.010}$	$0.336_{\pm 0.006}$	$0.529_{\pm 0.010}$	$0.309_{\pm 0.007}$	$0.239_{\pm 0.005}$	$1.601_{\pm 0.005}$
Weather	720	$0.366_{\pm 0.001}$	$0.403_{\pm 0.002}$	$0.605_{\pm 0.001}$	$0.365_{\pm 0.002}$	$0.281_{\pm 0.001}$	$1.816_{\pm 0.146}$
Traffic	96	$0.568_{\pm 0.004}$	$0.315_{\pm 0.002}$	$0.754_{\pm 0.003}$	$0.310_{\pm 0.001}$	$0.238_{\pm 0.001}$	$1.401_{\pm 0.467}$
Traffic	192	$0.601_{\pm 0.011}$	$0.323_{\pm 0.001}$	$0.775_{\pm 0.007}$	$0.319_{\pm 0.002}$	$0.246_{\pm 0.001}$	$1.242_{\pm 0.300}$
Traffic	336	$0.635_{\pm 0.003}$	$0.350_{\pm 0.002}$	$0.797_{\pm 0.002}$	$0.345_{\pm 0.001}$	$0.264_{\pm 0.001}$	$1.352_{\pm 0.093}$
Traffic	720	$0.668_{\pm 0.045}$	$0.361_{\pm 0.018}$	$0.817_{\pm 0.027}$	$0.357_{\pm 0.015}$	$0.273_{\pm 0.010}$	$1.827_{\pm 0.274}$
ILI	24	$0.461_{\pm 0.022}$	$0.408_{\pm 0.018}$	$0.679_{\pm 0.016}$	$0.447_{\pm 0.026}$	$0.348_{\pm 0.006}$	$6.972_{\pm 0.694}$
ILI	36	$2.771_{\pm 0.306}$	$0.929_{\pm 0.054}$	$1.663_{\pm 0.091}$	$1.014_{\pm 0.068}$	$0.834_{\pm 0.055}$	$7.259_{\pm 2.083}$
ILI	48	$2.614_{\pm 0.386}$	$0.921_{\pm 0.060}$	$1.614_{\pm 0.117}$	$0.955_{\pm 0.036}$	$0.819_{\pm 0.059}$	$7.772_{\pm 0.534}$
ILI	60	$2.431_{\pm 0.311}$	$0.891_{\pm 0.061}$	$1.557_{\pm 0.099}$	$0.939_{\pm 0.041}$	$0.796_{\pm 0.052}$	$8.421_{\pm 0.657}$

TABLE S9: K^2VAE: full FSPC coverage/event results on all 36 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
ETTh1	96	$8.962_{\pm 0.925}$	$7.231_{\pm 0.754}$	$0.133_{\pm 0.002}$	$0.062_{\pm 0.002}$	$0.019_{\pm 0.001}$	$0.072_{\pm 0.002}$
ETTh1	192	$9.771_{\pm 1.063}$	$8.112_{\pm 0.531}$	$0.148_{\pm 0.002}$	$0.067_{\pm 0.005}$	$0.022_{\pm 0.003}$	$0.079_{\pm 0.003}$
ETTh1	336	$11.095_{\pm 0.243}$	$9.423_{\pm 0.498}$	$0.154_{\pm 0.006}$	$0.065_{\pm 0.006}$	$0.021_{\pm 0.003}$	$0.080_{\pm 0.005}$
ETTh1	720	$11.757_{\pm 1.072}$	$9.307_{\pm 1.681}$	$0.150_{\pm 0.008}$	$0.058_{\pm 0.007}$	$0.017_{\pm 0.005}$	$0.075_{\pm 0.006}$
ETTh2	96	$4.667_{\pm 0.473}$	$2.835_{\pm 0.477}$	$0.075_{\pm 0.002}$	$0.056_{\pm 0.003}$	$0.037_{\pm 0.003}$	$0.056_{\pm 0.002}$
ETTh2	192	$5.521_{\pm 0.998}$	$3.148_{\pm 1.517}$	$0.086_{\pm 0.004}$	$0.062_{\pm 0.007}$	$0.044_{\pm 0.002}$	$0.064_{\pm 0.004}$
ETTh2	336	$8.326_{\pm 0.118}$	$4.533_{\pm 2.281}$	$0.111_{\pm 0.005}$	$0.073_{\pm 0.002}$	$0.060_{\pm 0.003}$	$0.081_{\pm 0.002}$
ETTh2	720	$8.861_{\pm 1.693}$	$6.615_{\pm 0.823}$	$0.116_{\pm 0.006}$	$0.082_{\pm 0.008}$	$0.069_{\pm 0.007}$	$0.089_{\pm 0.007}$
ETTm1	96	$5.513_{\pm 0.367}$	$2.597_{\pm 0.433}$	$0.118_{\pm 0.001}$	$0.057_{\pm 0.002}$	$0.018_{\pm 0.002}$	$0.064_{\pm 0.002}$
ETTm1	192	$5.999_{\pm 0.070}$	$3.307_{\pm 0.127}$	$0.127_{\pm 0.001}$	$0.058_{\pm 0.005}$	$0.017_{\pm 0.003}$	$0.067_{\pm 0.003}$
ETTm1	336	$6.909_{\pm 0.094}$	$4.295_{\pm 0.235}$	$0.135_{\pm 0.002}$	$0.061_{\pm 0.002}$	$0.019_{\pm 0.002}$	$0.072_{\pm 0.001}$
ETTm1	720	$8.456_{\pm 0.256}$	$6.717_{\pm 0.215}$	$0.143_{\pm 0.003}$	$0.061_{\pm 0.004}$	$0.018_{\pm 0.002}$	$0.074_{\pm 0.003}$
ETTm2	96	$4.102_{\pm 0.530}$	$2.048_{\pm 0.921}$	$0.050_{\pm 0.002}$	$0.039_{\pm 0.002}$	$0.026_{\pm 0.001}$	$0.038_{\pm 0.002}$
ETTm2	192	$5.057_{\pm 0.403}$	$2.329_{\pm 1.016}$	$0.056_{\pm 0.002}$	$0.042_{\pm 0.002}$	$0.029_{\pm 0.002}$	$0.042_{\pm 0.002}$
ETTm2	336	$5.593_{\pm 1.022}$	$2.879_{\pm 1.736}$	$0.065_{\pm 0.001}$	$0.049_{\pm 0.001}$	$0.032_{\pm 0.001}$	$0.049_{\pm 0.001}$
ETTm2	720	$6.696_{\pm 0.787}$	$5.220_{\pm 0.675}$	$0.090_{\pm 0.003}$	$0.063_{\pm 0.002}$	$0.042_{\pm 0.002}$	$0.065_{\pm 0.002}$
Electricity	96	$3.000_{\pm 0.834}$	$0.851_{\pm 0.591}$	$0.079_{\pm 0.001}$	$0.070_{\pm 0.000}$	$0.060_{\pm 0.000}$	$0.070_{\pm 0.000}$
Electricity	192	$3.050_{\pm 1.465}$	$1.795_{\pm 1.180}$	$0.084_{\pm 0.003}$	$0.076_{\pm 0.003}$	$0.065_{\pm 0.003}$	$0.075_{\pm 0.003}$
Electricity	336	$2.886_{\pm 0.211}$	$1.613_{\pm 0.484}$	$0.088_{\pm 0.002}$	$0.079_{\pm 0.002}$	$0.070_{\pm 0.002}$	$0.079_{\pm 0.002}$
Electricity	720	$3.864_{\pm 0.436}$	$1.980_{\pm 0.633}$	$0.102_{\pm 0.001}$	$0.095_{\pm 0.001}$	$0.084_{\pm 0.001}$	$0.094_{\pm 0.001}$
Exchange	96	$10.888_{\pm 3.479}$	$8.062_{\pm 3.877}$	$0.085_{\pm 0.002}$	$0.070_{\pm 0.003}$	$0.060_{\pm 0.004}$	$0.072_{\pm 0.003}$
Exchange	192	$14.677_{\pm 0.456}$	$9.819_{\pm 0.851}$	$0.101_{\pm 0.012}$	$0.087_{\pm 0.010}$	$0.078_{\pm 0.010}$	$0.089_{\pm 0.010}$
Exchange	336	$16.313_{\pm 2.996}$	$13.198_{\pm 2.616}$	$0.119_{\pm 0.005}$	$0.104_{\pm 0.003}$	$0.094_{\pm 0.003}$	$0.106_{\pm 0.004}$
Exchange	720	$24.893_{\pm 6.095}$	$23.149_{\pm 7.335}$	$0.220_{\pm 0.026}$	$0.187_{\pm 0.025}$	$0.168_{\pm 0.026}$	$0.192_{\pm 0.025}$
Weather	96	$5.501_{\pm 0.692}$	$2.587_{\pm 0.764}$	$0.095_{\pm 0.001}$	$0.050_{\pm 0.002}$	$0.022_{\pm 0.001}$	$0.056_{\pm 0.001}$
Weather	192	$4.805_{\pm 0.380}$	$2.829_{\pm 0.261}$	$0.105_{\pm 0.003}$	$0.055_{\pm 0.002}$	$0.025_{\pm 0.001}$	$0.062_{\pm 0.002}$
Weather	336	$5.600_{\pm 0.623}$	$4.048_{\pm 1.015}$	$0.116_{\pm 0.002}$	$0.062_{\pm 0.003}$	$0.031_{\pm 0.002}$	$0.070_{\pm 0.002}$
Weather	720	$6.582_{\pm 0.542}$	$4.867_{\pm 0.509}$	$0.124_{\pm 0.001}$	$0.067_{\pm 0.002}$	$0.036_{\pm 0.002}$	$0.076_{\pm 0.001}$
Traffic	96	$4.425_{\pm 0.933}$	$3.239_{\pm 1.315}$	$0.079_{\pm 0.001}$	$0.054_{\pm 0.000}$	$0.038_{\pm 0.000}$	$0.057_{\pm 0.000}$
Traffic	192	$4.425_{\pm 0.524}$	$3.201_{\pm 1.061}$	$0.082_{\pm 0.000}$	$0.056_{\pm 0.000}$	$0.039_{\pm 0.000}$	$0.059_{\pm 0.000}$
Traffic	336	$4.092_{\pm 0.109}$	$3.180_{\pm 0.100}$	$0.089_{\pm 0.001}$	$0.060_{\pm 0.001}$	$0.041_{\pm 0.000}$	$0.063_{\pm 0.001}$
Traffic	720	$6.471_{\pm 2.928}$	$4.410_{\pm 1.371}$	$0.095_{\pm 0.009}$	$0.063_{\pm 0.004}$	$0.043_{\pm 0.002}$	$0.067_{\pm 0.005}$
ILI	24	–	–	$0.130_{\pm 0.011}$	$0.127_{\pm 0.007}$	$0.129_{\pm 0.006}$	$0.129_{\pm 0.007}$
ILI	36	–	–	$0.132_{\pm 0.011}$	$0.163_{\pm 0.011}$	$0.177_{\pm 0.014}$	$0.157_{\pm 0.012}$
ILI	48	–	–	$0.110_{\pm 0.004}$	$0.135_{\pm 0.010}$	$0.153_{\pm 0.014}$	$0.133_{\pm 0.009}$
ILI	60	$20.446_{\pm 0.953}$	$20.446_{\pm 0.953}$	$0.126_{\pm 0.012}$	$0.132_{\pm 0.023}$	$0.148_{\pm 0.024}$	$0.135_{\pm 0.020}$

TABLE S10: PPM: full FSPC point/distribution results on all 7 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
ETTh1	192	$0.439_{\pm 0.001}$	$0.432_{\pm 0.001}$	$0.663_{\pm 0.001}$	$0.449_{\pm 0.001}$	$0.326_{\pm 0.001}$	$1.621_{\pm 0.013}$
ETTh2	192	$0.372_{\pm 0.001}$	$0.399_{\pm 0.003}$	$0.610_{\pm 0.000}$	$0.401_{\pm 0.002}$	$0.302_{\pm 0.001}$	$1.476_{\pm 0.089}$
ETTm1	192	$0.382_{\pm 0.001}$	$0.398_{\pm 0.001}$	$0.618_{\pm 0.001}$	$0.400_{\pm 0.001}$	$0.299_{\pm 0.000}$	$0.846_{\pm 0.192}$
ETTm2	192	$0.246_{\pm 0.001}$	$0.313_{\pm 0.002}$	$0.496_{\pm 0.001}$	$0.311_{\pm 0.002}$	$0.228_{\pm 0.001}$	$1.181_{\pm 0.058}$
Electricity	192	$0.183_{\pm 0.001}$	$0.268_{\pm 0.001}$	$0.428_{\pm 0.001}$	$0.267_{\pm 0.000}$	$0.201_{\pm 0.000}$	$1.294_{\pm 0.111}$
Weather	192	$0.238_{\pm 0.002}$	$0.273_{\pm 0.001}$	$0.488_{\pm 0.002}$	$0.287_{\pm 0.008}$	$0.201_{\pm 0.001}$	$2.690_{\pm 0.417}$
Traffic	192	$0.527_{\pm 0.001}$	$0.324_{\pm 0.000}$	$0.726_{\pm 0.000}$	$0.323_{\pm 0.000}$	$0.253_{\pm 0.000}$	$0.486_{\pm 0.040}$

TABLE S11: PPM: full FSPC coverage/event results on all 7 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
ETTh1	192	$7.460_{\pm 0.165}$	$4.492_{\pm 0.358}$	$0.083_{\pm 0.002}$	$0.063_{\pm 0.002}$	$0.042_{\pm 0.001}$	$0.063_{\pm 0.002}$
ETTh2	192	$5.808_{\pm 0.150}$	$2.899_{\pm 0.583}$	$0.066_{\pm 0.001}$	$0.039_{\pm 0.005}$	$0.026_{\pm 0.001}$	$0.044_{\pm 0.002}$
ETTm1	192	$5.517_{\pm 0.248}$	$1.130_{\pm 0.111}$	$0.129_{\pm 0.001}$	$0.054_{\pm 0.001}$	$0.016_{\pm 0.001}$	$0.066_{\pm 0.001}$
ETTm2	192	$6.828_{\pm 0.400}$	$6.393_{\pm 0.438}$	$0.042_{\pm 0.001}$	$0.031_{\pm 0.000}$	$0.020_{\pm 0.000}$	$0.031_{\pm 0.000}$
Electricity	192	$3.140_{\pm 0.359}$	$2.315_{\pm 0.352}$	$0.079_{\pm 0.000}$	$0.070_{\pm 0.000}$	$0.060_{\pm 0.000}$	$0.069_{\pm 0.000}$
Weather	192	$11.451_{\pm 0.658}$	$3.875_{\pm 0.635}$	$0.070_{\pm 0.001}$	$0.052_{\pm 0.000}$	$0.035_{\pm 0.001}$	$0.052_{\pm 0.001}$
Traffic	192	$2.665_{\pm 0.122}$	$1.453_{\pm 0.249}$	$0.088_{\pm 0.000}$	$0.059_{\pm 0.001}$	$0.040_{\pm 0.000}$	$0.062_{\pm 0.000}$

TABLE S12: TMDM: full FSPC point/distribution results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
ETTh1	192	$0.699_{\pm 0.067}$	$0.618_{\pm 0.042}$	$0.835_{\pm 0.040}$	$0.603_{\pm 0.029}$	$0.441_{\pm 0.021}$	$2.211_{\pm 1.036}$
ETTh2	192	$0.565_{\pm 0.089}$	$0.509_{\pm 0.034}$	$0.750_{\pm 0.058}$	$0.507_{\pm 0.033}$	$0.385_{\pm 0.026}$	$1.866_{\pm 0.517}$
ETTm1	192	$0.455_{\pm 0.053}$	$0.465_{\pm 0.024}$	$0.674_{\pm 0.039}$	$0.459_{\pm 0.021}$	$0.338_{\pm 0.016}$	$0.850_{\pm 0.284}$
ETTm2	192	$0.320_{\pm 0.060}$	$0.408_{\pm 0.042}$	$0.564_{\pm 0.052}$	$0.397_{\pm 0.043}$	$0.291_{\pm 0.030}$	$2.205_{\pm 0.388}$
Electricity	192	$0.236_{\pm 0.002}$	$0.336_{\pm 0.001}$	$0.486_{\pm 0.002}$	$0.337_{\pm 0.001}$	$0.248_{\pm 0.001}$	$1.199_{\pm 0.196}$
Exchange	192	$0.466_{\pm 0.117}$	$0.496_{\pm 0.086}$	$0.679_{\pm 0.085}$	$0.493_{\pm 0.082}$	$0.374_{\pm 0.063}$	$4.190_{\pm 0.669}$
Traffic	192	$0.629_{\pm 0.032}$	$0.414_{\pm 0.007}$	$0.793_{\pm 0.020}$	$0.409_{\pm 0.007}$	$0.309_{\pm 0.006}$	$1.650_{\pm 0.279}$
ILI	36	$2.742_{\pm 0.700}$	$0.990_{\pm 0.113}$	$1.647_{\pm 0.215}$	$0.965_{\pm 0.104}$	$0.820_{\pm 0.107}$	$6.026_{\pm 1.178}$
SolarEnergy	192	$0.233_{\pm 0.004}$	$0.262_{\pm 0.003}$	$0.483_{\pm 0.004}$	$0.257_{\pm 0.006}$	$0.191_{\pm 0.003}$	$0.991_{\pm 0.275}$

TABLE S13: TMDM: full FSPC coverage/event results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
ETTh1	192	$12.454_{\pm 1.606}$	$6.285_{\pm 3.450}$	$0.166_{\pm 0.007}$	$0.085_{\pm 0.002}$	$0.030_{\pm 0.001}$	$0.094_{\pm 0.002}$
ETTh2	192	$7.071_{\pm 1.663}$	$3.615_{\pm 1.803}$	$0.081_{\pm 0.005}$	$0.063_{\pm 0.006}$	$0.042_{\pm 0.008}$	$0.062_{\pm 0.006}$
ETTm1	192	$8.559_{\pm 1.152}$	$1.065_{\pm 0.310}$	$0.132_{\pm 0.005}$	$0.057_{\pm 0.003}$	$0.018_{\pm 0.002}$	$0.069_{\pm 0.003}$
ETTm2	192	$11.159_{\pm 1.709}$	$5.751_{\pm 1.317}$	$0.048_{\pm 0.006}$	$0.037_{\pm 0.004}$	$0.024_{\pm 0.003}$	$0.036_{\pm 0.004}$
Electricity	192	$3.692_{\pm 0.709}$	$3.097_{\pm 0.699}$	$0.097_{\pm 0.002}$	$0.085_{\pm 0.002}$	$0.073_{\pm 0.002}$	$0.085_{\pm 0.002}$
Exchange	192	$13.187_{\pm 1.683}$	$7.808_{\pm 2.677}$	$0.124_{\pm 0.010}$	$0.100_{\pm 0.011}$	$0.087_{\pm 0.010}$	$0.104_{\pm 0.010}$
Traffic	192	$4.478_{\pm 0.561}$	$4.119_{\pm 0.728}$	$0.103_{\pm 0.002}$	$0.065_{\pm 0.002}$	$0.042_{\pm 0.001}$	$0.070_{\pm 0.002}$
ILI	36	–	–	$0.146_{\pm 0.037}$	$0.171_{\pm 0.074}$	$0.186_{\pm 0.080}$	$0.168_{\pm 0.063}$
SolarEnergy	192	$7.884_{\pm 0.548}$	$1.813_{\pm 0.601}$	$0.069_{\pm 0.001}$	$0.048_{\pm 0.000}$	$0.023_{\pm 0.001}$	$0.047_{\pm 0.001}$

TABLE S14: Diffusion-TS: full FSPC point/distribution results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
ETTh1	192	$1.146_{\pm 0.161}$	$0.847_{\pm 0.074}$	$1.069_{\pm 0.076}$	$0.811_{\pm 0.049}$	$0.586_{\pm 0.033}$	$2.540_{\pm 1.074}$
ETTh2	192	$1.989_{\pm 0.677}$	$1.106_{\pm 0.163}$	$1.398_{\pm 0.232}$	$1.117_{\pm 0.218}$	$0.794_{\pm 0.147}$	$3.926_{\pm 2.965}$
ETTm1	192	$1.005_{\pm 0.120}$	$0.768_{\pm 0.044}$	$1.001_{\pm 0.060}$	$0.770_{\pm 0.048}$	$0.565_{\pm 0.035}$	$2.456_{\pm 0.686}$
ETTm2	192	$1.171_{\pm 0.424}$	$0.847_{\pm 0.172}$	$1.070_{\pm 0.202}$	$0.846_{\pm 0.177}$	$0.616_{\pm 0.132}$	$4.394_{\pm 1.292}$
Electricity	192	$0.539_{\pm 0.009}$	$0.532_{\pm 0.006}$	$0.734_{\pm 0.006}$	$0.542_{\pm 0.008}$	$0.395_{\pm 0.005}$	$2.745_{\pm 0.262}$
Exchange	192	$1.468_{\pm 0.481}$	$0.934_{\pm 0.151}$	$1.201_{\pm 0.192}$	$0.949_{\pm 0.173}$	$0.691_{\pm 0.103}$	$4.461_{\pm 0.577}$
Traffic	192	$0.737_{\pm 0.025}$	$0.447_{\pm 0.022}$	$0.859_{\pm 0.014}$	$0.442_{\pm 0.018}$	$0.334_{\pm 0.013}$	$1.256_{\pm 0.338}$
ILI	36	$4.154_{\pm 0.142}$	$1.341_{\pm 0.047}$	$2.038_{\pm 0.035}$	$1.362_{\pm 0.053}$	$1.115_{\pm 0.038}$	$8.312_{\pm 1.332}$
SolarEnergy	192	$0.289_{\pm 0.043}$	$0.347_{\pm 0.010}$	$0.537_{\pm 0.039}$	$0.331_{\pm 0.010}$	$0.237_{\pm 0.007}$	$4.167_{\pm 1.194}$

TABLE S15: Diffusion-TS: full FSPC coverage/event results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
ETTh1	192	$9.785_{\pm 2.304}$	$6.256_{\pm 3.936}$	$0.205_{\pm 0.001}$	$0.102_{\pm 0.014}$	$0.041_{\pm 0.015}$	$0.116_{\pm 0.010}$
ETTh2	192	$11.791_{\pm 8.381}$	$11.133_{\pm 8.971}$	$0.134_{\pm 0.059}$	$0.101_{\pm 0.050}$	$0.070_{\pm 0.045}$	$0.101_{\pm 0.051}$
ETTm1	192	$8.102_{\pm 1.655}$	$4.853_{\pm 1.688}$	$0.172_{\pm 0.002}$	$0.076_{\pm 0.004}$	$0.035_{\pm 0.004}$	$0.094_{\pm 0.003}$
ETTm2	192	$13.199_{\pm 3.284}$	$9.630_{\pm 2.106}$	$0.108_{\pm 0.018}$	$0.074_{\pm 0.017}$	$0.047_{\pm 0.014}$	$0.077_{\pm 0.016}$
Electricity	192	$7.585_{\pm 0.601}$	$7.391_{\pm 0.665}$	$0.157_{\pm 0.004}$	$0.138_{\pm 0.003}$	$0.117_{\pm 0.002}$	$0.137_{\pm 0.003}$
Exchange	192	$10.803_{\pm 3.170}$	$9.902_{\pm 3.529}$	$0.185_{\pm 0.024}$	$0.151_{\pm 0.021}$	$0.138_{\pm 0.020}$	$0.158_{\pm 0.021}$
Traffic	192	$4.077_{\pm 0.748}$	$2.566_{\pm 0.512}$	$0.113_{\pm 0.005}$	$0.074_{\pm 0.004}$	$0.049_{\pm 0.002}$	$0.079_{\pm 0.004}$
ILI	36	$21.373_{\pm 3.925}$	$21.373_{\pm 3.925}$	$0.304_{\pm 0.036}$	$0.333_{\pm 0.026}$	$0.333_{\pm 0.021}$	$0.323_{\pm 0.028}$
SolarEnergy	192	$13.689_{\pm 2.959}$	$12.004_{\pm 4.424}$	$0.070_{\pm 0.006}$	$0.049_{\pm 0.005}$	$0.031_{\pm 0.008}$	$0.050_{\pm 0.006}$

TABLE S16: TimeDiff: full FSPC point/distribution results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
ETTh1	192	$0.598_{\pm 0.016}$	$0.532_{\pm 0.008}$	$0.773_{\pm 0.010}$	$0.523_{\pm 0.005}$	$0.398_{\pm 0.002}$	$2.332_{\pm 0.544}$
ETTh2	192	$0.549_{\pm 0.167}$	$0.529_{\pm 0.102}$	$0.735_{\pm 0.110}$	$0.555_{\pm 0.109}$	$0.415_{\pm 0.078}$	$4.288_{\pm 1.217}$
ETTm1	192	$0.627_{\pm 0.021}$	$0.520_{\pm 0.007}$	$0.792_{\pm 0.013}$	$0.509_{\pm 0.003}$	$0.387_{\pm 0.004}$	$1.500_{\pm 0.283}$
ETTm2	192	$0.269_{\pm 0.007}$	$0.343_{\pm 0.002}$	$0.519_{\pm 0.007}$	$0.337_{\pm 0.003}$	$0.258_{\pm 0.005}$	$1.573_{\pm 0.331}$
Electricity	192	$0.864_{\pm 0.022}$	$0.765_{\pm 0.011}$	$0.929_{\pm 0.012}$	$0.766_{\pm 0.007}$	$0.525_{\pm 0.006}$	$0.773_{\pm 0.118}$
Exchange	192	$0.432_{\pm 0.023}$	$0.464_{\pm 0.017}$	$0.657_{\pm 0.018}$	$0.464_{\pm 0.045}$	$0.365_{\pm 0.024}$	$5.771_{\pm 1.500}$
Traffic	192	$1.344_{\pm 0.002}$	$0.785_{\pm 0.002}$	$1.159_{\pm 0.001}$	$0.780_{\pm 0.002}$	$0.565_{\pm 0.001}$	$0.791_{\pm 0.046}$
ILI	36	$3.173_{\pm 0.628}$	$1.033_{\pm 0.097}$	$1.776_{\pm 0.172}$	$1.055_{\pm 0.079}$	$0.881_{\pm 0.105}$	$7.328_{\pm 0.816}$
SolarEnergy	192	$0.732_{\pm 0.004}$	$0.719_{\pm 0.009}$	$0.855_{\pm 0.002}$	$0.514_{\pm 0.002}$	$0.396_{\pm 0.002}$	$7.018_{\pm 0.617}$

TABLE S17: TimeDiff: full FSPC coverage/event results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
ETTh1	192	$8.234_{\pm 0.521}$	$4.788_{\pm 1.815}$	$0.160_{\pm 0.005}$	$0.084_{\pm 0.003}$	$0.033_{\pm 0.002}$	$0.092_{\pm 0.003}$
ETTh2	192	$12.624_{\pm 3.862}$	$9.220_{\pm 3.129}$	$0.101_{\pm 0.004}$	$0.072_{\pm 0.005}$	$0.041_{\pm 0.004}$	$0.071_{\pm 0.004}$
ETTm1	192	$6.014_{\pm 0.431}$	$2.906_{\pm 0.844}$	$0.149_{\pm 0.001}$	$0.058_{\pm 0.002}$	$0.017_{\pm 0.001}$	$0.075_{\pm 0.001}$
ETTm2	192	$4.709_{\pm 0.060}$	$2.748_{\pm 1.047}$	$0.057_{\pm 0.002}$	$0.044_{\pm 0.002}$	$0.029_{\pm 0.002}$	$0.044_{\pm 0.002}$
Electricity	192	$2.931_{\pm 0.149}$	$1.338_{\pm 0.296}$	$0.193_{\pm 0.003}$	$0.152_{\pm 0.003}$	$0.121_{\pm 0.002}$	$0.155_{\pm 0.003}$
Exchange	192	$18.078_{\pm 6.185}$	$15.155_{\pm 6.526}$	$0.143_{\pm 0.023}$	$0.105_{\pm 0.012}$	$0.084_{\pm 0.004}$	$0.111_{\pm 0.013}$
Traffic	192	$1.660_{\pm 0.218}$	$1.155_{\pm 0.276}$	$0.165_{\pm 0.001}$	$0.095_{\pm 0.000}$	$0.055_{\pm 0.000}$	$0.105_{\pm 0.000}$
ILI	36	–	–	$0.167_{\pm 0.066}$	$0.227_{\pm 0.066}$	$0.241_{\pm 0.061}$	$0.212_{\pm 0.063}$
SolarEnergy	192	$8.363_{\pm 0.182}$	$8.019_{\pm 0.280}$	$0.136_{\pm 0.000}$	$0.062_{\pm 0.000}$	$0.023_{\pm 0.000}$	$0.073_{\pm 0.000}$

TABLE S18: NsDiff: full FSPC point/distribution results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
ETTh1	192	$0.484_{\pm 0.039}$	$0.446_{\pm 0.006}$	$0.695_{\pm 0.028}$	$0.470_{\pm 0.012}$	$0.389_{\pm 0.008}$	$1.510_{\pm 0.508}$
ETTh2	192	$0.550_{\pm 0.071}$	$0.496_{\pm 0.032}$	$0.740_{\pm 0.049}$	$0.505_{\pm 0.032}$	$0.372_{\pm 0.022}$	$2.403_{\pm 0.774}$
ETTm1	192	$0.471_{\pm 0.059}$	$0.448_{\pm 0.022}$	$0.685_{\pm 0.042}$	$0.456_{\pm 0.017}$	$0.342_{\pm 0.012}$	$1.425_{\pm 0.155}$
ETTm2	192	$0.273_{\pm 0.016}$	$0.350_{\pm 0.009}$	$0.523_{\pm 0.015}$	$0.346_{\pm 0.013}$	$0.264_{\pm 0.008}$	$1.750_{\pm 0.155}$
Electricity	192	$0.222_{\pm 0.006}$	$0.316_{\pm 0.006}$	$0.471_{\pm 0.006}$	$0.316_{\pm 0.006}$	$0.235_{\pm 0.004}$	$1.337_{\pm 0.205}$
Exchange	192	$0.326_{\pm 0.087}$	$0.407_{\pm 0.052}$	$0.567_{\pm 0.077}$	$0.405_{\pm 0.046}$	$0.325_{\pm 0.039}$	$4.360_{\pm 1.136}$
Traffic	192	$0.613_{\pm 0.022}$	$0.365_{\pm 0.014}$	$0.783_{\pm 0.014}$	$0.359_{\pm 0.011}$	$0.276_{\pm 0.009}$	$1.960_{\pm 0.422}$
ILI	36	$2.911_{\pm 0.804}$	$1.018_{\pm 0.224}$	$1.695_{\pm 0.238}$	$1.018_{\pm 0.237}$	$0.908_{\pm 0.226}$	$8.534_{\pm 1.345}$
SolarEnergy	192	$0.229_{\pm 0.023}$	$0.263_{\pm 0.008}$	$0.479_{\pm 0.024}$	$0.255_{\pm 0.010}$	$0.188_{\pm 0.008}$	$1.551_{\pm 0.564}$

TABLE S19: NsDiff: full FSPC coverage/event results on all 9 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
ETTh1	192	$10.180_{\pm 1.238}$	$3.241_{\pm 1.394}$	$0.153_{\pm 0.004}$	$0.078_{\pm 0.002}$	$0.027_{\pm 0.000}$	$0.086_{\pm 0.002}$
ETTh2	192	$7.951_{\pm 3.104}$	$6.287_{\pm 3.957}$	$0.083_{\pm 0.006}$	$0.065_{\pm 0.002}$	$0.042_{\pm 0.006}$	$0.063_{\pm 0.002}$
ETTm1	192	$8.253_{\pm 1.071}$	$3.381_{\pm 1.105}$	$0.136_{\pm 0.002}$	$0.060_{\pm 0.004}$	$0.019_{\pm 0.003}$	$0.072_{\pm 0.002}$
ETTm2	192	$6.058_{\pm 1.034}$	$4.821_{\pm 0.432}$	$0.056_{\pm 0.003}$	$0.042_{\pm 0.002}$	$0.028_{\pm 0.002}$	$0.042_{\pm 0.002}$
Electricity	192	$3.540_{\pm 0.875}$	$2.608_{\pm 0.797}$	$0.092_{\pm 0.002}$	$0.081_{\pm 0.002}$	$0.070_{\pm 0.002}$	$0.081_{\pm 0.002}$
Exchange	192	$15.428_{\pm 3.414}$	$11.262_{\pm 5.947}$	$0.120_{\pm 0.019}$	$0.096_{\pm 0.019}$	$0.081_{\pm 0.017}$	$0.099_{\pm 0.018}$
Traffic	192	$5.198_{\pm 1.603}$	$4.856_{\pm 1.748}$	$0.093_{\pm 0.004}$	$0.060_{\pm 0.003}$	$0.041_{\pm 0.002}$	$0.064_{\pm 0.003}$
ILI	36	–	–	$0.174_{\pm 0.147}$	$0.228_{\pm 0.138}$	$0.244_{\pm 0.126}$	$0.215_{\pm 0.135}$
SolarEnergy	192	$10.131_{\pm 1.532}$	$3.210_{\pm 1.564}$	$0.068_{\pm 0.004}$	$0.045_{\pm 0.001}$	$0.020_{\pm 0.001}$	$0.044_{\pm 0.002}$

TABLE S20: DeepSTUQ: full FSPC point/distribution results on all 4 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	MSE ↓	MAE ↓	RMSE ↓	Median MAE ↓	CRPS ↓	QICE (pp) ↓
PEMS03	12	$728.965_{\pm 10.562}$	$15.149_{\pm 0.058}$	$26.999_{\pm 0.195}$	$15.152_{\pm 0.060}$	$11.093_{\pm 0.044}$	$0.909_{\pm 0.070}$
PEMS04	12	$1008.753_{\pm 6.045}$	$19.125_{\pm 0.041}$	$31.761_{\pm 0.095}$	$19.088_{\pm 0.028}$	$14.015_{\pm 0.025}$	$0.373_{\pm 0.042}$
PEMS07	12	$1159.562_{\pm 7.452}$	$20.462_{\pm 0.041}$	$34.052_{\pm 0.109}$	$20.464_{\pm 0.043}$	$15.031_{\pm 0.023}$	$0.255_{\pm 0.093}$
PEMS08	12	$619.167_{\pm 4.659}$	$15.701_{\pm 0.055}$	$24.883_{\pm 0.094}$	$15.683_{\pm 0.058}$	$11.617_{\pm 0.043}$	$0.220_{\pm 0.156}$

TABLE S21: DeepSTUQ: full FSPC coverage/event results on all 4 configurations. FSPC uses the affine-plus-state mean. Subscripts: three-run sample SD. All metrics are minimized. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and coverage errors are in percentage points; Brier scores are dimensionless.

Dataset	H	State (pp) ↓	Pooled (pp) ↓	$\mathrm{BS}_{.80}$ ↓	$\mathrm{BS}_{.90}$ ↓	$\mathrm{BS}_{.95}$ ↓	Mean BS ↓
PEMS03	12	$2.162_{\pm 0.248}$	$1.920_{\pm 0.210}$	$0.058_{\pm 0.000}$	$0.039_{\pm 0.000}$	$0.025_{\pm 0.000}$	$0.041_{\pm 0.000}$
PEMS04	12	$1.144_{\pm 0.168}$	$0.856_{\pm 0.083}$	$0.060_{\pm 0.001}$	$0.044_{\pm 0.001}$	$0.028_{\pm 0.000}$	$0.044_{\pm 0.001}$
PEMS07	12	$1.633_{\pm 0.117}$	$0.678_{\pm 0.400}$	$0.067_{\pm 0.000}$	$0.046_{\pm 0.000}$	$0.031_{\pm 0.000}$	$0.048_{\pm 0.000}$
PEMS08	12	$0.943_{\pm 0.174}$	$0.435_{\pm 0.479}$	$0.060_{\pm 0.001}$	$0.042_{\pm 0.001}$	$0.029_{\pm 0.001}$	$0.044_{\pm 0.001}$

External calibration methods

Tables S22–S28 compare Native, EMOS, learned PIT, QRF and FSPC for each backbone and dataset–horizon configuration. These comparisons cover all 83 configurations and 249 runs, with three training runs per configuration. All methods use identical complete native test forecasts within each configuration. CRPS retains original traffic-flow units for DeepSTUQ and model-normalized units elsewhere; QICE is in percentage points. Main-paper Table IV also compares MSE, state coverage error and mean Brier score on this matched set.

Diffusion-TS disables its native denoised-value clamp under standard-score target normalization; Native and every correction share this sampler. Each external method fits the same validation reservoir of up to 50,000 coordinates from at most 100 native batches or windows. Forecasting parameters are fixed. FSPC uses validation-fitted templates and an affine-plus-state mean blended with a training-bias reference. The earlier two thirds of validation origins fit the candidates and the final third selects the blend, purging overlapping targets. If either block contains fewer than 128 retained coordinates, training forecasts fit the candidates and validation selects. The chosen blend is fixed before refitting both means and templates on all calibration coordinates. All calibration parameters are estimated before testing. Mean-model and calibration comparisons use the same benchmark test splits. Backbone averages first average configurations within each training run; the mean and sample standard deviation then use the three run averages. Cross-backbone summaries divide each run's backbone score by its paired Native score and equally average the seven ratios.

Gaussian EMOS uses an affine native mean and a nonnegative affine native variance, fitted by analytic Gaussian CRPS. Learned PIT follows the soft-weighted CRPS construction of Qian et al. (Technometrics, 2025), using Gaussian input CDFs and Epanechnikov weights based on predictive-mean proximity. For a reservoir of N coordinates, its nearest-neighbor bandwidth uses $\lfloor N/K \rfloor$ neighbors. It chooses K from 1 through 20 by the smallest mean squared probability calibration error over 3000 fixed uniform probability levels on the fitting reservoir. All fitting sources are retained. Exact kernel-weighted residual CDF and quantile queries implement the Gaussian-family solution. Narrow mean neighborhoods use direct nonnegative kernel sums. Coincident nearest neighbors on the kernel boundary use the continuous outward-bandwidth limit.

QRF adapts the ensemble postprocessing of Taillardat et al. (Monthly Weather Review, 2016) to the available native forecasts. Its eleven predictors are the native mean, sample standard deviation and nine deciles; no additional weather predictors are supplied. The forest uses 1000 bootstrap trees, a minimum leaf size of five and three candidate predictors per split. All observations assigned to a leaf contribute to its predictive distribution, with inverse leaf-size weighting and linear quantile interpolation. Each external method outputs the same 100 midpoint probability levels used for ensemble scoring.

TABLE S22: External calibrators on K^2VAE: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTh1 / 96	CRPS	$0.337_{\pm 0.009}$	$0.330_{\pm 0.003}$	$\mathbf{0.320}_{\pm 0.001}$	$0.361_{\pm 0.023}$	$0.353_{\pm 0.016}$
	QICE (pp)	$3.009_{\pm 1.009}$	$3.466_{\pm 0.226}$	$1.559_{\pm 0.246}$	$\mathbf{1.261}_{\pm 0.611}$	$3.130_{\pm 0.298}$
ETTh1 / 192	CRPS	$0.372_{\pm 0.005}$	$0.369_{\pm 0.003}$	$\mathbf{0.350}_{\pm 0.008}$	$0.399_{\pm 0.012}$	$0.420_{\pm 0.044}$
	QICE (pp)	$3.175_{\pm 1.022}$	$3.346_{\pm 0.402}$	$2.041_{\pm 0.753}$	$\mathbf{1.044}_{\pm 0.134}$	$3.780_{\pm 0.076}$
ETTh1 / 336	CRPS	$0.384_{\pm 0.005}$	$0.406_{\pm 0.006}$	$\mathbf{0.370}_{\pm 0.010}$	$0.462_{\pm 0.021}$	$0.441_{\pm 0.027}$
	QICE (pp)	$3.010_{\pm 0.786}$	$3.284_{\pm 0.035}$	$2.830_{\pm 0.289}$	$\mathbf{1.939}_{\pm 0.065}$	$4.359_{\pm 0.170}$
ETTh1 / 720	CRPS	$0.409_{\pm 0.003}$	$0.441_{\pm 0.008}$	$\mathbf{0.391}_{\pm 0.014}$	$0.495_{\pm 0.010}$	$0.459_{\pm 0.067}$
	QICE (pp)	$3.817_{\pm 0.673}$	$3.121_{\pm 0.135}$	$3.275_{\pm 0.562}$	$\mathbf{2.724}_{\pm 0.127}$	$4.284_{\pm 0.738}$
ETTh2 / 96	CRPS	$0.300_{\pm 0.007}$	$0.285_{\pm 0.006}$	$\mathbf{0.284}_{\pm 0.004}$	$0.298_{\pm 0.004}$	$0.285_{\pm 0.007}$
	QICE (pp)	$3.939_{\pm 0.175}$	$3.010_{\pm 0.150}$	$1.852_{\pm 0.418}$	$2.106_{\pm 0.252}$	$\mathbf{1.356}_{\pm 0.125}$
ETTh2 / 192	CRPS	$0.334_{\pm 0.002}$	$\mathbf{0.313}_{\pm 0.004}$	$0.323_{\pm 0.005}$	$0.325_{\pm 0.004}$	$0.315_{\pm 0.004}$
	QICE (pp)	$4.584_{\pm 0.306}$	$2.902_{\pm 0.142}$	$2.386_{\pm 1.360}$	$2.602_{\pm 0.166}$	$\mathbf{1.381}_{\pm 0.316}$
ETTh2 / 336	CRPS	$0.678_{\pm 0.194}$	$0.400_{\pm 0.043}$	$0.495_{\pm 0.123}$	$\mathbf{0.384}_{\pm 0.029}$	$0.436_{\pm 0.064}$
	QICE (pp)	$5.194_{\pm 0.219}$	$2.425_{\pm 0.107}$	$4.168_{\pm 1.927}$	$2.256_{\pm 0.302}$	$\mathbf{2.192}_{\pm 0.950}$
ETTh2 / 720	CRPS	$0.502_{\pm 0.135}$	$\mathbf{0.381}_{\pm 0.033}$	$0.404_{\pm 0.036}$	$0.387_{\pm 0.014}$	$0.396_{\pm 0.030}$
	QICE (pp)	$5.422_{\pm 0.460}$	$2.487_{\pm 0.264}$	$4.523_{\pm 2.922}$	$\mathbf{1.417}_{\pm 0.387}$	$2.635_{\pm 0.614}$
ETTm1 / 96	CRPS	$0.298_{\pm 0.003}$	$0.276_{\pm 0.003}$	$\mathbf{0.275}_{\pm 0.004}$	$0.286_{\pm 0.006}$	$0.280_{\pm 0.005}$
	QICE (pp)	$3.248_{\pm 0.568}$	$2.282_{\pm 0.153}$	$0.752_{\pm 0.378}$	$\mathbf{0.655}_{\pm 0.109}$	$1.217_{\pm 0.064}$
ETTm1 / 192	CRPS	$0.326_{\pm 0.006}$	$0.300_{\pm 0.002}$	$\mathbf{0.296}_{\pm 0.004}$	$0.308_{\pm 0.009}$	$0.304_{\pm 0.005}$
	QICE (pp)	$3.731_{\pm 0.794}$	$2.526_{\pm 0.156}$	$\mathbf{0.447}_{\pm 0.133}$	$0.710_{\pm 0.268}$	$1.392_{\pm 0.090}$
ETTm1 / 336	CRPS	$0.344_{\pm 0.011}$	$0.322_{\pm 0.006}$	$\mathbf{0.317}_{\pm 0.006}$	$0.332_{\pm 0.006}$	$0.333_{\pm 0.005}$
	QICE (pp)	$3.242_{\pm 0.494}$	$2.957_{\pm 0.166}$	$\mathbf{0.571}_{\pm 0.009}$	$0.880_{\pm 0.199}$	$1.861_{\pm 0.111}$
ETTm1 / 720	CRPS	$0.373_{\pm 0.006}$	$0.355_{\pm 0.004}$	$\mathbf{0.341}_{\pm 0.005}$	$0.370_{\pm 0.005}$	$0.382_{\pm 0.009}$
	QICE (pp)	$3.522_{\pm 0.497}$	$3.222_{\pm 0.200}$	$1.611_{\pm 0.144}$	$\mathbf{1.376}_{\pm 0.154}$	$2.982_{\pm 0.122}$
ETTm2 / 96	CRPS	$0.235_{\pm 0.005}$	$0.215_{\pm 0.004}$	$0.218_{\pm 0.007}$	$0.219_{\pm 0.001}$	$\mathbf{0.214}_{\pm 0.004}$
	QICE (pp)	$4.319_{\pm 0.114}$	$2.633_{\pm 0.095}$	$2.012_{\pm 0.532}$	$2.187_{\pm 0.141}$	$\mathbf{1.507}_{\pm 0.123}$

TABLE S22 continued

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTm2 / 192	CRPS	$0.276_{\pm 0.006}$	$\mathbf{0.252}_{\pm 0.004}$	$0.260_{\pm 0.007}$	$0.252_{\pm 0.002}$	$0.253_{\pm 0.003}$
	QICE (pp)	$4.334_{\pm 0.245}$	$2.758_{\pm 0.086}$	$2.040_{\pm 0.305}$	$2.229_{\pm 0.134}$	$\mathbf{1.598}_{\pm 0.215}$
ETTm2 / 336	CRPS	$0.306_{\pm 0.005}$	$0.279_{\pm 0.003}$	$0.283_{\pm 0.006}$	$\mathbf{0.271}_{\pm 0.006}$	$0.280_{\pm 0.003}$
	QICE (pp)	$4.138_{\pm 0.130}$	$2.845_{\pm 0.112}$	$1.753_{\pm 0.238}$	$2.184_{\pm 0.266}$	$\mathbf{1.690}_{\pm 0.554}$
ETTm2 / 720	CRPS	$0.462_{\pm 0.088}$	$0.346_{\pm 0.012}$	$0.354_{\pm 0.021}$	$\mathbf{0.328}_{\pm 0.007}$	$0.351_{\pm 0.018}$
	QICE (pp)	$4.559_{\pm 0.641}$	$2.390_{\pm 0.399}$	$3.182_{\pm 1.849}$	$2.648_{\pm 0.137}$	$\mathbf{2.276}_{\pm 0.150}$
Electricity / 96	CRPS	$0.209_{\pm 0.002}$	$0.205_{\pm 0.001}$	$0.206_{\pm 0.002}$	$0.206_{\pm 0.001}$	$\mathbf{0.204}_{\pm 0.001}$
	QICE (pp)	$1.267_{\pm 0.110}$	$1.260_{\pm 0.049}$	$1.083_{\pm 0.142}$	$0.728_{\pm 0.149}$	$\mathbf{0.408}_{\pm 0.154}$
Electricity / 192	CRPS	$0.219_{\pm 0.005}$	$0.215_{\pm 0.007}$	$0.215_{\pm 0.006}$	$0.215_{\pm 0.007}$	$\mathbf{0.214}_{\pm 0.005}$
	QICE (pp)	$1.363_{\pm 0.153}$	$1.367_{\pm 0.213}$	$1.607_{\pm 0.792}$	$1.027_{\pm 0.574}$	$\mathbf{0.685}_{\pm 0.503}$
Electricity / 336	CRPS	$0.240_{\pm 0.006}$	$0.228_{\pm 0.003}$	$0.229_{\pm 0.004}$	$\mathbf{0.227}_{\pm 0.003}$	$0.231_{\pm 0.004}$
	QICE (pp)	$1.569_{\pm 0.238}$	$1.534_{\pm 0.071}$	$1.917_{\pm 0.371}$	$0.909_{\pm 0.297}$	$\mathbf{0.632}_{\pm 0.125}$
Electricity / 720	CRPS	$0.283_{\pm 0.004}$	$0.259_{\pm 0.003}$	$0.264_{\pm 0.003}$	$\mathbf{0.259}_{\pm 0.003}$	$0.266_{\pm 0.003}$
	QICE (pp)	$1.771_{\pm 0.252}$	$1.629_{\pm 0.031}$	$3.146_{\pm 0.335}$	$1.582_{\pm 0.189}$	$\mathbf{1.012}_{\pm 0.235}$
Exchange / 96	CRPS	$0.271_{\pm 0.010}$	$0.243_{\pm 0.003}$	$0.263_{\pm 0.005}$	$0.291_{\pm 0.012}$	$\mathbf{0.242}_{\pm 0.029}$
	QICE (pp)	$5.156_{\pm 0.599}$	$6.755_{\pm 0.201}$	$7.379_{\pm 0.767}$	$7.200_{\pm 0.620}$	$\mathbf{4.064}_{\pm 0.295}$
Exchange / 192	CRPS	$0.343_{\pm 0.014}$	$0.360_{\pm 0.002}$	$0.380_{\pm 0.017}$	$0.445_{\pm 0.039}$	$\mathbf{0.287}_{\pm 0.013}$
	QICE (pp)	$5.097_{\pm 0.404}$	$7.955_{\pm 0.112}$	$8.158_{\pm 0.396}$	$8.459_{\pm 0.808}$	$\mathbf{4.119}_{\pm 0.335}$
Exchange / 336	CRPS	$0.462_{\pm 0.026}$	$0.522_{\pm 0.016}$	$0.533_{\pm 0.014}$	$0.595_{\pm 0.016}$	$\mathbf{0.388}_{\pm 0.018}$
	QICE (pp)	$5.938_{\pm 1.010}$	$9.322_{\pm 0.409}$	$9.155_{\pm 0.302}$	$9.846_{\pm 0.256}$	$\mathbf{4.106}_{\pm 0.892}$
Exchange / 720	CRPS	$0.690_{\pm 0.058}$	$1.209_{\pm 0.110}$	$1.053_{\pm 0.185}$	$0.937_{\pm 0.019}$	$\mathbf{0.580}_{\pm 0.020}$
	QICE (pp)	$6.643_{\pm 1.832}$	$13.131_{\pm 1.467}$	$12.050_{\pm 2.412}$	$11.391_{\pm 0.238}$	$\mathbf{6.488}_{\pm 0.715}$
Weather / 96	CRPS	$0.204_{\pm 0.004}$	$0.176_{\pm 0.003}$	$\mathbf{0.170}_{\pm 0.003}$	$0.176_{\pm 0.003}$	$0.172_{\pm 0.002}$
	QICE (pp)	$6.176_{\pm 0.143}$	$2.848_{\pm 0.125}$	$\mathbf{0.938}_{\pm 0.039}$	$1.721_{\pm 0.287}$	$1.167_{\pm 0.083}$
Weather / 192	CRPS	$0.233_{\pm 0.004}$	$0.208_{\pm 0.002}$	$\mathbf{0.203}_{\pm 0.002}$	$0.209_{\pm 0.001}$	$0.204_{\pm 0.002}$
	QICE (pp)	$5.236_{\pm 0.153}$	$2.885_{\pm 0.155}$	$\mathbf{0.967}_{\pm 0.115}$	$1.963_{\pm 0.047}$	$1.268_{\pm 0.106}$
Weather / 336	CRPS	$0.266_{\pm 0.006}$	$0.246_{\pm 0.007}$	$\mathbf{0.238}_{\pm 0.006}$	$0.247_{\pm 0.002}$	$0.239_{\pm 0.005}$
	QICE (pp)	$4.654_{\pm 0.427}$	$2.996_{\pm 0.086}$	$\mathbf{1.452}_{\pm 0.083}$	$2.302_{\pm 0.228}$	$1.601_{\pm 0.005}$
Weather / 720	CRPS	$0.309_{\pm 0.005}$	$0.294_{\pm 0.002}$	$\mathbf{0.277}_{\pm 0.002}$	$0.285_{\pm 0.002}$	$0.281_{\pm 0.001}$
	QICE (pp)	$4.314_{\pm 0.327}$	$2.952_{\pm 0.319}$	$\mathbf{1.736}_{\pm 0.136}$	$2.615_{\pm 0.117}$	$1.816_{\pm 0.146}$
Traffic / 96	CRPS	$0.248_{\pm 0.001}$	$0.245_{\pm 0.001}$	$0.243_{\pm 0.000}$	$\mathbf{0.231}_{\pm 0.001}$	$0.238_{\pm 0.001}$
	QICE (pp)	$2.598_{\pm 0.240}$	$\mathbf{1.287}_{\pm 0.310}$	$1.367_{\pm 0.313}$	$1.435_{\pm 0.370}$	$1.401_{\pm 0.467}$
Traffic / 192	CRPS	$0.256_{\pm 0.002}$	$0.252_{\pm 0.001}$	$0.250_{\pm 0.002}$	$\mathbf{0.237}_{\pm 0.001}$	$0.246_{\pm 0.001}$
	QICE (pp)	$2.673_{\pm 0.113}$	$1.495_{\pm 0.178}$	$1.681_{\pm 0.212}$	$1.298_{\pm 0.195}$	$\mathbf{1.242}_{\pm 0.300}$
Traffic / 336	CRPS	$0.276_{\pm 0.001}$	$0.270_{\pm 0.001}$	$0.269_{\pm 0.002}$	$\mathbf{0.256}_{\pm 0.002}$	$0.264_{\pm 0.001}$
	QICE (pp)	$2.842_{\pm 0.420}$	$1.413_{\pm 0.099}$	$1.685_{\pm 0.439}$	$1.464_{\pm 0.108}$	$\mathbf{1.352}_{\pm 0.093}$
Traffic / 720	CRPS	$0.282_{\pm 0.005}$	$0.278_{\pm 0.008}$	$0.276_{\pm 0.007}$	$\mathbf{0.263}_{\pm 0.008}$	$0.273_{\pm 0.010}$
	QICE (pp)	$2.695_{\pm 0.452}$	$\mathbf{1.760}_{\pm 0.388}$	$2.089_{\pm 0.281}$	$1.777_{\pm 0.257}$	$1.827_{\pm 0.274}$
ILI / 24	CRPS	$\mathbf{0.345}_{\pm 0.016}$	$0.481_{\pm 0.022}$	$0.577_{\pm 0.056}$	$0.639_{\pm 0.009}$	$0.348_{\pm 0.006}$
	QICE (pp)	$8.865_{\pm 0.225}$	$\mathbf{4.968}_{\pm 0.496}$	$11.246_{\pm 0.356}$	$10.325_{\pm 0.810}$	$6.972_{\pm 0.694}$
ILI / 36	CRPS	$0.841_{\pm 0.056}$	$0.894_{\pm 0.102}$	$0.979_{\pm 0.076}$	$1.290_{\pm 0.008}$	$\mathbf{0.834}_{\pm 0.055}$
	QICE (pp)	$9.262_{\pm 0.238}$	$\mathbf{5.481}_{\pm 0.860}$	$8.008_{\pm 1.574}$	$10.206_{\pm 0.278}$	$7.259_{\pm 2.083}$
ILI / 48	CRPS	$0.838_{\pm 0.062}$	$0.887_{\pm 0.069}$	$1.039_{\pm 0.067}$	$1.314_{\pm 0.031}$	$\mathbf{0.819}_{\pm 0.059}$
	QICE (pp)	$10.335_{\pm 0.715}$	$\mathbf{6.550}_{\pm 1.266}$	$9.480_{\pm 0.881}$	$10.149_{\pm 0.481}$	$7.772_{\pm 0.534}$
ILI / 60	CRPS	$0.802_{\pm 0.059}$	$0.800_{\pm 0.054}$	$0.916_{\pm 0.082}$	$1.192_{\pm 0.008}$	$\mathbf{0.796}_{\pm 0.052}$
	QICE (pp)	$10.873_{\pm 1.133}$	$\mathbf{6.841}_{\pm 0.371}$	$9.643_{\pm 0.203}$	$10.508_{\pm 0.786}$	$8.421_{\pm 0.657}$

TABLE S23: External calibrators on PPM: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTh1 / 192	CRPS	$0.337_{\pm 0.000}$	$0.354_{\pm 0.001}$	$0.334_{\pm 0.000}$	$0.390_{\pm 0.012}$	$\mathbf{0.326}_{\pm 0.001}$
	QICE (pp)	$1.813_{\pm 0.011}$	$4.310_{\pm 0.074}$	$1.286_{\pm 0.070}$	$\mathbf{1.174}_{\pm 0.140}$	$1.621_{\pm 0.013}$
ETTh2 / 192	CRPS	$0.305_{\pm 0.001}$	$0.305_{\pm 0.001}$	$0.313_{\pm 0.012}$	$0.367_{\pm 0.009}$	$\mathbf{0.302}_{\pm 0.001}$
	QICE (pp)	$3.521_{\pm 0.114}$	$2.366_{\pm 0.047}$	$1.667_{\pm 0.207}$	$4.672_{\pm 0.219}$	$\mathbf{1.476}_{\pm 0.089}$
ETTm1 / 192	CRPS	$0.314_{\pm 0.001}$	$\mathbf{0.298}_{\pm 0.000}$	$0.332_{\pm 0.002}$	$0.331_{\pm 0.003}$	$0.299_{\pm 0.000}$
	QICE (pp)	$2.176_{\pm 0.023}$	$2.343_{\pm 0.001}$	$2.913_{\pm 0.061}$	$1.108_{\pm 0.110}$	$\mathbf{0.846}_{\pm 0.192}$
ETTm2 / 192	CRPS	$0.236_{\pm 0.000}$	$0.241_{\pm 0.001}$	$0.240_{\pm 0.000}$	$0.287_{\pm 0.011}$	$\mathbf{0.228}_{\pm 0.001}$
	QICE (pp)	$1.971_{\pm 0.119}$	$2.990_{\pm 0.011}$	$2.246_{\pm 0.039}$	$3.726_{\pm 0.239}$	$\mathbf{1.181}_{\pm 0.058}$
Electricity / 192	CRPS	$0.206_{\pm 0.000}$	$0.203_{\pm 0.000}$	$0.204_{\pm 0.000}$	$0.205_{\pm 0.000}$	$\mathbf{0.201}_{\pm 0.000}$
	QICE (pp)	$3.929_{\pm 0.064}$	$1.697_{\pm 0.013}$	$2.494_{\pm 0.119}$	$1.479_{\pm 0.104}$	$\mathbf{1.294}_{\pm 0.111}$
Weather / 192	CRPS	$0.215_{\pm 0.001}$	$0.216_{\pm 0.001}$	$0.220_{\pm 0.001}$	$0.255_{\pm 0.006}$	$\mathbf{0.201}_{\pm 0.001}$
	QICE (pp)	$4.454_{\pm 0.024}$	$4.002_{\pm 0.043}$	$6.394_{\pm 0.043}$	$\mathbf{2.214}_{\pm 0.259}$	$2.690_{\pm 0.417}$
Traffic / 192	CRPS	$0.255_{\pm 0.001}$	$0.258_{\pm 0.001}$	$0.254_{\pm 0.000}$	$0.256_{\pm 0.000}$	$\mathbf{0.253}_{\pm 0.000}$
	QICE (pp)	$3.073_{\pm 0.244}$	$2.471_{\pm 0.103}$	$1.405_{\pm 0.060}$	$0.596_{\pm 0.058}$	$\mathbf{0.486}_{\pm 0.040}$

TABLE S24: External calibrators on TMDM: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTh1 / 192	CRPS	$0.453_{\pm 0.039}$	$0.464_{\pm 0.026}$	$0.451_{\pm 0.027}$	$0.470_{\pm 0.027}$	$\mathbf{0.441}_{\pm 0.021}$
	QICE (pp)	$3.272_{\pm 1.558}$	$3.077_{\pm 0.723}$	$3.066_{\pm 0.860}$	$3.036_{\pm 0.740}$	$\mathbf{2.211}_{\pm 1.036}$
ETTh2 / 192	CRPS	$0.401_{\pm 0.022}$	$0.387_{\pm 0.025}$	$\mathbf{0.379}_{\pm 0.022}$	$0.386_{\pm 0.025}$	$0.385_{\pm 0.026}$
	QICE (pp)	$4.488_{\pm 0.828}$	$2.064_{\pm 0.583}$	$\mathbf{1.330}_{\pm 0.249}$	$2.261_{\pm 0.212}$	$1.866_{\pm 0.517}$
ETTm1 / 192	CRPS	$0.347_{\pm 0.017}$	$0.344_{\pm 0.016}$	$\mathbf{0.336}_{\pm 0.017}$	$0.351_{\pm 0.024}$	$0.338_{\pm 0.016}$
	QICE (pp)	$2.615_{\pm 0.547}$	$1.007_{\pm 0.158}$	$0.829_{\pm 0.106}$	$\mathbf{0.780}_{\pm 0.176}$	$0.850_{\pm 0.284}$
ETTm2 / 192	CRPS	$0.298_{\pm 0.032}$	$0.296_{\pm 0.039}$	$\mathbf{0.282}_{\pm 0.036}$	$0.284_{\pm 0.028}$	$0.291_{\pm 0.030}$
	QICE (pp)	$2.180_{\pm 0.751}$	$2.203_{\pm 1.032}$	$\mathbf{1.798}_{\pm 0.911}$	$3.440_{\pm 0.555}$	$2.205_{\pm 0.388}$
Electricity / 192	CRPS	$0.442_{\pm 0.002}$	$0.253_{\pm 0.002}$	$0.250_{\pm 0.001}$	$0.253_{\pm 0.001}$	$\mathbf{0.248}_{\pm 0.001}$
	QICE (pp)	$10.416_{\pm 0.025}$	$1.415_{\pm 0.192}$	$1.457_{\pm 0.154}$	$1.573_{\pm 0.217}$	$\mathbf{1.199}_{\pm 0.196}$
Exchange / 192	CRPS	$0.392_{\pm 0.047}$	$0.424_{\pm 0.069}$	$0.446_{\pm 0.093}$	$0.522_{\pm 0.082}$	$\mathbf{0.374}_{\pm 0.063}$
	QICE (pp)	$6.437_{\pm 1.957}$	$5.687_{\pm 0.729}$	$6.017_{\pm 1.371}$	$9.395_{\pm 0.802}$	$\mathbf{4.190}_{\pm 0.669}$
Traffic / 192	CRPS	$0.558_{\pm 0.003}$	$0.318_{\pm 0.007}$	$0.332_{\pm 0.009}$	$\mathbf{0.300}_{\pm 0.006}$	$0.309_{\pm 0.006}$
	QICE (pp)	$10.604_{\pm 0.091}$	$\mathbf{1.536}_{\pm 0.223}$	$1.637_{\pm 0.208}$	$1.642_{\pm 0.291}$	$1.650_{\pm 0.279}$
ILI / 36	CRPS	$\mathbf{0.810}_{\pm 0.100}$	$0.979_{\pm 0.050}$	$0.965_{\pm 0.057}$	$1.008_{\pm 0.037}$	$0.820_{\pm 0.107}$
	QICE (pp)	$\mathbf{5.201}_{\pm 0.869}$	$7.630_{\pm 0.200}$	$8.312_{\pm 0.760}$	$8.683_{\pm 0.998}$	$6.026_{\pm 1.178}$
SolarEnergy / 192	CRPS	$0.341_{\pm 0.004}$	$0.241_{\pm 0.003}$	$0.218_{\pm 0.031}$	$\mathbf{0.172}_{\pm 0.002}$	$0.191_{\pm 0.003}$
	QICE (pp)	$9.262_{\pm 0.122}$	$2.714_{\pm 0.093}$	$\mathbf{0.918}_{\pm 0.076}$	$1.025_{\pm 0.053}$	$0.991_{\pm 0.275}$

TABLE S25: External calibrators on Diffusion-TS: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTh1 / 192	CRPS	$0.659_{\pm 0.068}$	$\mathbf{0.577}_{\pm 0.033}$	$0.638_{\pm 0.114}$	$0.613_{\pm 0.040}$	$0.586_{\pm 0.033}$
	QICE (pp)	$2.816_{\pm 1.683}$	$3.062_{\pm 0.730}$	$3.083_{\pm 1.368}$	$3.375_{\pm 1.392}$	$\mathbf{2.540}_{\pm 1.074}$
ETTh2 / 192	CRPS	$1.147_{\pm 0.124}$	$0.871_{\pm 0.074}$	$0.793_{\pm 0.015}$	$\mathbf{0.641}_{\pm 0.072}$	$0.794_{\pm 0.147}$
	QICE (pp)	$7.742_{\pm 0.110}$	$5.188_{\pm 0.163}$	$4.021_{\pm 0.565}$	$\mathbf{3.702}_{\pm 0.692}$	$3.926_{\pm 2.965}$
ETTm1 / 192	CRPS	$0.873_{\pm 0.056}$	$\mathbf{0.548}_{\pm 0.016}$	$0.659_{\pm 0.097}$	$0.576_{\pm 0.014}$	$0.565_{\pm 0.035}$
	QICE (pp)	$13.310_{\pm 0.288}$	$3.027_{\pm 0.133}$	$\mathbf{1.826}_{\pm 0.752}$	$2.866_{\pm 0.458}$	$2.456_{\pm 0.686}$
ETTm2 / 192	CRPS	$0.940_{\pm 0.288}$	$0.641_{\pm 0.153}$	$0.749_{\pm 0.216}$	$\mathbf{0.613}_{\pm 0.116}$	$0.616_{\pm 0.132}$
	QICE (pp)	$15.118_{\pm 0.276}$	$4.969_{\pm 1.355}$	$8.692_{\pm 1.442}$	$6.290_{\pm 0.762}$	$\mathbf{4.394}_{\pm 1.292}$
Electricity / 192	CRPS	$0.453_{\pm 0.010}$	$0.404_{\pm 0.003}$	$0.407_{\pm 0.002}$	$0.406_{\pm 0.005}$	$\mathbf{0.395}_{\pm 0.005}$
	QICE (pp)	$10.318_{\pm 0.294}$	$\mathbf{2.566}_{\pm 0.223}$	$3.259_{\pm 0.316}$	$3.006_{\pm 0.051}$	$2.745_{\pm 0.262}$
Exchange / 192	CRPS	$0.998_{\pm 0.402}$	$\mathbf{0.668}_{\pm 0.063}$	$0.694_{\pm 0.056}$	$0.731_{\pm 0.081}$	$0.691_{\pm 0.103}$
	QICE (pp)	$9.985_{\pm 2.311}$	$\mathbf{3.898}_{\pm 0.109}$	$5.110_{\pm 0.359}$	$7.213_{\pm 0.978}$	$4.461_{\pm 0.577}$
Traffic / 192	CRPS	$0.432_{\pm 0.024}$	$0.346_{\pm 0.012}$	$0.338_{\pm 0.022}$	$0.339_{\pm 0.009}$	$\mathbf{0.334}_{\pm 0.013}$
	QICE (pp)	$14.337_{\pm 0.493}$	$1.594_{\pm 0.155}$	$\mathbf{1.185}_{\pm 0.282}$	$1.562_{\pm 0.370}$	$1.256_{\pm 0.338}$
ILI / 36	CRPS	$\mathbf{1.025}_{\pm 0.049}$	$1.125_{\pm 0.031}$	$1.098_{\pm 0.046}$	$1.155_{\pm 0.016}$	$1.115_{\pm 0.038}$
	QICE (pp)	$\mathbf{4.561}_{\pm 1.212}$	$9.122_{\pm 0.375}$	$8.063_{\pm 1.846}$	$10.709_{\pm 0.121}$	$8.312_{\pm 1.332}$
SolarEnergy / 192	CRPS	$0.365_{\pm 0.016}$	$0.313_{\pm 0.026}$	$0.255_{\pm 0.028}$	$\mathbf{0.216}_{\pm 0.002}$	$0.237_{\pm 0.007}$
	QICE (pp)	$8.069_{\pm 2.068}$	$4.664_{\pm 0.264}$	$2.858_{\pm 1.599}$	$\mathbf{2.380}_{\pm 0.456}$	$4.167_{\pm 1.194}$

TABLE S26: External calibrators on TimeDiff: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTh1 / 192	CRPS	$0.474_{\pm 0.008}$	$0.408_{\pm 0.003}$	$\mathbf{0.390}_{\pm 0.005}$	$0.445_{\pm 0.018}$	$0.398_{\pm 0.002}$
	QICE (pp)	$13.863_{\pm 0.515}$	$2.874_{\pm 0.139}$	$1.693_{\pm 0.461}$	$\mathbf{1.097}_{\pm 0.310}$	$2.332_{\pm 0.544}$
ETTh2 / 192	CRPS	$0.491_{\pm 0.098}$	$\mathbf{0.397}_{\pm 0.083}$	$0.412_{\pm 0.075}$	$0.424_{\pm 0.062}$	$0.415_{\pm 0.078}$
	QICE (pp)	$13.457_{\pm 0.655}$	$\mathbf{2.805}_{\pm 1.890}$	$4.765_{\pm 1.874}$	$5.743_{\pm 1.349}$	$4.288_{\pm 1.217}$
ETTm1 / 192	CRPS	$0.490_{\pm 0.008}$	$0.392_{\pm 0.005}$	$\mathbf{0.386}_{\pm 0.004}$	$0.419_{\pm 0.012}$	$0.387_{\pm 0.004}$
	QICE (pp)	$14.646_{\pm 0.337}$	$2.138_{\pm 0.130}$	$1.097_{\pm 0.067}$	$\mathbf{0.973}_{\pm 0.133}$	$1.500_{\pm 0.283}$
ETTm2 / 192	CRPS	$0.315_{\pm 0.005}$	$0.264_{\pm 0.004}$	$0.259_{\pm 0.007}$	$0.270_{\pm 0.003}$	$\mathbf{0.258}_{\pm 0.005}$
	QICE (pp)	$12.446_{\pm 0.847}$	$2.856_{\pm 0.310}$	$1.916_{\pm 0.115}$	$3.534_{\pm 0.430}$	$\mathbf{1.573}_{\pm 0.331}$
Electricity / 192	CRPS	$0.749_{\pm 0.011}$	$0.531_{\pm 0.007}$	$0.526_{\pm 0.005}$	$0.544_{\pm 0.004}$	$\mathbf{0.525}_{\pm 0.006}$
	QICE (pp)	$15.524_{\pm 0.024}$	$1.933_{\pm 0.143}$	$\mathbf{0.754}_{\pm 0.124}$	$1.202_{\pm 0.103}$	$0.773_{\pm 0.118}$
Exchange / 192	CRPS	$0.443_{\pm 0.010}$	$0.394_{\pm 0.025}$	$0.445_{\pm 0.041}$	$0.492_{\pm 0.066}$	$\mathbf{0.365}_{\pm 0.024}$
	QICE (pp)	$14.438_{\pm 0.680}$	$6.452_{\pm 0.483}$	$8.028_{\pm 0.538}$	$11.672_{\pm 0.177}$	$\mathbf{5.771}_{\pm 1.500}$
Traffic / 192	CRPS	$0.778_{\pm 0.001}$	$0.568_{\pm 0.001}$	$0.572_{\pm 0.001}$	$0.576_{\pm 0.001}$	$\mathbf{0.565}_{\pm 0.001}$
	QICE (pp)	$15.478_{\pm 0.009}$	$1.448_{\pm 0.047}$	$\mathbf{0.512}_{\pm 0.053}$	$1.147_{\pm 0.043}$	$0.791_{\pm 0.046}$
ILI / 36	CRPS	$1.123_{\pm 0.018}$	$0.910_{\pm 0.094}$	$0.900_{\pm 0.081}$	$1.009_{\pm 0.059}$	$\mathbf{0.881}_{\pm 0.105}$
	QICE (pp)	$13.460_{\pm 1.395}$	$\mathbf{7.328}_{\pm 1.066}$	$10.159_{\pm 1.772}$	$11.556_{\pm 0.496}$	$\mathbf{7.328}_{\pm 0.816}$
SolarEnergy / 192	CRPS	$0.700_{\pm 0.002}$	$0.463_{\pm 0.002}$	$0.402_{\pm 0.003}$	$0.406_{\pm 0.003}$	$\mathbf{0.396}_{\pm 0.002}$
	QICE (pp)	$15.123_{\pm 0.262}$	$11.681_{\pm 0.626}$	$1.636_{\pm 0.197}$	$\mathbf{1.269}_{\pm 0.232}$	$7.018_{\pm 0.617}$

TABLE S27: External calibrators on NsDiff: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
ETTh1 / 192	CRPS	$0.371_{\pm 0.010}$	$0.386_{\pm 0.006}$	$\mathbf{0.367}_{\pm 0.012}$	$0.431_{\pm 0.015}$	$0.389_{\pm 0.008}$
	QICE (pp)	$2.454_{\pm 0.074}$	$2.123_{\pm 0.154}$	$\mathbf{1.233}_{\pm 0.261}$	$2.191_{\pm 0.467}$	$1.510_{\pm 0.508}$
ETTh2 / 192	CRPS	$0.385_{\pm 0.026}$	$0.372_{\pm 0.027}$	$\mathbf{0.366}_{\pm 0.022}$	$0.399_{\pm 0.024}$	$0.372_{\pm 0.022}$
	QICE (pp)	$3.597_{\pm 1.031}$	$2.731_{\pm 1.410}$	$\mathbf{2.170}_{\pm 1.201}$	$3.056_{\pm 0.886}$	$2.403_{\pm 0.774}$
ETTm1 / 192	CRPS	$0.343_{\pm 0.018}$	$\mathbf{0.337}_{\pm 0.014}$	$0.338_{\pm 0.014}$	$0.337_{\pm 0.012}$	$0.342_{\pm 0.012}$
	QICE (pp)	$1.927_{\pm 0.105}$	$1.751_{\pm 0.168}$	$1.014_{\pm 0.226}$	$\mathbf{0.977}_{\pm 0.118}$	$1.425_{\pm 0.155}$
ETTm2 / 192	CRPS	$0.275_{\pm 0.006}$	$0.273_{\pm 0.003}$	$0.265_{\pm 0.001}$	$0.274_{\pm 0.006}$	$\mathbf{0.264}_{\pm 0.008}$
	QICE (pp)	$3.181_{\pm 0.203}$	$3.182_{\pm 0.388}$	$2.159_{\pm 0.439}$	$3.305_{\pm 0.294}$	$\mathbf{1.750}_{\pm 0.155}$
Electricity / 192	CRPS	$0.277_{\pm 0.002}$	$0.238_{\pm 0.004}$	$0.238_{\pm 0.003}$	$0.239_{\pm 0.004}$	$\mathbf{0.235}_{\pm 0.004}$
	QICE (pp)	$6.542_{\pm 0.086}$	$1.507_{\pm 0.103}$	$1.726_{\pm 0.252}$	$1.624_{\pm 0.229}$	$\mathbf{1.337}_{\pm 0.205}$
Exchange / 192	CRPS	$0.352_{\pm 0.052}$	$0.400_{\pm 0.060}$	$0.478_{\pm 0.097}$	$0.501_{\pm 0.050}$	$\mathbf{0.325}_{\pm 0.039}$
	QICE (pp)	$7.556_{\pm 1.304}$	$6.643_{\pm 2.916}$	$7.608_{\pm 1.030}$	$9.788_{\pm 0.732}$	$\mathbf{4.360}_{\pm 1.136}$
Traffic / 192	CRPS	$0.374_{\pm 0.005}$	$0.292_{\pm 0.009}$	$0.291_{\pm 0.009}$	$\mathbf{0.267}_{\pm 0.008}$	$0.276_{\pm 0.009}$
	QICE (pp)	$8.681_{\pm 0.185}$	$\mathbf{1.806}_{\pm 0.214}$	$1.816_{\pm 0.366}$	$1.911_{\pm 0.290}$	$1.960_{\pm 0.422}$
ILI / 36	CRPS	$\mathbf{0.851}_{\pm 0.184}$	$0.982_{\pm 0.154}$	$0.954_{\pm 0.185}$	$1.023_{\pm 0.146}$	$0.908_{\pm 0.226}$
	QICE (pp)	$\mathbf{6.016}_{\pm 0.936}$	$8.725_{\pm 0.746}$	$8.646_{\pm 1.189}$	$9.101_{\pm 1.440}$	$8.534_{\pm 1.345}$
SolarEnergy / 192	CRPS	$0.288_{\pm 0.015}$	$0.242_{\pm 0.007}$	$0.203_{\pm 0.010}$	$\mathbf{0.176}_{\pm 0.008}$	$0.188_{\pm 0.008}$
	QICE (pp)	$7.461_{\pm 0.130}$	$3.730_{\pm 0.691}$	$2.849_{\pm 0.543}$	$\mathbf{1.330}_{\pm 0.200}$	$1.551_{\pm 0.564}$

TABLE S28: External calibrators on DeepSTUQ: per-dataset complete-test results on the common comparison set. Lower-right subscripts give three-run sample SD; bold marks the lowest unrounded mean within each row.

Dataset / H	Metric ↓	Native	EMOS	Learned PIT	QRF	FSPC
PEMS03 / 12	CRPS	$11.181_{\pm 0.055}$	$11.130_{\pm 0.043}$	$\mathbf{11.088}_{\pm 0.042}$	$11.639_{\pm 0.041}$	$11.093_{\pm 0.044}$
	QICE (pp)	$1.338_{\pm 0.040}$	$0.839_{\pm 0.036}$	$\mathbf{0.815}_{\pm 0.047}$	$1.889_{\pm 0.073}$	$0.909_{\pm 0.070}$
PEMS04 / 12	CRPS	$14.166_{\pm 0.037}$	$14.074_{\pm 0.034}$	$\mathbf{14.006}_{\pm 0.016}$	$14.737_{\pm 0.029}$	$14.015_{\pm 0.025}$
	QICE (pp)	$1.989_{\pm 0.075}$	$1.001_{\pm 0.085}$	$\mathbf{0.236}_{\pm 0.042}$	$1.456_{\pm 0.037}$	$0.373_{\pm 0.042}$
PEMS07 / 12	CRPS	$15.235_{\pm 0.032}$	$15.091_{\pm 0.019}$	$15.051_{\pm 0.012}$	$15.854_{\pm 0.032}$	$\mathbf{15.031}_{\pm 0.023}$
	QICE (pp)	$2.448_{\pm 0.047}$	$0.963_{\pm 0.071}$	$0.424_{\pm 0.153}$	$1.138_{\pm 0.051}$	$\mathbf{0.255}_{\pm 0.093}$
PEMS08 / 12	CRPS	$11.865_{\pm 0.053}$	$11.659_{\pm 0.040}$	$11.617_{\pm 0.038}$	$12.242_{\pm 0.049}$	$\mathbf{11.617}_{\pm 0.043}$
	QICE (pp)	$2.829_{\pm 0.062}$	$1.188_{\pm 0.011}$	$0.258_{\pm 0.111}$	$1.092_{\pm 0.052}$	$\mathbf{0.220}_{\pm 0.156}$

RESIDUAL-TO-STATE ASSIGNMENT CONTROLS

The following table isolates state association before mean control, using the same retained test coordinates and source-filtered residual templates for genuine, shuffled and global assignments.

TABLE S29

STATE-ASSIGNMENT CONTROLS BEFORE MEAN ADJUSTMENT ON IDENTICAL RETAINED TEST COORDINATES. THE SHUFFLED COLUMN AVERAGES FIVE PERMUTATIONS PER TASK, PRESERVING SOURCE VALIDITY, BOUNDARIES, OCCUPANCY AND FALLBACK ELIGIBILITY. DIFFUSION-TS USES THE SAME SAMPLING SETTINGS AS THE MAIN COMPARISONS. SUBSCRIPTS: THREE-RUN SAMPLE SD; BOLD: LOWEST UNROUNDED ROW MEAN. DEEPSTUQ CRPS USES TRAFFIC-FLOW UNITS; OTHER BACKBONES USE MODEL-NORMALIZED UNITS.

Backbone	Metric	Native	Global template	Shuffled states	Genuine states
K^2VAE	CRPS ↓	$0.376_{\pm 0.002}$	$0.399_{\pm 0.003}$	$0.399_{\pm 0.003}$	$\mathbf{0.376}_{\pm 0.009}$
	QICE (pp) ↓	$4.516_{\pm 0.127}$	$3.462_{\pm 0.162}$	$3.468_{\pm 0.163}$	$\mathbf{2.881}_{\pm 0.153}$
PPM	CRPS ↓	$\mathbf{0.266}_{\pm 0.000}$	$0.269_{\pm 0.000}$	$0.269_{\pm 0.000}$	$0.267_{\pm 0.000}$
	QICE (pp) ↓	$2.975_{\pm 0.015}$	$1.434_{\pm 0.029}$	$\mathbf{1.431}_{\pm 0.044}$	$1.534_{\pm 0.029}$
TMDM	CRPS ↓	$0.449_{\pm 0.016}$	$0.403_{\pm 0.008}$	$0.404_{\pm 0.008}$	$\mathbf{0.395}_{\pm 0.008}$
	QICE (pp) ↓	$6.054_{\pm 0.502}$	$2.741_{\pm 0.105}$	$2.763_{\pm 0.115}$	$\mathbf{2.722}_{\pm 0.160}$
Diffusion-TS	CRPS ↓	$0.765_{\pm 0.026}$	$0.675_{\pm 0.001}$	$0.674_{\pm 0.001}$	$\mathbf{0.597}_{\pm 0.012}$
	QICE (pp) ↓	$9.582_{\pm 0.221}$	$4.100_{\pm 0.214}$	$4.085_{\pm 0.208}$	$\mathbf{3.382}_{\pm 0.163}$
TimeDiff	CRPS ↓	$0.618_{\pm 0.011}$	$0.486_{\pm 0.010}$	$0.487_{\pm 0.010}$	$\mathbf{0.478}_{\pm 0.012}$
	QICE (pp) ↓	$14.271_{\pm 0.281}$	$3.457_{\pm 0.461}$	$3.423_{\pm 0.447}$	$\mathbf{3.001}_{\pm 0.338}$
NsDiff	CRPS ↓	$0.391_{\pm 0.016}$	$0.389_{\pm 0.019}$	$0.390_{\pm 0.019}$	$\mathbf{0.380}_{\pm 0.021}$
	QICE (pp) ↓	$5.274_{\pm 0.140}$	$3.145_{\pm 0.289}$	$3.170_{\pm 0.299}$	$\mathbf{3.132}_{\pm 0.235}$
DeepSTUQ	CRPS ↓	$13.106_{\pm 0.047}$	$12.976_{\pm 0.047}$	$12.981_{\pm 0.046}$	$\mathbf{12.936}_{\pm 0.047}$
	QICE (pp) ↓	$2.136_{\pm 0.027}$	$\mathbf{0.454}_{\pm 0.043}$	$0.464_{\pm 0.042}$	$0.456_{\pm 0.055}$

TABLE S30

COMPONENT ABLATION ON IDENTICAL RETAINED TEST COORDINATES (249 TASKS; AT MOST 50K COORDINATES PER TASK). ALL FSPC MEANS USE THE AFFINE-PLUS-STATE MODEL. SUBSCRIPTS: THREE-RUN SAMPLE SD; BOLD: LOWEST UNROUNDED BACKBONE-WISE MEAN. STATE ERRORS USE COMMON ELIGIBLE CONFIGURATIONS. DEEPSTUQ USES TRAFFIC-FLOW UNITS (MSE SQUARED); OTHER BACKBONES USE MODEL-NORMALIZED UNITS. QICE/STATE ERRORS: PERCENTAGE POINTS; BRIER: DIMENSIONLESS.

Backbone	Method	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier score ↓
K^2VAE	FSPC	$\mathbf{0.623}_{\pm 0.025}$	$\mathbf{0.355}_{\pm 0.003}$	$\mathbf{2.831}_{\pm 0.090}$	$\mathbf{7.880}_{\pm 0.184}$	$\mathbf{0.081}_{\pm 0.001}$
	w/o Mean Control	$0.778_{\pm 0.049}$	$0.376_{\pm 0.009}$	$2.881_{\pm 0.153}$	$8.975_{\pm 0.356}$	$0.085_{\pm 0.004}$
	w/o Mean/Filter	$0.778_{\pm 0.049}$	$0.376_{\pm 0.009}$	$2.881_{\pm 0.153}$	$8.975_{\pm 0.356}$	$0.085_{\pm 0.004}$
PPM	FSPC	$\mathbf{0.339}_{\pm 0.002}$	$\mathbf{0.265}_{\pm 0.000}$	$1.679_{\pm 0.093}$	$6.148_{\pm 0.137}$	$0.063_{\pm 0.000}$
	w/o Mean Control	$0.359_{\pm 0.002}$	$0.267_{\pm 0.000}$	$1.534_{\pm 0.029}$	$5.892_{\pm 0.085}$	$\mathbf{0.060}_{\pm 0.000}$
	w/o Mean/Filter	$95.410_{\pm 7.596} \times 10^6$	$144.822_{\pm 6.665}$	$\mathbf{1.514}_{\pm 0.044}$	$\mathbf{5.869}_{\pm 0.048}$	$0.061_{\pm 0.000}$
TMDM	FSPC	$\mathbf{0.704}_{\pm 0.075}$	$\mathbf{0.377}_{\pm 0.014}$	$\mathbf{2.357}_{\pm 0.124}$	$\mathbf{8.601}_{\pm 0.318}$	$\mathbf{0.082}_{\pm 0.006}$
	w/o Mean Control	$0.773_{\pm 0.051}$	$0.395_{\pm 0.008}$	$2.722_{\pm 0.160}$	$8.854_{\pm 0.265}$	$0.090_{\pm 0.003}$
	w/o Mean/Filter	$0.773_{\pm 0.051}$	$0.395_{\pm 0.008}$	$2.722_{\pm 0.160}$	$8.854_{\pm 0.265}$	$0.090_{\pm 0.003}$
Diffusion-TS	FSPC	$\mathbf{1.390}_{\pm 0.121}$	$\mathbf{0.592}_{\pm 0.023}$	$3.796_{\pm 0.230}$	$11.165_{\pm 1.128}$	$0.126_{\pm 0.007}$
	w/o Mean Control	$1.428_{\pm 0.050}$	$0.597_{\pm 0.012}$	$\mathbf{3.382}_{\pm 0.163}$	$\mathbf{10.078}_{\pm 0.384}$	$\mathbf{0.123}_{\pm 0.005}$
	w/o Mean/Filter	$1.428_{\pm 0.050}$	$0.597_{\pm 0.012}$	$\mathbf{3.382}_{\pm 0.163}$	$\mathbf{10.078}_{\pm 0.384}$	$\mathbf{0.123}_{\pm 0.005}$
TimeDiff	FSPC	$\mathbf{0.955}_{\pm 0.070}$	$\mathbf{0.465}_{\pm 0.015}$	$3.483_{\pm 0.247}$	$7.858_{\pm 1.405}$	$\mathbf{0.104}_{\pm 0.007}$
	w/o Mean Control	$0.992_{\pm 0.060}$	$0.478_{\pm 0.012}$	$\mathbf{3.001}_{\pm 0.338}$	$\mathbf{7.589}_{\pm 0.276}$	$0.110_{\pm 0.004}$
	w/o Mean/Filter	$0.992_{\pm 0.060}$	$0.478_{\pm 0.012}$	$\mathbf{3.001}_{\pm 0.338}$	$\mathbf{7.589}_{\pm 0.276}$	$0.110_{\pm 0.004}$
NsDiff	FSPC	$\mathbf{0.712}_{\pm 0.109}$	$\mathbf{0.366}_{\pm 0.021}$	$\mathbf{2.757}_{\pm 0.011}$	$\mathbf{8.355}_{\pm 0.248}$	$\mathbf{0.085}_{\pm 0.013}$
	w/o Mean Control	$0.761_{\pm 0.106}$	$0.380_{\pm 0.021}$	$3.132_{\pm 0.235}$	$8.711_{\pm 0.444}$	$0.092_{\pm 0.012}$
	w/o Mean/Filter	$0.761_{\pm 0.106}$	$0.380_{\pm 0.021}$	$3.132_{\pm 0.235}$	$8.711_{\pm 0.444}$	$0.092_{\pm 0.012}$
DeepSTUQ	FSPC	$\mathbf{877.797}_{\pm 9.855}$	$\mathbf{12.934}_{\pm 0.044}$	$0.473_{\pm 0.047}$	$1.584_{\pm 0.034}$	$0.044_{\pm 0.000}$
	w/o Mean Control	$878.489_{\pm 10.201}$	$12.936_{\pm 0.047}$	$\mathbf{0.456}_{\pm 0.055}$	$\mathbf{1.540}_{\pm 0.058}$	$\mathbf{0.044}_{\pm 0.000}$
	w/o Mean/Filter	$878.489_{\pm 10.201}$	$12.936_{\pm 0.047}$	$\mathbf{0.456}_{\pm 0.055}$	$\mathbf{1.540}_{\pm 0.058}$	$\mathbf{0.044}_{\pm 0.000}$

MEAN AND RESIDUAL-SOURCE COMPONENT CONTROLS

The original mean/filter controls use the same retained test coordinates as the factorial study. The uncentered variants isolate residual-source filtering before mean control.

MEAN AND RESIDUAL-SHAPE FACTORIAL ABLATION

These four arms share the frozen full FSPC fit and the same retained test coordinates. Mean only shifts the Native ensemble without changing its spread or member spacings. Shape only replaces its centered shape while preserving the Native mean. FSPC applies both. The accompanying CSV reports all twelve metrics and paired component contrasts. These retained-cohort measurements use a different evaluation population from the complete-test main comparisons.

TABLE S31: K^2VAE: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTh1	96	Native	$\mathbf{0.423}_{\pm 0.006}$	$\mathbf{0.337}_{\pm 0.009}$	$3.009_{\pm 1.009}$	$\mathbf{7.120}_{\pm 0.139}$	$0.076_{\pm 0.001}$
		Mean only	$0.439_{\pm 0.010}$	$0.357_{\pm 0.014}$	$2.895_{\pm 0.773}$	$11.708_{\pm 0.460}$	$0.072_{\pm 0.002}$
		Shape only	$\mathbf{0.423}_{\pm 0.006}$	$0.372_{\pm 0.020}$	$\mathbf{2.555}_{\pm 0.097}$	$9.251_{\pm 0.385}$	$0.074_{\pm 0.002}$
		FSPC	$0.439_{\pm 0.010}$	$0.353_{\pm 0.016}$	$3.130_{\pm 0.298}$	$8.962_{\pm 0.925}$	$\mathbf{0.072}_{\pm 0.002}$
ETTh1	192	Native	$\mathbf{0.479}_{\pm 0.009}$	$\mathbf{0.372}_{\pm 0.005}$	$3.175_{\pm 1.022}$	$\mathbf{8.101}_{\pm 0.848}$	$0.083_{\pm 0.003}$
		Mean only	$0.543_{\pm 0.008}$	$0.424_{\pm 0.002}$	$3.455_{\pm 0.539}$	$14.155_{\pm 0.512}$	$\mathbf{0.077}_{\pm 0.001}$
		Shape only	$\mathbf{0.479}_{\pm 0.009}$	$0.441_{\pm 0.059}$	$\mathbf{2.884}_{\pm 0.266}$	$8.968_{\pm 0.391}$	$0.080_{\pm 0.005}$
		FSPC	$0.543_{\pm 0.008}$	$0.420_{\pm 0.044}$	$3.780_{\pm 0.076}$	$9.771_{\pm 1.063}$	$0.079_{\pm 0.003}$
ETTh1	336	Native	$\mathbf{0.532}_{\pm 0.021}$	$\mathbf{0.384}_{\pm 0.005}$	$\mathbf{3.009}_{\pm 0.789}$	$\mathbf{8.987}_{\pm 0.868}$	$0.083_{\pm 0.004}$
		Mean only	$0.637_{\pm 0.011}$	$0.462_{\pm 0.002}$	$3.570_{\pm 0.484}$	$14.976_{\pm 0.188}$	$\mathbf{0.076}_{\pm 0.004}$
		Shape only	$\mathbf{0.532}_{\pm 0.021}$	$0.455_{\pm 0.034}$	$3.570_{\pm 0.060}$	$9.807_{\pm 0.444}$	$0.081_{\pm 0.003}$
		FSPC	$0.637_{\pm 0.011}$	$0.441_{\pm 0.027}$	$4.358_{\pm 0.163}$	$11.094_{\pm 0.258}$	$0.080_{\pm 0.005}$
ETTh1	720	Native	$\mathbf{0.524}_{\pm 0.011}$	$\mathbf{0.410}_{\pm 0.003}$	$3.796_{\pm 0.673}$	$\mathbf{10.356}_{\pm 0.888}$	$0.083_{\pm 0.003}$
		Mean only	$0.533_{\pm 0.013}$	$0.421_{\pm 0.010}$	$\mathbf{3.453}_{\pm 0.654}$	$12.691_{\pm 0.902}$	$0.077_{\pm 0.003}$
		Shape only	$\mathbf{0.524}_{\pm 0.011}$	$0.458_{\pm 0.077}$	$3.491_{\pm 0.756}$	$10.553_{\pm 1.579}$	$0.077_{\pm 0.006}$
		FSPC	$0.533_{\pm 0.013}$	$0.460_{\pm 0.067}$	$4.269_{\pm 0.757}$	$11.781_{\pm 1.065}$	$\mathbf{0.075}_{\pm 0.006}$
ETTh2	96	Native	$0.319_{\pm 0.014}$	$0.300_{\pm 0.007}$	$3.939_{\pm 0.175}$	$9.626_{\pm 0.318}$	$0.053_{\pm 0.003}$
		Mean only	$\mathbf{0.317}_{\pm 0.014}$	$0.301_{\pm 0.007}$	$3.942_{\pm 0.067}$	$10.043_{\pm 0.641}$	$\mathbf{0.053}_{\pm 0.002}$
		Shape only	$0.319_{\pm 0.014}$	$0.285_{\pm 0.007}$	$\mathbf{1.204}_{\pm 0.088}$	$5.298_{\pm 0.624}$	$0.056_{\pm 0.002}$
		FSPC	$\mathbf{0.317}_{\pm 0.014}$	$\mathbf{0.285}_{\pm 0.007}$	$1.356_{\pm 0.125}$	$\mathbf{4.667}_{\pm 0.473}$	$0.056_{\pm 0.002}$
ETTh2	192	Native	$\mathbf{0.374}_{\pm 0.008}$	$0.334_{\pm 0.002}$	$4.584_{\pm 0.306}$	$10.385_{\pm 0.119}$	$\mathbf{0.056}_{\pm 0.004}$
		Mean only	$0.375_{\pm 0.011}$	$0.335_{\pm 0.003}$	$4.611_{\pm 0.334}$	$10.842_{\pm 0.662}$	$0.056_{\pm 0.004}$
		Shape only	$\mathbf{0.374}_{\pm 0.008}$	$0.315_{\pm 0.004}$	$\mathbf{1.273}_{\pm 0.191}$	$5.566_{\pm 0.831}$	$0.064_{\pm 0.004}$
		FSPC	$0.375_{\pm 0.011}$	$\mathbf{0.315}_{\pm 0.004}$	$1.381_{\pm 0.316}$	$\mathbf{5.521}_{\pm 0.998}$	$0.064_{\pm 0.004}$
ETTh2	336	Native	$0.976_{\pm 0.487}$	$0.678_{\pm 0.195}$	$5.199_{\pm 0.240}$	$11.050_{\pm 1.133}$	$0.087_{\pm 0.003}$
		Mean only	$\mathbf{0.919}_{\pm 0.429}$	$0.680_{\pm 0.196}$	$5.590_{\pm 0.596}$	$12.154_{\pm 1.822}$	$0.088_{\pm 0.003}$
		Shape only	$0.976_{\pm 0.487}$	$0.437_{\pm 0.064}$	$\mathbf{1.407}_{\pm 0.206}$	$8.691_{\pm 0.175}$	$0.082_{\pm 0.003}$
		FSPC	$\mathbf{0.919}_{\pm 0.429}$	$\mathbf{0.435}_{\pm 0.063}$	$2.193_{\pm 0.949}$	$\mathbf{8.298}_{\pm 0.077}$	$\mathbf{0.081}_{\pm 0.002}$
ETTh2	720	Native	$\mathbf{0.480}_{\pm 0.051}$	$0.501_{\pm 0.134}$	$5.424_{\pm 0.483}$	$12.480_{\pm 1.241}$	$0.084_{\pm 0.012}$
		Mean only	$0.489_{\pm 0.040}$	$0.523_{\pm 0.139}$	$6.474_{\pm 0.324}$	$15.994_{\pm 1.470}$	$\mathbf{0.083}_{\pm 0.014}$
		Shape only	$\mathbf{0.480}_{\pm 0.051}$	$\mathbf{0.394}_{\pm 0.032}$	$\mathbf{1.633}_{\pm 0.190}$	$\mathbf{8.790}_{\pm 0.916}$	$0.092_{\pm 0.008}$
		FSPC	$0.489_{\pm 0.040}$	$0.396_{\pm 0.030}$	$2.650_{\pm 0.623}$	$8.811_{\pm 1.656}$	$0.089_{\pm 0.006}$
ETTm1	96	Native	$0.338_{\pm 0.012}$	$0.297_{\pm 0.003}$	$3.274_{\pm 0.571}$	$6.841_{\pm 0.272}$	$0.072_{\pm 0.003}$
		Mean only	$\mathbf{0.334}_{\pm 0.011}$	$0.297_{\pm 0.002}$	$3.083_{\pm 0.570}$	$7.608_{\pm 0.555}$	$0.070_{\pm 0.002}$
		Shape only	$0.338_{\pm 0.012}$	$0.285_{\pm 0.006}$	$1.221_{\pm 0.194}$	$6.531_{\pm 0.068}$	$0.065_{\pm 0.003}$
		FSPC	$\mathbf{0.334}_{\pm 0.011}$	$\mathbf{0.279}_{\pm 0.006}$	$\mathbf{1.212}_{\pm 0.097}$	$\mathbf{5.534}_{\pm 0.439}$	$\mathbf{0.064}_{\pm 0.002}$
ETTm1	192	Native	$0.392_{\pm 0.007}$	$0.324_{\pm 0.005}$	$3.766_{\pm 0.836}$	$6.899_{\pm 0.100}$	$0.077_{\pm 0.001}$
		Mean only	$\mathbf{0.384}_{\pm 0.011}$	$0.326_{\pm 0.004}$	$3.580_{\pm 0.818}$	$8.287_{\pm 0.577}$	$0.073_{\pm 0.002}$
		Shape only	$0.392_{\pm 0.007}$	$0.312_{\pm 0.004}$	$\mathbf{1.182}_{\pm 0.119}$	$6.885_{\pm 0.071}$	$0.071_{\pm 0.003}$
		FSPC	$\mathbf{0.384}_{\pm 0.011}$	$\mathbf{0.302}_{\pm 0.005}$	$1.428_{\pm 0.085}$	$\mathbf{6.014}_{\pm 0.050}$	$\mathbf{0.068}_{\pm 0.003}$
ETTm1	336	Native	$0.451_{\pm 0.015}$	$0.343_{\pm 0.009}$	$3.252_{\pm 0.512}$	$6.967_{\pm 0.376}$	$0.080_{\pm 0.001}$
		Mean only	$\mathbf{0.439}_{\pm 0.012}$	$0.348_{\pm 0.014}$	$2.959_{\pm 0.357}$	$8.963_{\pm 0.881}$	$0.075_{\pm 0.001}$
		Shape only	$0.451_{\pm 0.015}$	$0.347_{\pm 0.002}$	$\mathbf{1.517}_{\pm 0.035}$	$7.394_{\pm 0.234}$	$0.075_{\pm 0.001}$
		FSPC	$\mathbf{0.439}_{\pm 0.012}$	$\mathbf{0.332}_{\pm 0.004}$	$1.860_{\pm 0.139}$	$\mathbf{6.886}_{\pm 0.114}$	$\mathbf{0.071}_{\pm 0.001}$
ETTm1	720	Native	$0.513_{\pm 0.029}$	$\mathbf{0.373}_{\pm 0.006}$	$3.517_{\pm 0.438}$	$\mathbf{7.615}_{\pm 0.655}$	$0.085_{\pm 0.007}$
		Mean only	$\mathbf{0.508}_{\pm 0.017}$	$0.397_{\pm 0.005}$	$3.330_{\pm 0.280}$	$12.279_{\pm 0.706}$	$0.076_{\pm 0.004}$
		Shape only	$0.513_{\pm 0.029}$	$0.406_{\pm 0.011}$	$\mathbf{2.292}_{\pm 0.173}$	$8.717_{\pm 0.513}$	$0.080_{\pm 0.006}$
		FSPC	$\mathbf{0.508}_{\pm 0.017}$	$0.382_{\pm 0.009}$	$3.005_{\pm 0.085}$	$8.510_{\pm 0.268}$	$\mathbf{0.074}_{\pm 0.003}$
ETTm2	96	Native	$0.191_{\pm 0.007}$	$0.236_{\pm 0.006}$	$4.325_{\pm 0.121}$	$9.020_{\pm 0.895}$	$0.042_{\pm 0.002}$
		Mean only	$\mathbf{0.188}_{\pm 0.005}$	$0.235_{\pm 0.005}$	$3.750_{\pm 0.217}$	$9.547_{\pm 0.128}$	$0.041_{\pm 0.002}$
		Shape only	$0.191_{\pm 0.007}$	$0.218_{\pm 0.007}$	$1.638_{\pm 0.421}$	$5.006_{\pm 0.740}$	$0.039_{\pm 0.001}$
		FSPC	$\mathbf{0.188}_{\pm 0.005}$	$\mathbf{0.215}_{\pm 0.005}$	$\mathbf{1.514}_{\pm 0.127}$	$\mathbf{4.111}_{\pm 0.574}$	$\mathbf{0.038}_{\pm 0.002}$
ETTm2	192	Native	$0.259_{\pm 0.005}$	$0.275_{\pm 0.007}$	$4.301_{\pm 0.274}$	$9.133_{\pm 0.691}$	$0.045_{\pm 0.003}$
		Mean only	$\mathbf{0.257}_{\pm 0.006}$	$0.275_{\pm 0.006}$	$3.759_{\pm 0.284}$	$9.414_{\pm 0.377}$	$0.045_{\pm 0.003}$
		Shape only	$0.259_{\pm 0.005}$	$0.254_{\pm 0.003}$	$1.629_{\pm 0.314}$	$5.548_{\pm 0.221}$	$0.042_{\pm 0.002}$
		FSPC	$\mathbf{0.257}_{\pm 0.006}$	$\mathbf{0.252}_{\pm 0.004}$	$\mathbf{1.577}_{\pm 0.171}$	$\mathbf{4.962}_{\pm 0.470}$	$\mathbf{0.042}_{\pm 0.002}$

Continued

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTm2	336	Native	$0.315_{\pm 0.003}$	$0.308_{\pm 0.004}$	$4.128_{\pm 0.120}$	$8.247_{\pm 0.505}$	$0.052_{\pm 0.002}$
		Mean only	$\mathbf{0.314}_{\pm 0.003}$	$0.309_{\pm 0.005}$	$3.816_{\pm 0.206}$	$9.729_{\pm 1.203}$	$0.052_{\pm 0.003}$
		Shape only	$0.315_{\pm 0.003}$	$\mathbf{0.280}_{\pm 0.003}$	$\mathbf{1.389}_{\pm 0.173}$	$5.653_{\pm 0.098}$	$\mathbf{0.049}_{\pm 0.001}$
		FSPC	$\mathbf{0.314}_{\pm 0.003}$	$0.281_{\pm 0.003}$	$1.685_{\pm 0.567}$	$\mathbf{5.523}_{\pm 0.945}$	$0.049_{\pm 0.001}$
ETTm2	720	Native	$0.707_{\pm 0.420}$	$0.465_{\pm 0.091}$	$4.618_{\pm 0.707}$	$7.916_{\pm 0.629}$	$0.080_{\pm 0.008}$
		Mean only	$\mathbf{0.686}_{\pm 0.387}$	$0.471_{\pm 0.094}$	$4.561_{\pm 1.014}$	$11.004_{\pm 1.446}$	$0.081_{\pm 0.009}$
		Shape only	$0.707_{\pm 0.420}$	$\mathbf{0.350}_{\pm 0.019}$	$\mathbf{1.626}_{\pm 0.292}$	$7.319_{\pm 0.182}$	$0.067_{\pm 0.003}$
		FSPC	$\mathbf{0.686}_{\pm 0.387}$	$0.351_{\pm 0.020}$	$2.272_{\pm 0.122}$	$\mathbf{6.713}_{\pm 0.805}$	$\mathbf{0.065}_{\pm 0.003}$
Electricity	96	Native	$0.167_{\pm 0.003}$	$0.208_{\pm 0.002}$	$1.258_{\pm 0.123}$	$3.678_{\pm 0.936}$	$0.071_{\pm 0.000}$
		Mean only	$\mathbf{0.166}_{\pm 0.002}$	$0.207_{\pm 0.002}$	$1.212_{\pm 0.125}$	$3.455_{\pm 0.648}$	$0.071_{\pm 0.001}$
		Shape only	$0.167_{\pm 0.003}$	$0.203_{\pm 0.002}$	$0.469_{\pm 0.141}$	$3.480_{\pm 0.992}$	$0.070_{\pm 0.000}$
		FSPC	$\mathbf{0.166}_{\pm 0.002}$	$\mathbf{0.203}_{\pm 0.002}$	$\mathbf{0.383}_{\pm 0.188}$	$\mathbf{3.095}_{\pm 0.940}$	$\mathbf{0.069}_{\pm 0.000}$
Electricity	192	Native	$0.181_{\pm 0.013}$	$0.218_{\pm 0.008}$	$1.362_{\pm 0.152}$	$3.848_{\pm 1.055}$	$0.076_{\pm 0.003}$
		Mean only	$\mathbf{0.181}_{\pm 0.015}$	$0.218_{\pm 0.008}$	$1.339_{\pm 0.087}$	$3.700_{\pm 1.177}$	$0.076_{\pm 0.003}$
		Shape only	$0.181_{\pm 0.013}$	$0.213_{\pm 0.007}$	$\mathbf{0.654}_{\pm 0.469}$	$3.629_{\pm 1.254}$	$\mathbf{0.075}_{\pm 0.003}$
		FSPC	$\mathbf{0.181}_{\pm 0.015}$	$\mathbf{0.213}_{\pm 0.008}$	$0.708_{\pm 0.513}$	$\mathbf{3.248}_{\pm 1.421}$	$0.075_{\pm 0.003}$
Electricity	336	Native	$0.207_{\pm 0.007}$	$0.237_{\pm 0.007}$	$1.577_{\pm 0.248}$	$3.574_{\pm 0.081}$	$0.080_{\pm 0.002}$
		Mean only	$\mathbf{0.204}_{\pm 0.007}$	$0.237_{\pm 0.008}$	$1.457_{\pm 0.223}$	$3.494_{\pm 0.340}$	$0.081_{\pm 0.002}$
		Shape only	$0.207_{\pm 0.007}$	$0.229_{\pm 0.005}$	$0.640_{\pm 0.135}$	$3.297_{\pm 0.283}$	$\mathbf{0.079}_{\pm 0.002}$
		FSPC	$\mathbf{0.204}_{\pm 0.007}$	$\mathbf{0.229}_{\pm 0.005}$	$\mathbf{0.613}_{\pm 0.151}$	$\mathbf{2.974}_{\pm 0.214}$	$0.079_{\pm 0.002}$
Electricity	720	Native	$0.266_{\pm 0.003}$	$0.283_{\pm 0.004}$	$1.747_{\pm 0.211}$	$3.902_{\pm 0.294}$	$0.093_{\pm 0.001}$
		Mean only	$\mathbf{0.260}_{\pm 0.001}$	$0.284_{\pm 0.004}$	$1.546_{\pm 0.300}$	$4.033_{\pm 0.471}$	$0.095_{\pm 0.001}$
		Shape only	$0.266_{\pm 0.003}$	$\mathbf{0.265}_{\pm 0.002}$	$\mathbf{0.975}_{\pm 0.231}$	$\mathbf{3.785}_{\pm 0.548}$	$\mathbf{0.092}_{\pm 0.000}$
		FSPC	$\mathbf{0.260}_{\pm 0.001}$	$0.266_{\pm 0.002}$	$1.011_{\pm 0.219}$	$3.924_{\pm 0.400}$	$0.094_{\pm 0.001}$
Exchange	96	Native	$\mathbf{0.138}_{\pm 0.006}$	$0.271_{\pm 0.010}$	$5.156_{\pm 0.599}$	$12.356_{\pm 0.237}$	$0.078_{\pm 0.006}$
		Mean only	$0.144_{\pm 0.014}$	$0.275_{\pm 0.016}$	$5.354_{\pm 0.933}$	$13.460_{\pm 2.118}$	$0.078_{\pm 0.006}$
		Shape only	$\mathbf{0.138}_{\pm 0.006}$	$\mathbf{0.238}_{\pm 0.033}$	$\mathbf{3.752}_{\pm 0.789}$	$\mathbf{9.797}_{\pm 1.592}$	$0.073_{\pm 0.002}$
		FSPC	$0.144_{\pm 0.014}$	$0.242_{\pm 0.029}$	$4.064_{\pm 0.295}$	$10.888_{\pm 3.479}$	$\mathbf{0.072}_{\pm 0.003}$
Exchange	192	Native	$\mathbf{0.277}_{\pm 0.018}$	$0.343_{\pm 0.014}$	$5.097_{\pm 0.404}$	$\mathbf{14.045}_{\pm 0.347}$	$0.094_{\pm 0.009}$
		Mean only	$\mathbf{0.277}_{\pm 0.018}$	$0.343_{\pm 0.014}$	$5.097_{\pm 0.404}$	$\mathbf{14.045}_{\pm 0.347}$	$0.094_{\pm 0.009}$
		Shape only	$\mathbf{0.277}_{\pm 0.018}$	$\mathbf{0.287}_{\pm 0.013}$	$\mathbf{4.119}_{\pm 0.335}$	$14.677_{\pm 0.456}$	$\mathbf{0.089}_{\pm 0.010}$
		FSPC	$\mathbf{0.277}_{\pm 0.018}$	$\mathbf{0.287}_{\pm 0.013}$	$\mathbf{4.119}_{\pm 0.335}$	$14.677_{\pm 0.456}$	$\mathbf{0.089}_{\pm 0.010}$
Exchange	336	Native	$\mathbf{0.500}_{\pm 0.033}$	$0.462_{\pm 0.026}$	$5.938_{\pm 1.010}$	$\mathbf{14.017}_{\pm 1.413}$	$0.122_{\pm 0.006}$
		Mean only	$\mathbf{0.500}_{\pm 0.033}$	$0.462_{\pm 0.026}$	$5.938_{\pm 1.010}$	$\mathbf{14.017}_{\pm 1.413}$	$0.122_{\pm 0.006}$
		Shape only	$\mathbf{0.500}_{\pm 0.033}$	$\mathbf{0.388}_{\pm 0.018}$	$\mathbf{4.106}_{\pm 0.892}$	$16.313_{\pm 2.996}$	$\mathbf{0.106}_{\pm 0.004}$
		FSPC	$\mathbf{0.500}_{\pm 0.033}$	$\mathbf{0.388}_{\pm 0.018}$	$\mathbf{4.106}_{\pm 0.892}$	$16.313_{\pm 2.996}$	$\mathbf{0.106}_{\pm 0.004}$
Exchange	720	Native	$\mathbf{1.272}_{\pm 0.147}$	$0.690_{\pm 0.058}$	$6.640_{\pm 1.835}$	$\mathbf{15.833}_{\pm 1.141}$	$0.186_{\pm 0.005}$
		Mean only	$1.603_{\pm 0.496}$	$0.791_{\pm 0.134}$	$7.473_{\pm 1.700}$	$19.473_{\pm 5.567}$	$0.204_{\pm 0.031}$
		Shape only	$\mathbf{1.272}_{\pm 0.147}$	$\mathbf{0.626}_{\pm 0.021}$	$\mathbf{5.545}_{\pm 1.289}$	$20.917_{\pm 4.016}$	$\mathbf{0.180}_{\pm 0.007}$
		FSPC	$1.603_{\pm 0.496}$	$0.733_{\pm 0.166}$	$7.001_{\pm 2.142}$	$24.880_{\pm 6.075}$	$0.192_{\pm 0.025}$
Weather	96	Native	$0.172_{\pm 0.008}$	$0.204_{\pm 0.003}$	$6.166_{\pm 0.143}$	$6.443_{\pm 0.121}$	$0.065_{\pm 0.001}$
		Mean only	$\mathbf{0.167}_{\pm 0.005}$	$0.207_{\pm 0.003}$	$5.839_{\pm 0.083}$	$9.902_{\pm 0.705}$	$0.066_{\pm 0.001}$
		Shape only	$0.172_{\pm 0.008}$	$0.176_{\pm 0.003}$	$2.013_{\pm 0.253}$	$6.980_{\pm 0.428}$	$\mathbf{0.050}_{\pm 0.002}$
		FSPC	$\mathbf{0.167}_{\pm 0.005}$	$\mathbf{0.172}_{\pm 0.002}$	$\mathbf{1.232}_{\pm 0.106}$	$\mathbf{5.606}_{\pm 0.686}$	$0.056_{\pm 0.001}$
Weather	192	Native	$0.226_{\pm 0.002}$	$0.232_{\pm 0.004}$	$5.242_{\pm 0.149}$	$6.226_{\pm 0.262}$	$0.069_{\pm 0.002}$
		Mean only	$\mathbf{0.218}_{\pm 0.004}$	$0.235_{\pm 0.003}$	$5.009_{\pm 0.086}$	$9.558_{\pm 0.089}$	$0.071_{\pm 0.002}$
		Shape only	$0.226_{\pm 0.002}$	$0.214_{\pm 0.003}$	$2.627_{\pm 0.204}$	$7.682_{\pm 0.408}$	$\mathbf{0.055}_{\pm 0.002}$
		FSPC	$\mathbf{0.218}_{\pm 0.004}$	$\mathbf{0.204}_{\pm 0.002}$	$\mathbf{1.301}_{\pm 0.085}$	$\mathbf{4.813}_{\pm 0.341}$	$0.062_{\pm 0.002}$
Weather	336	Native	$0.303_{\pm 0.013}$	$0.265_{\pm 0.006}$	$4.612_{\pm 0.382}$	$6.535_{\pm 0.743}$	$0.077_{\pm 0.002}$
		Mean only	$\mathbf{0.281}_{\pm 0.008}$	$0.269_{\pm 0.005}$	$4.256_{\pm 0.272}$	$10.693_{\pm 0.496}$	$0.078_{\pm 0.001}$
		Shape only	$0.303_{\pm 0.013}$	$0.257_{\pm 0.003}$	$2.877_{\pm 0.434}$	$9.314_{\pm 0.155}$	$\mathbf{0.065}_{\pm 0.004}$
		FSPC	$\mathbf{0.281}_{\pm 0.008}$	$\mathbf{0.239}_{\pm 0.004}$	$\mathbf{1.609}_{\pm 0.029}$	$\mathbf{5.678}_{\pm 0.671}$	$0.069_{\pm 0.002}$
Weather	720	Native	$0.407_{\pm 0.005}$	$0.309_{\pm 0.003}$	$4.276_{\pm 0.320}$	$7.113_{\pm 0.137}$	$0.086_{\pm 0.002}$
		Mean only	$\mathbf{0.367}_{\pm 0.005}$	$0.315_{\pm 0.001}$	$3.853_{\pm 0.058}$	$11.079_{\pm 0.456}$	$0.084_{\pm 0.002}$
		Shape only	$0.407_{\pm 0.005}$	$0.301_{\pm 0.002}$	$2.469_{\pm 0.234}$	$9.952_{\pm 0.437}$	$0.077_{\pm 0.002}$
		FSPC	$\mathbf{0.367}_{\pm 0.005}$	$\mathbf{0.282}_{\pm 0.002}$	$\mathbf{1.836}_{\pm 0.148}$	$\mathbf{6.572}_{\pm 0.497}$	$\mathbf{0.075}_{\pm 0.002}$
Traffic	96	Native	$0.560_{\pm 0.035}$	$0.247_{\pm 0.002}$	$2.634_{\pm 0.278}$	$5.541_{\pm 0.425}$	$0.058_{\pm 0.001}$
		Mean only	$\mathbf{0.539}_{\pm 0.032}$	$0.245_{\pm 0.003}$	$2.624_{\pm 0.291}$	$5.797_{\pm 0.662}$	$0.059_{\pm 0.001}$
		Shape only	$0.560_{\pm 0.035}$	$0.241_{\pm 0.001}$	$\mathbf{1.166}_{\pm 0.461}$	$5.441_{\pm 0.702}$	$0.058_{\pm 0.001}$
		FSPC	$\mathbf{0.539}_{\pm 0.032}$	$\mathbf{0.238}_{\pm 0.002}$	$1.429_{\pm 0.465}$	$\mathbf{4.559}_{\pm 0.978}$	$\mathbf{0.057}_{\pm 0.000}$
Traffic	192	Native	$0.615_{\pm 0.035}$	$0.257_{\pm 0.003}$	$2.668_{\pm 0.111}$	$6.075_{\pm 0.323}$	$0.061_{\pm 0.001}$
		Mean only	$\mathbf{0.596}_{\pm 0.041}$	$0.254_{\pm 0.003}$	$2.934_{\pm 0.127}$	$6.179_{\pm 0.942}$	$0.061_{\pm 0.000}$
		Shape only	$0.615_{\pm 0.035}$	$0.250_{\pm 0.002}$	$\mathbf{1.078}_{\pm 0.217}$	$5.744_{\pm 0.315}$	$0.060_{\pm 0.001}$
		FSPC	$\mathbf{0.596}_{\pm 0.041}$	$\mathbf{0.246}_{\pm 0.003}$	$1.269_{\pm 0.328}$	$\mathbf{4.555}_{\pm 0.538}$	$\mathbf{0.059}_{\pm 0.000}$

Continued

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
Traffic	336	Native	$0.650_{\pm 0.043}$	$0.275_{\pm 0.001}$	$2.852_{\pm 0.410}$	$5.897_{\pm 0.169}$	$0.065_{\pm 0.001}$
		Mean only	$\mathbf{0.630}_{\pm 0.039}$	$0.272_{\pm 0.001}$	$3.036_{\pm 0.479}$	$5.717_{\pm 0.562}$	$0.065_{\pm 0.001}$
		Shape only	$0.650_{\pm 0.043}$	$0.268_{\pm 0.002}$	$1.331_{\pm 0.365}$	$5.577_{\pm 0.431}$	$0.065_{\pm 0.001}$
		FSPC	$\mathbf{0.630}_{\pm 0.039}$	$\mathbf{0.263}_{\pm 0.001}$	$\mathbf{1.322}_{\pm 0.156}$	$\mathbf{4.125}_{\pm 0.167}$	$\mathbf{0.064}_{\pm 0.001}$
Traffic	720	Native	$0.651_{\pm 0.050}$	$0.280_{\pm 0.006}$	$2.715_{\pm 0.471}$	$6.665_{\pm 1.012}$	$0.068_{\pm 0.003}$
		Mean only	$\mathbf{0.639}_{\pm 0.073}$	$0.280_{\pm 0.012}$	$2.726_{\pm 0.355}$	$6.700_{\pm 2.844}$	$0.069_{\pm 0.006}$
		Shape only	$0.651_{\pm 0.050}$	$0.273_{\pm 0.004}$	$\mathbf{1.579}_{\pm 0.416}$	$\mathbf{6.495}_{\pm 0.285}$	$\mathbf{0.067}_{\pm 0.002}$
		FSPC	$\mathbf{0.639}_{\pm 0.073}$	$\mathbf{0.272}_{\pm 0.011}$	$1.822_{\pm 0.242}$	$6.540_{\pm 3.015}$	$0.067_{\pm 0.005}$
ILI	24	Native	$\mathbf{0.461}_{\pm 0.022}$	$\mathbf{0.345}_{\pm 0.016}$	$8.865_{\pm 0.225}$	–	$0.130_{\pm 0.002}$
		Mean only	$\mathbf{0.461}_{\pm 0.022}$	$\mathbf{0.345}_{\pm 0.016}$	$8.865_{\pm 0.225}$	–	$0.130_{\pm 0.002}$
		Shape only	$\mathbf{0.461}_{\pm 0.022}$	$0.348_{\pm 0.006}$	$\mathbf{6.972}_{\pm 0.694}$	–	$\mathbf{0.129}_{\pm 0.007}$
		FSPC	$\mathbf{0.461}_{\pm 0.022}$	$0.348_{\pm 0.006}$	$\mathbf{6.972}_{\pm 0.694}$	–	$\mathbf{0.129}_{\pm 0.007}$
ILI	36	Native	$\mathbf{2.771}_{\pm 0.306}$	$0.841_{\pm 0.056}$	$9.262_{\pm 0.238}$	–	$\mathbf{0.154}_{\pm 0.015}$
		Mean only	$\mathbf{2.771}_{\pm 0.306}$	$0.841_{\pm 0.056}$	$9.262_{\pm 0.238}$	–	$\mathbf{0.154}_{\pm 0.015}$
		Shape only	$\mathbf{2.771}_{\pm 0.306}$	$\mathbf{0.834}_{\pm 0.055}$	$\mathbf{7.259}_{\pm 2.083}$	–	$0.157_{\pm 0.012}$
		FSPC	$\mathbf{2.771}_{\pm 0.306}$	$\mathbf{0.834}_{\pm 0.055}$	$\mathbf{7.259}_{\pm 2.083}$	–	$0.157_{\pm 0.012}$
ILI	48	Native	$\mathbf{2.595}_{\pm 0.368}$	$0.838_{\pm 0.062}$	$10.335_{\pm 0.715}$	–	$0.131_{\pm 0.013}$
		Mean only	$2.614_{\pm 0.386}$	$0.843_{\pm 0.068}$	$10.405_{\pm 0.819}$	–	$0.134_{\pm 0.017}$
		Shape only	$\mathbf{2.595}_{\pm 0.368}$	$\mathbf{0.814}_{\pm 0.053}$	$7.790_{\pm 0.355}$	–	$\mathbf{0.130}_{\pm 0.006}$
		FSPC	$2.614_{\pm 0.386}$	$0.819_{\pm 0.059}$	$\mathbf{7.772}_{\pm 0.534}$	–	$0.133_{\pm 0.009}$
ILI	60	Native	$\mathbf{2.392}_{\pm 0.294}$	$0.802_{\pm 0.059}$	$10.873_{\pm 1.133}$	$31.224_{\pm 3.398}$	$\mathbf{0.123}_{\pm 0.021}$
		Mean only	$2.431_{\pm 0.311}$	$0.814_{\pm 0.067}$	$11.063_{\pm 1.251}$	$33.195_{\pm 4.681}$	$0.130_{\pm 0.026}$
		Shape only	$\mathbf{2.392}_{\pm 0.294}$	$\mathbf{0.785}_{\pm 0.046}$	$\mathbf{8.286}_{\pm 0.486}$	$\mathbf{19.800}_{\pm 0.194}$	$0.129_{\pm 0.016}$
		FSPC	$2.431_{\pm 0.311}$	$0.796_{\pm 0.052}$	$8.421_{\pm 0.657}$	$20.446_{\pm 0.953}$	$0.135_{\pm 0.020}$

TABLE S32: PPM: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTh1	192	Native	$0.447_{\pm 0.005}$	$0.337_{\pm 0.002}$	$\mathbf{1.792}_{\pm 0.095}$	$\mathbf{6.421}_{\pm 0.094}$	$\mathbf{0.073}_{\pm 0.001}$
		Mean only	$\mathbf{0.439}_{\pm 0.005}$	$\mathbf{0.336}_{\pm 0.001}$	$1.899_{\pm 0.084}$	$7.091_{\pm 0.166}$	$0.073_{\pm 0.001}$
		Shape only	$0.447_{\pm 0.005}$	$0.352_{\pm 0.002}$	$2.532_{\pm 0.101}$	$8.304_{\pm 0.107}$	$0.075_{\pm 0.001}$
		FSPC	$\mathbf{0.439}_{\pm 0.005}$	$0.346_{\pm 0.001}$	$2.695_{\pm 0.107}$	$7.379_{\pm 0.117}$	$0.073_{\pm 0.001}$
ETTh2	192	Native	$0.373_{\pm 0.008}$	$0.305_{\pm 0.004}$	$3.534_{\pm 0.177}$	$7.817_{\pm 0.524}$	$0.045_{\pm 0.001}$
		Mean only	$\mathbf{0.371}_{\pm 0.006}$	$0.309_{\pm 0.003}$	$4.299_{\pm 0.208}$	$9.794_{\pm 0.613}$	$\mathbf{0.042}_{\pm 0.001}$
		Shape only	$0.373_{\pm 0.008}$	$0.303_{\pm 0.003}$	$2.213_{\pm 0.135}$	$6.505_{\pm 0.183}$	$0.047_{\pm 0.001}$
		FSPC	$\mathbf{0.371}_{\pm 0.006}$	$\mathbf{0.302}_{\pm 0.002}$	$\mathbf{1.482}_{\pm 0.129}$	$\mathbf{5.783}_{\pm 0.205}$	$0.044_{\pm 0.001}$
ETTm1	192	Native	$\mathbf{0.379}_{\pm 0.004}$	$0.313_{\pm 0.002}$	$2.136_{\pm 0.036}$	$7.125_{\pm 0.107}$	$0.071_{\pm 0.001}$
		Mean only	$0.380_{\pm 0.002}$	$0.317_{\pm 0.001}$	$2.074_{\pm 0.038}$	$8.225_{\pm 0.077}$	$0.070_{\pm 0.001}$
		Shape only	$\mathbf{0.379}_{\pm 0.004}$	$0.300_{\pm 0.001}$	$1.026_{\pm 0.143}$	$\mathbf{5.373}_{\pm 0.185}$	$0.070_{\pm 0.001}$
		FSPC	$0.380_{\pm 0.002}$	$\mathbf{0.298}_{\pm 0.001}$	$\mathbf{0.832}_{\pm 0.179}$	$5.523_{\pm 0.134}$	$\mathbf{0.066}_{\pm 0.001}$
ETTm2	192	Native	$\mathbf{0.248}_{\pm 0.002}$	$0.237_{\pm 0.001}$	$1.962_{\pm 0.199}$	$5.632_{\pm 0.015}$	$\mathbf{0.037}_{\pm 0.000}$
		Mean only	$0.248_{\pm 0.002}$	$0.242_{\pm 0.002}$	$3.093_{\pm 0.158}$	$9.879_{\pm 0.356}$	$0.047_{\pm 0.001}$
		Shape only	$\mathbf{0.248}_{\pm 0.002}$	$\mathbf{0.236}_{\pm 0.002}$	$\mathbf{1.838}_{\pm 0.057}$	$\mathbf{4.774}_{\pm 0.183}$	$0.037_{\pm 0.000}$
		FSPC	$0.248_{\pm 0.002}$	$0.239_{\pm 0.002}$	$2.212_{\pm 0.104}$	$6.944_{\pm 0.363}$	$0.046_{\pm 0.001}$
Electricity	192	Native	$0.179_{\pm 0.003}$	$0.205_{\pm 0.001}$	$3.907_{\pm 0.059}$	$7.646_{\pm 0.163}$	$0.072_{\pm 0.000}$
		Mean only	$\mathbf{0.178}_{\pm 0.003}$	$0.205_{\pm 0.002}$	$4.069_{\pm 0.114}$	$8.076_{\pm 0.642}$	$0.071_{\pm 0.000}$
		Shape only	$0.179_{\pm 0.003}$	$0.200_{\pm 0.001}$	$\mathbf{1.075}_{\pm 0.030}$	$3.386_{\pm 0.215}$	$0.070_{\pm 0.000}$
		FSPC	$\mathbf{0.178}_{\pm 0.003}$	$\mathbf{0.199}_{\pm 0.001}$	$1.267_{\pm 0.106}$	$\mathbf{3.245}_{\pm 0.347}$	$\mathbf{0.069}_{\pm 0.000}$
Weather	192	Native	$0.237_{\pm 0.006}$	$\mathbf{0.214}_{\pm 0.001}$	$4.417_{\pm 0.019}$	$\mathbf{10.769}_{\pm 0.120}$	$0.062_{\pm 0.001}$
		Mean only	$\mathbf{0.234}_{\pm 0.006}$	$0.216_{\pm 0.001}$	$5.488_{\pm 0.180}$	$14.601_{\pm 0.573}$	$0.085_{\pm 0.002}$
		Shape only	$0.237_{\pm 0.006}$	$0.225_{\pm 0.005}$	$2.951_{\pm 0.262}$	$11.431_{\pm 0.437}$	$\mathbf{0.056}_{\pm 0.000}$
		FSPC	$\mathbf{0.234}_{\pm 0.006}$	$0.220_{\pm 0.004}$	$\mathbf{2.742}_{\pm 0.426}$	$11.436_{\pm 0.703}$	$0.082_{\pm 0.001}$
Traffic	192	Native	$\mathbf{0.523}_{\pm 0.012}$	$0.255_{\pm 0.004}$	$3.073_{\pm 0.180}$	$6.382_{\pm 0.684}$	$\mathbf{0.062}_{\pm 0.001}$
		Mean only	$0.524_{\pm 0.017}$	$0.261_{\pm 0.004}$	$3.425_{\pm 0.171}$	$8.073_{\pm 0.414}$	$0.063_{\pm 0.001}$
		Shape only	$\mathbf{0.523}_{\pm 0.012}$	$\mathbf{0.253}_{\pm 0.003}$	$0.523_{\pm 0.027}$	$3.159_{\pm 0.175}$	$0.063_{\pm 0.001}$
		FSPC	$0.524_{\pm 0.017}$	$0.254_{\pm 0.003}$	$\mathbf{0.520}_{\pm 0.071}$	$\mathbf{2.724}_{\pm 0.052}$	$0.062_{\pm 0.001}$

TABLE S33: TMDM: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTh1	192	Native	$\mathbf{0.689}_{\pm 0.075}$	$0.453_{\pm 0.039}$	$3.272_{\pm 1.558}$	$15.868_{\pm 3.651}$	$0.098_{\pm 0.004}$
		Mean only	$0.699_{\pm 0.067}$	$0.459_{\pm 0.035}$	$3.582_{\pm 1.226}$	$16.982_{\pm 2.736}$	$0.098_{\pm 0.005}$
		Shape only	$\mathbf{0.689}_{\pm 0.075}$	$0.442_{\pm 0.021}$	$2.474_{\pm 0.868}$	$12.597_{\pm 0.994}$	$0.097_{\pm 0.003}$
		FSPC	$0.699_{\pm 0.067}$	$\mathbf{0.441}_{\pm 0.021}$	$\mathbf{2.211}_{\pm 1.036}$	$\mathbf{12.454}_{\pm 1.606}$	$\mathbf{0.094}_{\pm 0.002}$
ETTh2	192	Native	$0.573_{\pm 0.084}$	$0.401_{\pm 0.022}$	$4.488_{\pm 0.828}$	$11.209_{\pm 1.734}$	$0.061_{\pm 0.009}$
		Mean only	$\mathbf{0.565}_{\pm 0.089}$	$0.399_{\pm 0.025}$	$4.565_{\pm 0.881}$	$11.244_{\pm 2.451}$	$\mathbf{0.059}_{\pm 0.007}$
		Shape only	$0.573_{\pm 0.084}$	$0.387_{\pm 0.024}$	$1.880_{\pm 0.432}$	$7.128_{\pm 0.799}$	$0.064_{\pm 0.007}$
		FSPC	$\mathbf{0.565}_{\pm 0.089}$	$\mathbf{0.385}_{\pm 0.026}$	$\mathbf{1.866}_{\pm 0.517}$	$\mathbf{7.071}_{\pm 1.663}$	$0.062_{\pm 0.006}$
ETTm1	192	Native	$0.470_{\pm 0.061}$	$0.347_{\pm 0.017}$	$2.615_{\pm 0.547}$	$11.537_{\pm 2.121}$	$0.070_{\pm 0.003}$
		Mean only	$\mathbf{0.455}_{\pm 0.053}$	$0.344_{\pm 0.015}$	$2.695_{\pm 0.532}$	$10.935_{\pm 1.633}$	$\mathbf{0.068}_{\pm 0.002}$
		Shape only	$0.470_{\pm 0.061}$	$0.346_{\pm 0.018}$	$1.195_{\pm 0.366}$	$9.333_{\pm 1.829}$	$0.074_{\pm 0.005}$
		FSPC	$\mathbf{0.455}_{\pm 0.053}$	$\mathbf{0.338}_{\pm 0.016}$	$\mathbf{0.850}_{\pm 0.284}$	$\mathbf{8.559}_{\pm 1.152}$	$0.069_{\pm 0.003}$
ETTm2	192	Native	$0.320_{\pm 0.061}$	$0.298_{\pm 0.032}$	$2.180_{\pm 0.751}$	$11.368_{\pm 1.949}$	$0.036_{\pm 0.004}$
		Mean only	$\mathbf{0.320}_{\pm 0.060}$	$0.298_{\pm 0.032}$	$\mathbf{2.163}_{\pm 0.755}$	$11.331_{\pm 1.924}$	$0.037_{\pm 0.004}$
		Shape only	$0.320_{\pm 0.061}$	$0.291_{\pm 0.030}$	$2.213_{\pm 0.390}$	$11.188_{\pm 1.737}$	$\mathbf{0.036}_{\pm 0.004}$
		FSPC	$\mathbf{0.320}_{\pm 0.060}$	$\mathbf{0.291}_{\pm 0.030}$	$2.205_{\pm 0.388}$	$\mathbf{11.159}_{\pm 1.709}$	$0.036_{\pm 0.004}$
Electricity	192	Native	$0.242_{\pm 0.003}$	$0.442_{\pm 0.002}$	$10.418_{\pm 0.010}$	$8.265_{\pm 0.196}$	$0.115_{\pm 0.000}$
		Mean only	$\mathbf{0.237}_{\pm 0.003}$	$0.441_{\pm 0.002}$	$10.430_{\pm 0.014}$	$7.697_{\pm 0.330}$	$0.117_{\pm 0.001}$
		Shape only	$0.242_{\pm 0.003}$	$0.251_{\pm 0.001}$	$1.305_{\pm 0.160}$	$5.109_{\pm 0.498}$	$\mathbf{0.084}_{\pm 0.001}$
		FSPC	$\mathbf{0.237}_{\pm 0.003}$	$\mathbf{0.248}_{\pm 0.002}$	$\mathbf{1.213}_{\pm 0.224}$	$\mathbf{3.846}_{\pm 0.763}$	$0.085_{\pm 0.002}$
Exchange	192	Native	$\mathbf{0.457}_{\pm 0.106}$	$0.392_{\pm 0.047}$	$6.437_{\pm 1.957}$	$16.803_{\pm 3.092}$	$0.104_{\pm 0.009}$
		Mean only	$0.466_{\pm 0.117}$	$0.395_{\pm 0.050}$	$6.360_{\pm 2.024}$	$17.094_{\pm 2.948}$	$\mathbf{0.100}_{\pm 0.007}$
		Shape only	$\mathbf{0.457}_{\pm 0.106}$	$\mathbf{0.372}_{\pm 0.059}$	$\mathbf{3.982}_{\pm 0.514}$	$\mathbf{12.814}_{\pm 1.292}$	$0.107_{\pm 0.013}$
		FSPC	$0.466_{\pm 0.117}$	$0.374_{\pm 0.063}$	$4.190_{\pm 0.669}$	$13.187_{\pm 1.683}$	$0.104_{\pm 0.010}$
Traffic	192	Native	$0.660_{\pm 0.078}$	$0.557_{\pm 0.004}$	$10.588_{\pm 0.107}$	$9.387_{\pm 0.293}$	$0.109_{\pm 0.002}$
		Mean only	$\mathbf{0.623}_{\pm 0.071}$	$0.552_{\pm 0.002}$	$11.015_{\pm 0.072}$	$8.967_{\pm 0.278}$	$0.110_{\pm 0.001}$
		Shape only	$0.660_{\pm 0.078}$	$0.321_{\pm 0.010}$	$1.790_{\pm 0.255}$	$6.843_{\pm 0.527}$	$0.073_{\pm 0.003}$
		FSPC	$\mathbf{0.623}_{\pm 0.071}$	$\mathbf{0.309}_{\pm 0.007}$	$\mathbf{1.652}_{\pm 0.290}$	$\mathbf{4.600}_{\pm 0.579}$	$\mathbf{0.071}_{\pm 0.002}$
ILI	36	Native	$2.754_{\pm 0.682}$	$0.810_{\pm 0.100}$	$5.201_{\pm 0.869}$	–	$0.172_{\pm 0.038}$
		Mean only	$\mathbf{2.742}_{\pm 0.700}$	$\mathbf{0.808}_{\pm 0.103}$	$\mathbf{5.196}_{\pm 0.878}$	–	$0.171_{\pm 0.039}$
		Shape only	$2.754_{\pm 0.682}$	$0.823_{\pm 0.103}$	$5.995_{\pm 1.151}$	–	$0.168_{\pm 0.063}$
		FSPC	$\mathbf{2.742}_{\pm 0.700}$	$0.820_{\pm 0.107}$	$6.026_{\pm 1.178}$	–	$\mathbf{0.168}_{\pm 0.063}$
SolarEnergy	192	Native	$0.242_{\pm 0.003}$	$0.341_{\pm 0.003}$	$9.285_{\pm 0.087}$	$12.888_{\pm 0.197}$	$0.051_{\pm 0.001}$
		Mean only	$\mathbf{0.231}_{\pm 0.003}$	$0.337_{\pm 0.003}$	$10.396_{\pm 0.035}$	$8.130_{\pm 0.128}$	$0.050_{\pm 0.000}$
		Shape only	$0.242_{\pm 0.003}$	$0.223_{\pm 0.002}$	$6.065_{\pm 0.118}$	$23.194_{\pm 1.439}$	$0.047_{\pm 0.001}$
		FSPC	$\mathbf{0.231}_{\pm 0.003}$	$\mathbf{0.190}_{\pm 0.002}$	$\mathbf{0.996}_{\pm 0.274}$	$\mathbf{7.934}_{\pm 0.624}$	$\mathbf{0.046}_{\pm 0.001}$

TABLE S34: Diffusion-TS: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTh1	192	Native	$1.423_{\pm 0.332}$	$0.659_{\pm 0.068}$	$2.816_{\pm 1.683}$	$11.420_{\pm 1.778}$	$0.154_{\pm 0.039}$
		Mean only	$\mathbf{1.146}_{\pm 0.161}$	$0.604_{\pm 0.048}$	$2.944_{\pm 2.053}$	$\mathbf{9.283}_{\pm 2.220}$	$\mathbf{0.115}_{\pm 0.006}$
		Shape only	$1.423_{\pm 0.332}$	$0.677_{\pm 0.077}$	$4.136_{\pm 2.218}$	$13.944_{\pm 2.748}$	$0.168_{\pm 0.046}$
		FSPC	$\mathbf{1.146}_{\pm 0.161}$	$\mathbf{0.586}_{\pm 0.033}$	$\mathbf{2.540}_{\pm 1.074}$	$9.785_{\pm 2.304}$	$0.116_{\pm 0.010}$
ETTh2	192	Native	$3.539_{\pm 1.304}$	$1.147_{\pm 0.124}$	$7.742_{\pm 0.110}$	$17.065_{\pm 5.940}$	$0.190_{\pm 0.045}$
		Mean only	$\mathbf{1.989}_{\pm 0.677}$	$0.923_{\pm 0.036}$	$7.115_{\pm 1.584}$	$12.194_{\pm 4.666}$	$0.130_{\pm 0.022}$
		Shape only	$3.539_{\pm 1.304}$	$1.136_{\pm 0.257}$	$7.098_{\pm 2.827}$	$22.338_{\pm 7.026}$	$0.196_{\pm 0.059}$
		FSPC	$\mathbf{1.989}_{\pm 0.677}$	$\mathbf{0.794}_{\pm 0.147}$	$\mathbf{3.926}_{\pm 2.965}$	$\mathbf{11.791}_{\pm 8.381}$	$\mathbf{0.101}_{\pm 0.051}$
ETTm1	192	Native	$1.539_{\pm 0.256}$	$0.873_{\pm 0.056}$	$13.310_{\pm 0.288}$	$30.581_{\pm 0.394}$	$0.156_{\pm 0.014}$
		Mean only	$\mathbf{1.005}_{\pm 0.120}$	$0.696_{\pm 0.032}$	$12.754_{\pm 0.516}$	$26.036_{\pm 1.208}$	$0.107_{\pm 0.006}$
		Shape only	$1.539_{\pm 0.256}$	$0.691_{\pm 0.054}$	$2.750_{\pm 0.542}$	$14.977_{\pm 0.817}$	$0.124_{\pm 0.009}$
		FSPC	$\mathbf{1.005}_{\pm 0.120}$	$\mathbf{0.565}_{\pm 0.035}$	$\mathbf{2.456}_{\pm 0.686}$	$\mathbf{8.102}_{\pm 1.655}$	$\mathbf{0.094}_{\pm 0.003}$
ETTm2	192	Native	$1.599_{\pm 0.824}$	$0.940_{\pm 0.288}$	$15.118_{\pm 0.276}$	$32.289_{\pm 3.682}$	$0.085_{\pm 0.030}$
		Mean only	$\mathbf{1.171}_{\pm 0.424}$	$0.817_{\pm 0.163}$	$14.970_{\pm 0.331}$	$30.018_{\pm 0.735}$	$0.082_{\pm 0.027}$
		Shape only	$1.599_{\pm 0.824}$	$0.730_{\pm 0.238}$	$5.421_{\pm 2.285}$	$17.422_{\pm 7.656}$	$0.080_{\pm 0.015}$
		FSPC	$\mathbf{1.171}_{\pm 0.424}$	$\mathbf{0.616}_{\pm 0.132}$	$\mathbf{4.394}_{\pm 1.292}$	$\mathbf{13.199}_{\pm 3.284}$	$\mathbf{0.077}_{\pm 0.016}$

Continued

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
Electricity	192	Native	$\mathbf{0.514}_{\pm 0.018}$	$0.453_{\pm 0.012}$	$10.308_{\pm 0.315}$	$19.220_{\pm 1.154}$	$0.166_{\pm 0.006}$
		Mean only	$0.536_{\pm 0.010}$	$0.455_{\pm 0.007}$	$10.011_{\pm 0.372}$	$19.308_{\pm 1.056}$	$0.164_{\pm 0.003}$
		Shape only	$\mathbf{0.514}_{\pm 0.018}$	$\mathbf{0.384}_{\pm 0.009}$	$\mathbf{2.023}_{\pm 0.342}$	$\mathbf{4.653}_{\pm 1.318}$	$\mathbf{0.136}_{\pm 0.005}$
		FSPC	$0.536_{\pm 0.010}$	$0.394_{\pm 0.005}$	$2.704_{\pm 0.314}$	$7.566_{\pm 0.727}$	$0.137_{\pm 0.003}$
Exchange	192	Native	$2.379_{\pm 1.803}$	$0.998_{\pm 0.402}$	$9.985_{\pm 2.311}$	$22.856_{\pm 10.914}$	$0.228_{\pm 0.145}$
		Mean only	$\mathbf{1.468}_{\pm 0.481}$	$0.802_{\pm 0.129}$	$8.980_{\pm 1.794}$	$16.243_{\pm 7.041}$	$0.179_{\pm 0.035}$
		Shape only	$2.379_{\pm 1.803}$	$0.894_{\pm 0.325}$	$7.095_{\pm 1.341}$	$20.339_{\pm 9.837}$	$0.204_{\pm 0.103}$
		FSPC	$\mathbf{1.468}_{\pm 0.481}$	$\mathbf{0.691}_{\pm 0.103}$	$\mathbf{4.461}_{\pm 0.577}$	$\mathbf{10.803}_{\pm 3.170}$	$\mathbf{0.158}_{\pm 0.021}$
Traffic	192	Native	$0.764_{\pm 0.069}$	$0.432_{\pm 0.023}$	$14.333_{\pm 0.478}$	$26.845_{\pm 0.958}$	$0.110_{\pm 0.009}$
		Mean only	$\mathbf{0.758}_{\pm 0.074}$	$0.431_{\pm 0.022}$	$14.346_{\pm 0.431}$	$26.417_{\pm 0.768}$	$0.114_{\pm 0.010}$
		Shape only	$0.764_{\pm 0.069}$	$0.341_{\pm 0.015}$	$2.049_{\pm 0.092}$	$6.443_{\pm 0.355}$	$\mathbf{0.077}_{\pm 0.004}$
		FSPC	$\mathbf{0.758}_{\pm 0.074}$	$\mathbf{0.335}_{\pm 0.013}$	$\mathbf{1.221}_{\pm 0.335}$	$\mathbf{4.183}_{\pm 0.671}$	$0.079_{\pm 0.004}$
ILI	36	Native	$\mathbf{4.040}_{\pm 0.301}$	$\mathbf{1.025}_{\pm 0.049}$	$\mathbf{4.561}_{\pm 1.212}$	$\mathbf{11.356}_{\pm 3.208}$	$\mathbf{0.249}_{\pm 0.031}$
		Mean only	$4.154_{\pm 0.142}$	$1.043_{\pm 0.030}$	$4.709_{\pm 1.258}$	$12.611_{\pm 2.186}$	$0.261_{\pm 0.022}$
		Shape only	$\mathbf{4.040}_{\pm 0.301}$	$1.087_{\pm 0.057}$	$7.228_{\pm 1.029}$	$18.236_{\pm 4.348}$	$0.295_{\pm 0.044}$
		FSPC	$4.154_{\pm 0.142}$	$1.115_{\pm 0.038}$	$8.312_{\pm 1.332}$	$21.373_{\pm 3.925}$	$0.323_{\pm 0.028}$
SolarEnergy	192	Native	$0.403_{\pm 0.002}$	$0.363_{\pm 0.015}$	$8.065_{\pm 2.029}$	$29.175_{\pm 4.209}$	$0.065_{\pm 0.015}$
		Mean only	$\mathbf{0.286}_{\pm 0.043}$	$0.291_{\pm 0.020}$	$4.519_{\pm 1.581}$	$19.789_{\pm 4.687}$	$0.055_{\pm 0.008}$
		Shape only	$0.403_{\pm 0.002}$	$0.327_{\pm 0.039}$	$7.676_{\pm 1.500}$	$28.543_{\pm 2.813}$	$0.057_{\pm 0.008}$
		FSPC	$\mathbf{0.286}_{\pm 0.043}$	$\mathbf{0.235}_{\pm 0.007}$	$\mathbf{4.148}_{\pm 1.231}$	$\mathbf{13.679}_{\pm 2.991}$	$\mathbf{0.050}_{\pm 0.006}$

TABLE S35: TimeDiff: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTh1	192	Native	$\mathbf{0.595}_{\pm 0.011}$	$0.474_{\pm 0.008}$	$13.863_{\pm 0.515}$	$25.823_{\pm 0.733}$	$0.111_{\pm 0.003}$
		Mean only	$0.598_{\pm 0.016}$	$0.496_{\pm 0.019}$	$13.948_{\pm 0.577}$	$28.408_{\pm 1.822}$	$0.118_{\pm 0.002}$
		Shape only	$\mathbf{0.595}_{\pm 0.011}$	$0.410_{\pm 0.012}$	$2.368_{\pm 0.183}$	$9.035_{\pm 1.206}$	$0.102_{\pm 0.005}$
		FSPC	$0.598_{\pm 0.016}$	$\mathbf{0.398}_{\pm 0.002}$	$\mathbf{2.332}_{\pm 0.544}$	$\mathbf{8.234}_{\pm 0.521}$	$\mathbf{0.092}_{\pm 0.003}$
ETTh2	192	Native	$\mathbf{0.547}_{\pm 0.166}$	$0.491_{\pm 0.098}$	$13.457_{\pm 0.655}$	$27.037_{\pm 1.717}$	$0.078_{\pm 0.020}$
		Mean only	$0.549_{\pm 0.167}$	$0.492_{\pm 0.099}$	$13.511_{\pm 0.679}$	$27.226_{\pm 2.305}$	$0.075_{\pm 0.015}$
		Shape only	$\mathbf{0.547}_{\pm 0.166}$	$\mathbf{0.415}_{\pm 0.078}$	$\mathbf{4.258}_{\pm 1.271}$	$\mathbf{12.519}_{\pm 3.976}$	$0.073_{\pm 0.005}$
		FSPC	$0.549_{\pm 0.167}$	$0.415_{\pm 0.078}$	$4.288_{\pm 1.217}$	$12.624_{\pm 3.862}$	$\mathbf{0.071}_{\pm 0.004}$
ETTm1	192	Native	$\mathbf{0.624}_{\pm 0.011}$	$0.490_{\pm 0.008}$	$14.646_{\pm 0.337}$	$27.286_{\pm 0.502}$	$0.095_{\pm 0.001}$
		Mean only	$0.627_{\pm 0.021}$	$0.501_{\pm 0.012}$	$14.811_{\pm 0.315}$	$28.719_{\pm 0.553}$	$0.095_{\pm 0.001}$
		Shape only	$\mathbf{0.624}_{\pm 0.011}$	$0.390_{\pm 0.003}$	$\mathbf{1.488}_{\pm 0.517}$	$6.069_{\pm 0.500}$	$0.079_{\pm 0.001}$
		FSPC	$0.627_{\pm 0.021}$	$\mathbf{0.387}_{\pm 0.004}$	$1.500_{\pm 0.283}$	$\mathbf{6.014}_{\pm 0.431}$	$\mathbf{0.075}_{\pm 0.001}$
ETTm2	192	Native	$0.276_{\pm 0.008}$	$0.315_{\pm 0.005}$	$12.446_{\pm 0.847}$	$22.986_{\pm 1.535}$	$0.049_{\pm 0.002}$
		Mean only	$\mathbf{0.269}_{\pm 0.007}$	$0.320_{\pm 0.008}$	$12.920_{\pm 0.984}$	$24.903_{\pm 1.694}$	$0.051_{\pm 0.003}$
		Shape only	$0.276_{\pm 0.008}$	$0.260_{\pm 0.001}$	$1.889_{\pm 0.297}$	$5.345_{\pm 1.808}$	$\mathbf{0.043}_{\pm 0.003}$
		FSPC	$\mathbf{0.269}_{\pm 0.007}$	$\mathbf{0.258}_{\pm 0.005}$	$\mathbf{1.573}_{\pm 0.331}$	$\mathbf{4.709}_{\pm 0.060}$	$0.044_{\pm 0.002}$
Electricity	192	Native	$0.873_{\pm 0.029}$	$0.748_{\pm 0.011}$	$15.526_{\pm 0.048}$	$26.696_{\pm 0.323}$	$0.225_{\pm 0.004}$
		Mean only	$\mathbf{0.862}_{\pm 0.021}$	$0.751_{\pm 0.011}$	$15.575_{\pm 0.039}$	$27.154_{\pm 0.087}$	$0.228_{\pm 0.003}$
		Shape only	$0.873_{\pm 0.029}$	$0.527_{\pm 0.008}$	$1.220_{\pm 0.179}$	$3.688_{\pm 0.389}$	$\mathbf{0.154}_{\pm 0.004}$
		FSPC	$\mathbf{0.862}_{\pm 0.021}$	$\mathbf{0.524}_{\pm 0.006}$	$\mathbf{0.742}_{\pm 0.107}$	$\mathbf{2.986}_{\pm 0.203}$	$0.155_{\pm 0.003}$
Exchange	192	Native	$\mathbf{0.430}_{\pm 0.027}$	$0.443_{\pm 0.010}$	$14.438_{\pm 0.680}$	$32.471_{\pm 2.966}$	$0.136_{\pm 0.014}$
		Mean only	$0.432_{\pm 0.023}$	$0.444_{\pm 0.009}$	$14.409_{\pm 0.653}$	$32.372_{\pm 3.106}$	$0.136_{\pm 0.014}$
		Shape only	$\mathbf{0.430}_{\pm 0.027}$	$\mathbf{0.364}_{\pm 0.025}$	$5.803_{\pm 1.483}$	$\mathbf{18.054}_{\pm 6.209}$	$0.111_{\pm 0.013}$
		FSPC	$0.432_{\pm 0.023}$	$0.365_{\pm 0.024}$	$\mathbf{5.771}_{\pm 1.500}$	$18.078_{\pm 6.185}$	$\mathbf{0.111}_{\pm 0.013}$
Traffic	192	Native	$1.366_{\pm 0.047}$	$0.778_{\pm 0.002}$	$15.473_{\pm 0.021}$	$27.707_{\pm 0.072}$	$0.127_{\pm 0.001}$
		Mean only	$\mathbf{1.355}_{\pm 0.048}$	$0.772_{\pm 0.003}$	$15.466_{\pm 0.010}$	$27.326_{\pm 0.042}$	$0.124_{\pm 0.001}$
		Shape only	$1.366_{\pm 0.047}$	$0.569_{\pm 0.002}$	$0.831_{\pm 0.093}$	$3.383_{\pm 0.108}$	$0.106_{\pm 0.001}$
		FSPC	$\mathbf{1.355}_{\pm 0.048}$	$\mathbf{0.565}_{\pm 0.002}$	$\mathbf{0.782}_{\pm 0.082}$	$\mathbf{1.808}_{\pm 0.206}$	$\mathbf{0.105}_{\pm 0.001}$
ILI	36	Native	$3.781_{\pm 0.232}$	$1.123_{\pm 0.018}$	$13.460_{\pm 1.395}$	–	$0.408_{\pm 0.072}$
		Mean only	$\mathbf{3.173}_{\pm 0.628}$	$0.977_{\pm 0.086}$	$12.931_{\pm 1.151}$	–	$0.263_{\pm 0.063}$
		Shape only	$3.781_{\pm 0.232}$	$1.025_{\pm 0.009}$	$9.492_{\pm 0.620}$	–	$0.322_{\pm 0.032}$
		FSPC	$\mathbf{3.173}_{\pm 0.628}$	$\mathbf{0.881}_{\pm 0.105}$	$\mathbf{7.328}_{\pm 0.816}$	–	$\mathbf{0.212}_{\pm 0.063}$
SolarEnergy	192	Native	$0.820_{\pm 0.009}$	$0.701_{\pm 0.003}$	$15.131_{\pm 0.254}$	$32.032_{\pm 0.976}$	$0.082_{\pm 0.001}$
		Mean only	$\mathbf{0.730}_{\pm 0.003}$	$0.706_{\pm 0.008}$	$15.829_{\pm 0.016}$	$34.335_{\pm 0.141}$	$0.082_{\pm 0.001}$
		Shape only	$0.820_{\pm 0.009}$	$0.508_{\pm 0.008}$	$7.719_{\pm 0.694}$	$21.950_{\pm 1.052}$	$0.074_{\pm 0.000}$
		FSPC	$\mathbf{0.730}_{\pm 0.003}$	$\mathbf{0.395}_{\pm 0.001}$	$\mathbf{7.030}_{\pm 0.619}$	$\mathbf{8.413}_{\pm 0.265}$	$\mathbf{0.073}_{\pm 0.000}$

TABLE S36: NsDiff: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
ETTh1	192	Native	$\mathbf{0.503}_{\pm 0.046}$	$\mathbf{0.371}_{\pm 0.010}$	$2.454_{\pm 0.074}$	$\mathbf{7.582}_{\pm 0.521}$	$0.089_{\pm 0.004}$
		Mean only	$0.588_{\pm 0.041}$	$0.417_{\pm 0.006}$	$2.468_{\pm 0.211}$	$14.481_{\pm 0.509}$	$0.094_{\pm 0.001}$
		Shape only	$\mathbf{0.503}_{\pm 0.046}$	$0.387_{\pm 0.016}$	$1.990_{\pm 0.183}$	$10.043_{\pm 0.579}$	$0.100_{\pm 0.006}$
		FSPC	$0.588_{\pm 0.041}$	$0.389_{\pm 0.008}$	$\mathbf{1.510}_{\pm 0.508}$	$10.180_{\pm 1.238}$	$\mathbf{0.086}_{\pm 0.002}$
ETTh2	192	Native	$0.565_{\pm 0.069}$	$0.385_{\pm 0.026}$	$3.597_{\pm 1.031}$	$10.921_{\pm 2.533}$	$0.061_{\pm 0.004}$
		Mean only	$\mathbf{0.550}_{\pm 0.071}$	$0.378_{\pm 0.028}$	$3.361_{\pm 1.221}$	$9.713_{\pm 3.338}$	$\mathbf{0.061}_{\pm 0.003}$
		Shape only	$0.565_{\pm 0.069}$	$0.378_{\pm 0.020}$	$2.897_{\pm 0.379}$	$9.372_{\pm 2.263}$	$0.064_{\pm 0.003}$
		FSPC	$\mathbf{0.550}_{\pm 0.071}$	$\mathbf{0.372}_{\pm 0.022}$	$\mathbf{2.403}_{\pm 0.774}$	$\mathbf{7.951}_{\pm 3.104}$	$0.063_{\pm 0.002}$
ETTm1	192	Native	$0.474_{\pm 0.061}$	$0.343_{\pm 0.018}$	$1.927_{\pm 0.105}$	$8.970_{\pm 1.267}$	$0.070_{\pm 0.001}$
		Mean only	$\mathbf{0.471}_{\pm 0.059}$	$\mathbf{0.341}_{\pm 0.015}$	$1.746_{\pm 0.105}$	$8.706_{\pm 0.707}$	$\mathbf{0.069}_{\pm 0.001}$
		Shape only	$0.474_{\pm 0.061}$	$0.344_{\pm 0.014}$	$1.458_{\pm 0.253}$	$8.447_{\pm 1.593}$	$0.073_{\pm 0.003}$
		FSPC	$\mathbf{0.471}_{\pm 0.059}$	$0.342_{\pm 0.012}$	$\mathbf{1.425}_{\pm 0.155}$	$\mathbf{8.253}_{\pm 1.071}$	$0.072_{\pm 0.002}$
ETTm2	192	Native	$\mathbf{0.272}_{\pm 0.015}$	$0.275_{\pm 0.006}$	$3.181_{\pm 0.203}$	$7.442_{\pm 0.528}$	$0.052_{\pm 0.002}$
		Mean only	$0.273_{\pm 0.016}$	$0.276_{\pm 0.006}$	$3.212_{\pm 0.236}$	$7.848_{\pm 0.157}$	$0.053_{\pm 0.002}$
		Shape only	$\mathbf{0.272}_{\pm 0.015}$	$\mathbf{0.263}_{\pm 0.007}$	$\mathbf{1.557}_{\pm 0.234}$	$\mathbf{5.643}_{\pm 0.725}$	$0.042_{\pm 0.002}$
		FSPC	$0.273_{\pm 0.016}$	$0.264_{\pm 0.008}$	$1.750_{\pm 0.155}$	$6.058_{\pm 1.034}$	$\mathbf{0.042}_{\pm 0.002}$
Electricity	192	Native	$\mathbf{0.222}_{\pm 0.005}$	$0.277_{\pm 0.003}$	$6.563_{\pm 0.114}$	$6.284_{\pm 0.238}$	$0.088_{\pm 0.001}$
		Mean only	$0.222_{\pm 0.006}$	$0.277_{\pm 0.003}$	$6.548_{\pm 0.143}$	$6.354_{\pm 0.374}$	$0.088_{\pm 0.001}$
		Shape only	$\mathbf{0.222}_{\pm 0.005}$	$\mathbf{0.235}_{\pm 0.005}$	$\mathbf{1.297}_{\pm 0.201}$	$\mathbf{3.514}_{\pm 0.692}$	$\mathbf{0.081}_{\pm 0.001}$
		FSPC	$0.222_{\pm 0.006}$	$0.235_{\pm 0.005}$	$1.309_{\pm 0.244}$	$3.528_{\pm 0.951}$	$0.081_{\pm 0.002}$
Exchange	192	Native	$\mathbf{0.342}_{\pm 0.081}$	$0.352_{\pm 0.052}$	$7.556_{\pm 1.304}$	$21.358_{\pm 2.387}$	$0.108_{\pm 0.026}$
		Mean only	$0.366_{\pm 0.087}$	$0.368_{\pm 0.055}$	$8.099_{\pm 1.244}$	$23.109_{\pm 4.492}$	$0.110_{\pm 0.024}$
		Shape only	$\mathbf{0.342}_{\pm 0.081}$	$\mathbf{0.315}_{\pm 0.038}$	$\mathbf{3.964}_{\pm 1.006}$	$\mathbf{15.238}_{\pm 1.651}$	$0.102_{\pm 0.016}$
		FSPC	$0.366_{\pm 0.087}$	$0.325_{\pm 0.039}$	$4.360_{\pm 1.136}$	$15.428_{\pm 3.414}$	$\mathbf{0.099}_{\pm 0.018}$
Traffic	192	Native	$0.611_{\pm 0.053}$	$0.374_{\pm 0.008}$	$8.684_{\pm 0.205}$	$7.996_{\pm 0.493}$	$0.080_{\pm 0.003}$
		Mean only	$\mathbf{0.596}_{\pm 0.060}$	$0.373_{\pm 0.009}$	$8.878_{\pm 0.227}$	$8.226_{\pm 1.205}$	$0.082_{\pm 0.003}$
		Shape only	$0.611_{\pm 0.053}$	$0.280_{\pm 0.010}$	$\mathbf{1.864}_{\pm 0.223}$	$5.820_{\pm 0.355}$	$\mathbf{0.065}_{\pm 0.003}$
		FSPC	$\mathbf{0.596}_{\pm 0.060}$	$\mathbf{0.276}_{\pm 0.012}$	$2.005_{\pm 0.412}$	$\mathbf{5.290}_{\pm 1.603}$	$0.065_{\pm 0.003}$
ILI	36	Native	$\mathbf{3.051}_{\pm 0.901}$	$\mathbf{0.851}_{\pm 0.184}$	$\mathbf{6.016}_{\pm 0.936}$	–	$\mathbf{0.185}_{\pm 0.085}$
		Mean only	$3.111_{\pm 1.004}$	$0.860_{\pm 0.200}$	$\mathbf{6.016}_{\pm 0.936}$	–	$0.192_{\pm 0.097}$
		Shape only	$\mathbf{3.051}_{\pm 0.901}$	$0.898_{\pm 0.209}$	$8.614_{\pm 1.478}$	–	$0.205_{\pm 0.118}$
		FSPC	$3.111_{\pm 1.004}$	$0.908_{\pm 0.226}$	$8.534_{\pm 1.345}$	–	$0.215_{\pm 0.135}$
SolarEnergy	192	Native	$0.234_{\pm 0.023}$	$0.287_{\pm 0.015}$	$7.485_{\pm 0.180}$	$13.549_{\pm 0.211}$	$0.046_{\pm 0.002}$
		Mean only	$\mathbf{0.228}_{\pm 0.023}$	$0.283_{\pm 0.014}$	$8.792_{\pm 0.433}$	$\mathbf{9.763}_{\pm 0.627}$	$0.046_{\pm 0.002}$
		Shape only	$0.234_{\pm 0.023}$	$0.212_{\pm 0.008}$	$5.923_{\pm 0.564}$	$21.775_{\pm 1.091}$	$0.044_{\pm 0.003}$
		FSPC	$\mathbf{0.228}_{\pm 0.023}$	$\mathbf{0.187}_{\pm 0.009}$	$\mathbf{1.518}_{\pm 0.585}$	$10.156_{\pm 1.484}$	$\mathbf{0.044}_{\pm 0.002}$

TABLE S37: DeepSTUQ: mean/shape factorial ablation on the same retained test coordinates. Subscripts: three-run sample SD; bold: lowest unrounded mean for the same configuration and metric. Undefined state errors are marked –. DeepSTUQ retains traffic-flow units; other backbones use model-normalized units. QICE and state errors use percentage points.

Dataset	H	Arm	MSE ↓	CRPS ↓	QICE (pp) ↓	State coverage error (pp) ↓	Mean Brier ↓
PEMS03	12	Native	$747.541_{\pm 11.622}$	$11.237_{\pm 0.035}$	$1.304_{\pm 0.062}$	$2.950_{\pm 0.238}$	$0.042_{\pm 0.001}$
		Mean only	$\mathbf{740.747}_{\pm 13.258}$	$11.174_{\pm 0.053}$	$1.309_{\pm 0.079}$	$2.480_{\pm 0.248}$	$0.041_{\pm 0.001}$
		Shape only	$747.541_{\pm 11.622}$	$11.235_{\pm 0.034}$	$0.926_{\pm 0.031}$	$3.137_{\pm 0.045}$	$0.041_{\pm 0.000}$
		FSPC	$\mathbf{740.747}_{\pm 13.258}$	$\mathbf{11.150}_{\pm 0.052}$	$\mathbf{0.921}_{\pm 0.106}$	$\mathbf{2.325}_{\pm 0.178}$	$\mathbf{0.041}_{\pm 0.001}$
PEMS04	12	Native	$1027.390_{\pm 35.394}$	$14.223_{\pm 0.123}$	$1.980_{\pm 0.094}$	$2.455_{\pm 0.153}$	$0.045_{\pm 0.001}$
		Mean only	$\mathbf{1025.945}_{\pm 35.153}$	$14.219_{\pm 0.126}$	$1.945_{\pm 0.090}$	$2.719_{\pm 0.234}$	$0.045_{\pm 0.001}$
		Shape only	$1027.390_{\pm 35.394}$	$14.094_{\pm 0.105}$	$0.645_{\pm 0.016}$	$2.138_{\pm 0.193}$	$0.045_{\pm 0.001}$
		FSPC	$\mathbf{1025.945}_{\pm 35.153}$	$\mathbf{14.075}_{\pm 0.108}$	$\mathbf{0.439}_{\pm 0.035}$	$\mathbf{1.309}_{\pm 0.206}$	$\mathbf{0.045}_{\pm 0.001}$
PEMS07	12	Native	$1139.260_{\pm 10.301}$	$15.173_{\pm 0.085}$	$2.405_{\pm 0.015}$	$3.054_{\pm 0.165}$	$0.049_{\pm 0.001}$
		Mean only	$\mathbf{1135.164}_{\pm 11.039}$	$15.162_{\pm 0.086}$	$2.405_{\pm 0.034}$	$3.103_{\pm 0.045}$	$0.049_{\pm 0.001}$
		Shape only	$1139.260_{\pm 10.301}$	$15.000_{\pm 0.078}$	$\mathbf{0.177}_{\pm 0.069}$	$2.374_{\pm 0.124}$	$0.049_{\pm 0.001}$
		FSPC	$\mathbf{1135.164}_{\pm 11.039}$	$\mathbf{14.967}_{\pm 0.078}$	$0.297_{\pm 0.145}$	$\mathbf{1.674}_{\pm 0.162}$	$\mathbf{0.048}_{\pm 0.001}$
PEMS08	12	Native	$\mathbf{609.287}_{\pm 4.856}$	$11.793_{\pm 0.077}$	$2.855_{\pm 0.074}$	$3.088_{\pm 0.070}$	$0.045_{\pm 0.001}$
		Mean only	$609.332_{\pm 4.320}$	$11.793_{\pm 0.068}$	$2.849_{\pm 0.056}$	$3.214_{\pm 0.348}$	$0.045_{\pm 0.001}$
		Shape only	$\mathbf{609.287}_{\pm 4.856}$	$11.553_{\pm 0.080}$	$0.471_{\pm 0.305}$	$1.372_{\pm 0.571}$	$\mathbf{0.044}_{\pm 0.001}$
		FSPC	$609.332_{\pm 4.320}$	$\mathbf{11.546}_{\pm 0.072}$	$\mathbf{0.235}_{\pm 0.112}$	$\mathbf{1.030}_{\pm 0.124}$	$0.044_{\pm 0.001}$

FSPC RELIABILITY ACROSS MODELS AND DATASETS

The figures compare FSPC with Native on identical complete test forecasts and training-defined diagnostic groups. Aggregate comparisons include all evaluated horizons; state maps and reliability curves cover all 56 backbone–dataset pairs at a shared forecast length where available. State maps average all three training runs. The source-ablation figures report source-filtering score ratios and compare Native, FSPC and w/o Mean/Filter at selected test coordinates. The fourteen examples use validation-based dataset selection and forecasts from the first training run. Coordinate selection uses forecasts without test targets. FSPC curves combine the affine-plus-state mean and centered residual templates with parameters fixed before testing.

Detailed FSPC reliability across models and datasets

The following figures use the full FSPC output and paired Native predictions.

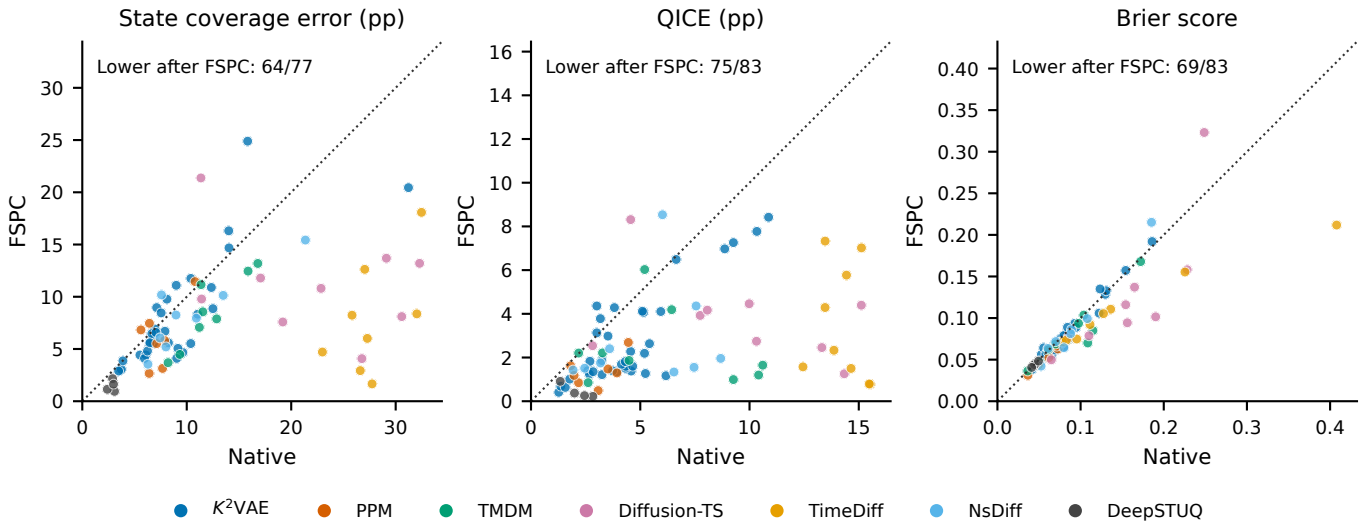


Fig. S1. Paired Native/FSPC performance across all 83 dataset–horizon configurations on complete test iterators. Each point averages three training runs. State coverage excludes configurations lacking eligible states in a run; QICE and Brier retain all configurations. Points below the diagonal improve after FSPC. Colors identify backbones.

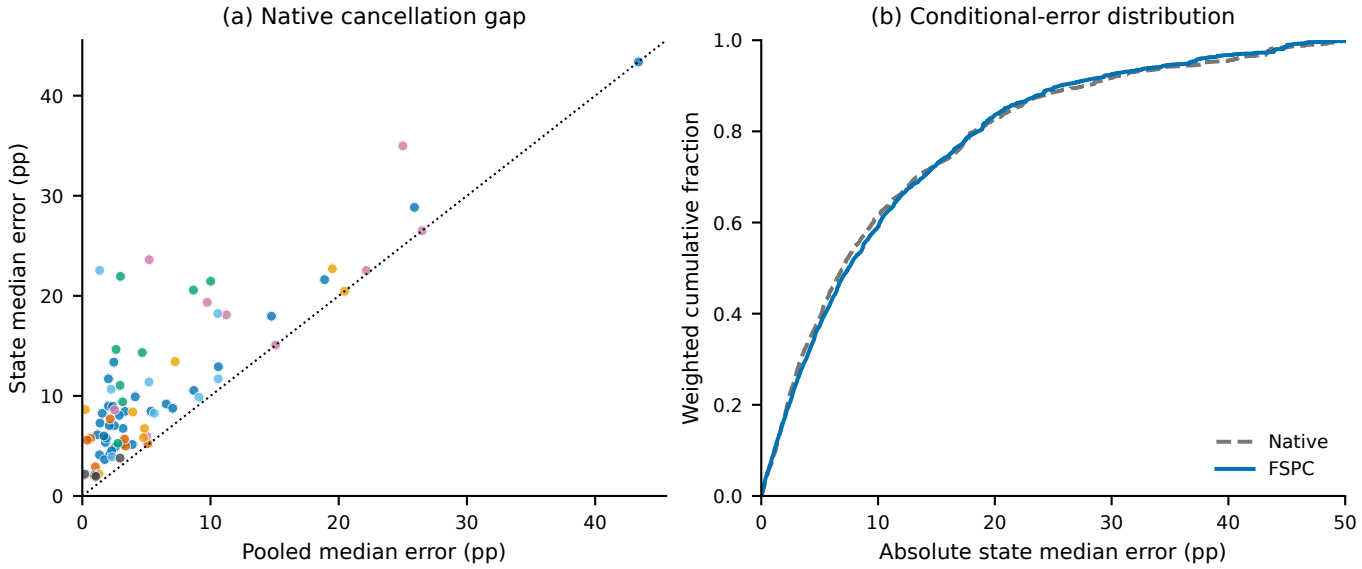


Fig. S2. Hidden conditional error and FSPC reliability on complete test iterators. Left: Native pooled median error versus count-weighted state median error on identical eligible coordinates. Right: cumulative distributions of absolute state median errors before and after FSPC. Coordinates weight states within a configuration/run; configurations and runs receive equal weight. The curves describe conditional reliability.

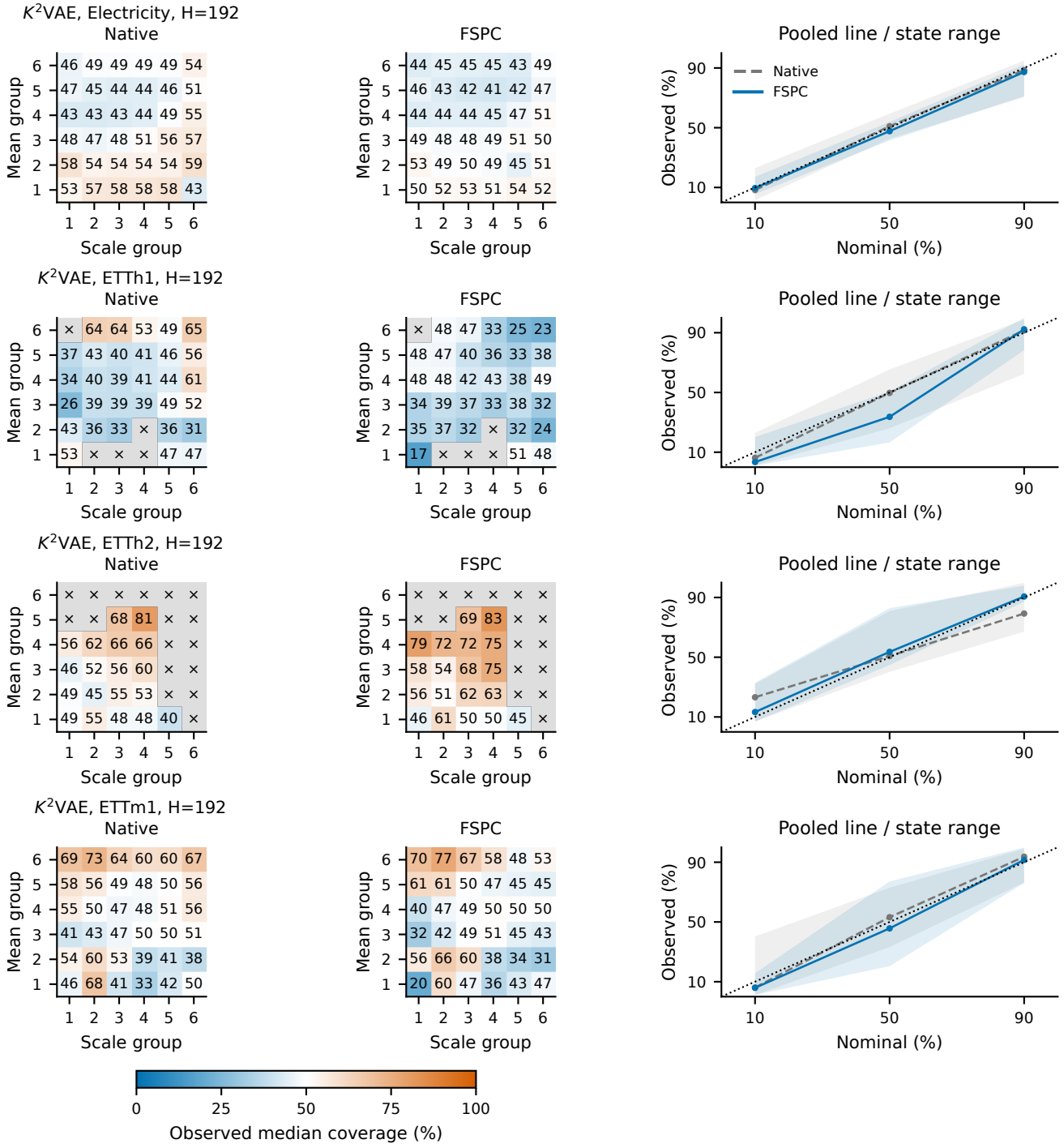


Fig. S3. Test-set extension of Figure 2: K^2VAE, Electricity, $H = 192$; K^2VAE, ETTh1, $H = 192$; K^2VAE, ETTh2, $H = 192$; K^2VAE, ETTm1, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

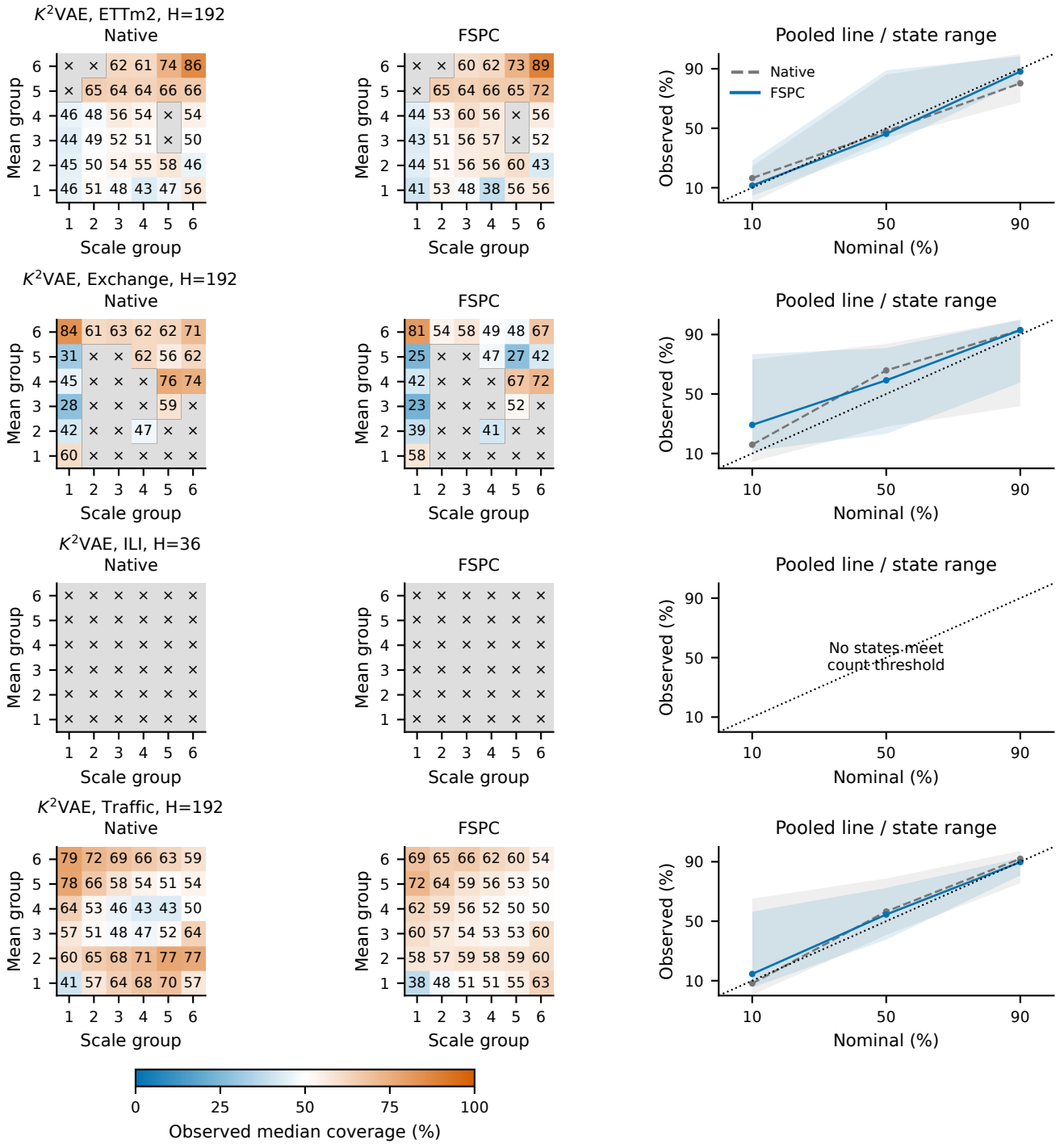


Fig. S4. Test-set extension of Figure 2: K^2VAE, ETTm2, $H = 192$; K^2VAE, Exchange, $H = 192$; K^2VAE, ILI, $H = 36$; K^2VAE, Traffic, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

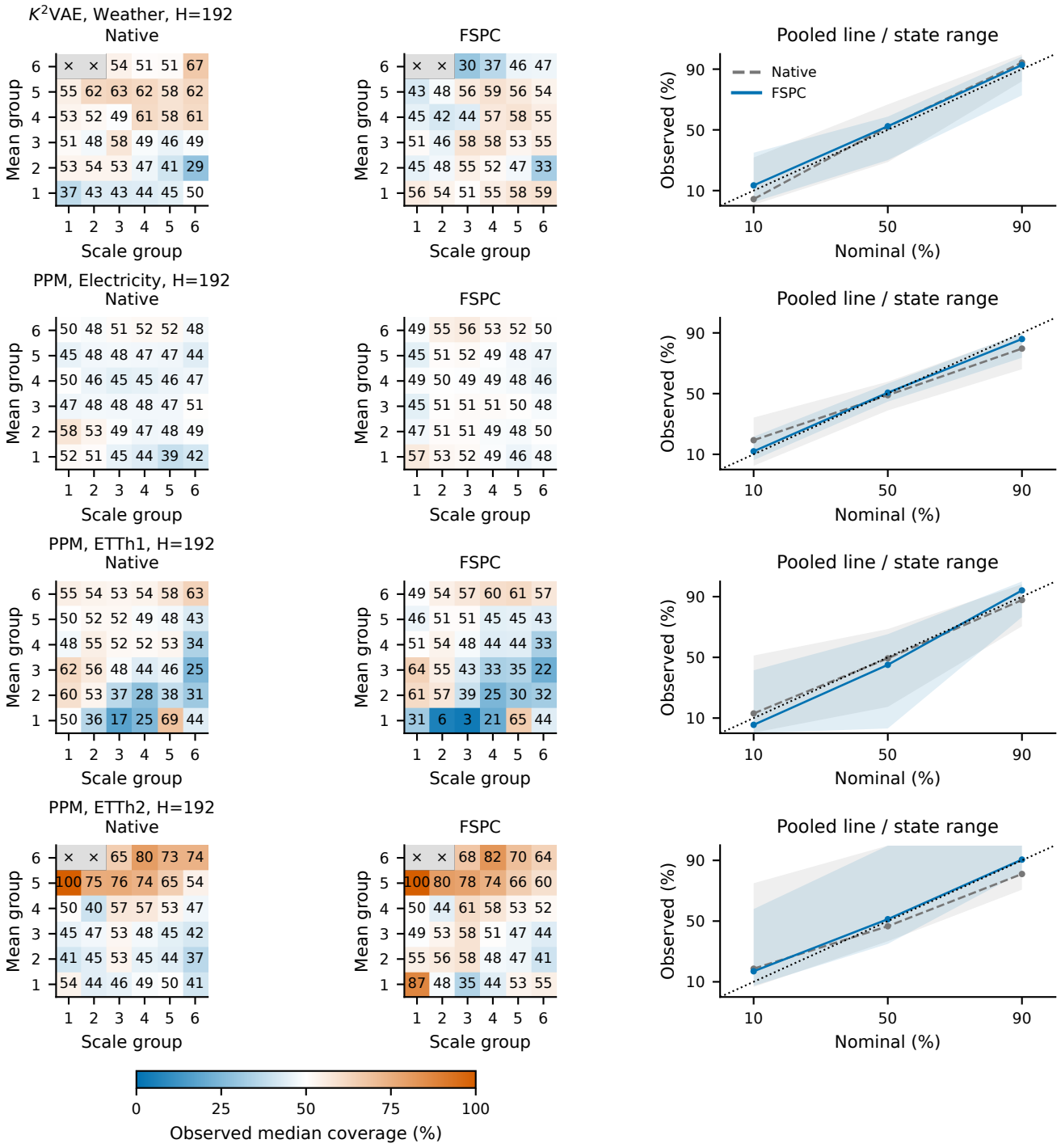


Fig. S5. Test-set extension of Figure 2: K^2VAE, Weather, $H = 192$; PPM, Electricity, $H = 192$; PPM, ETTh1, $H = 192$; PPM, ETTh2, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

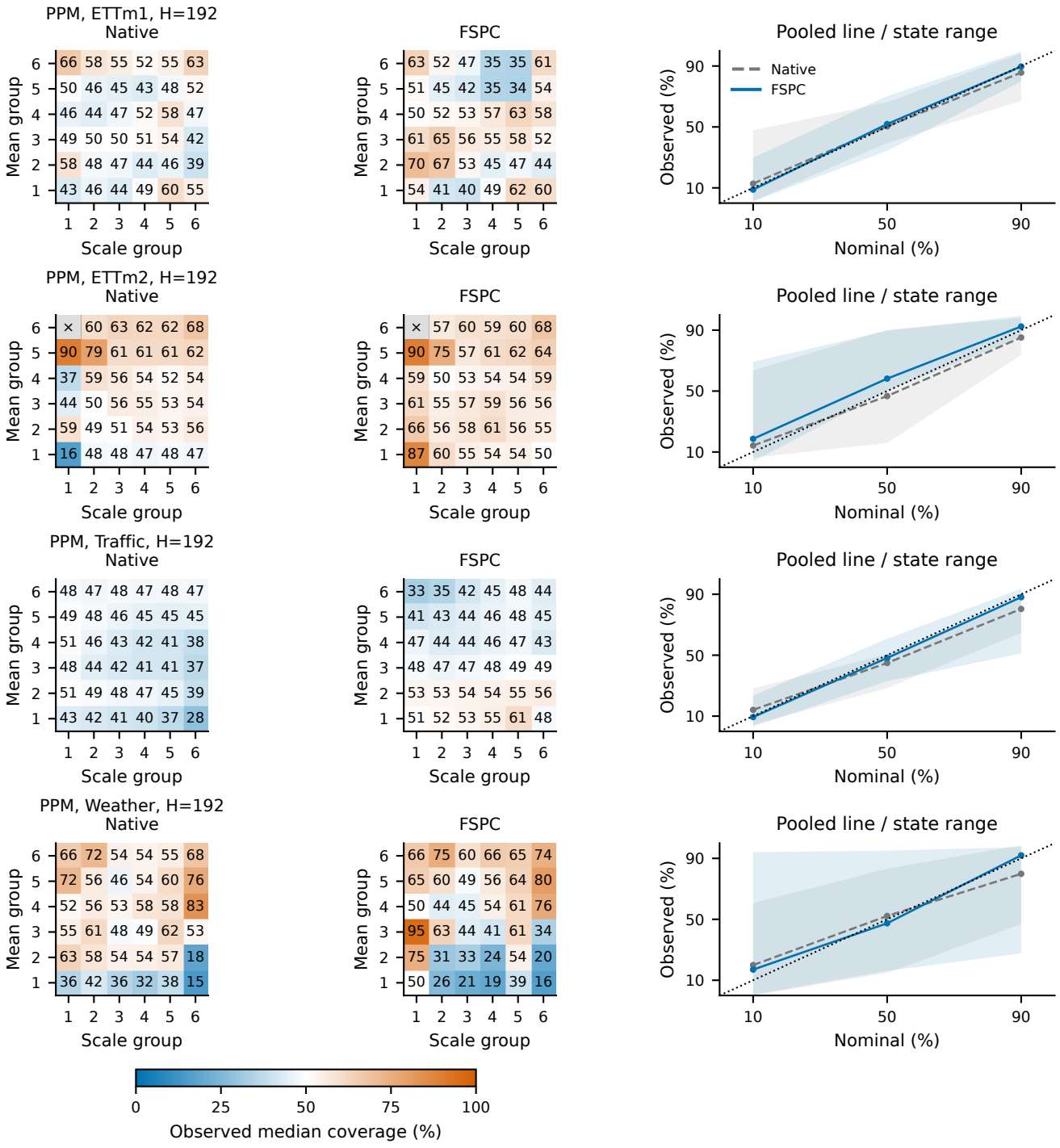


Fig. S6. Test-set extension of Figure 2: PPM, ETTm1, $H = 192$; PPM, ETTm2, $H = 192$; PPM, Traffic, $H = 192$; PPM, Weather, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

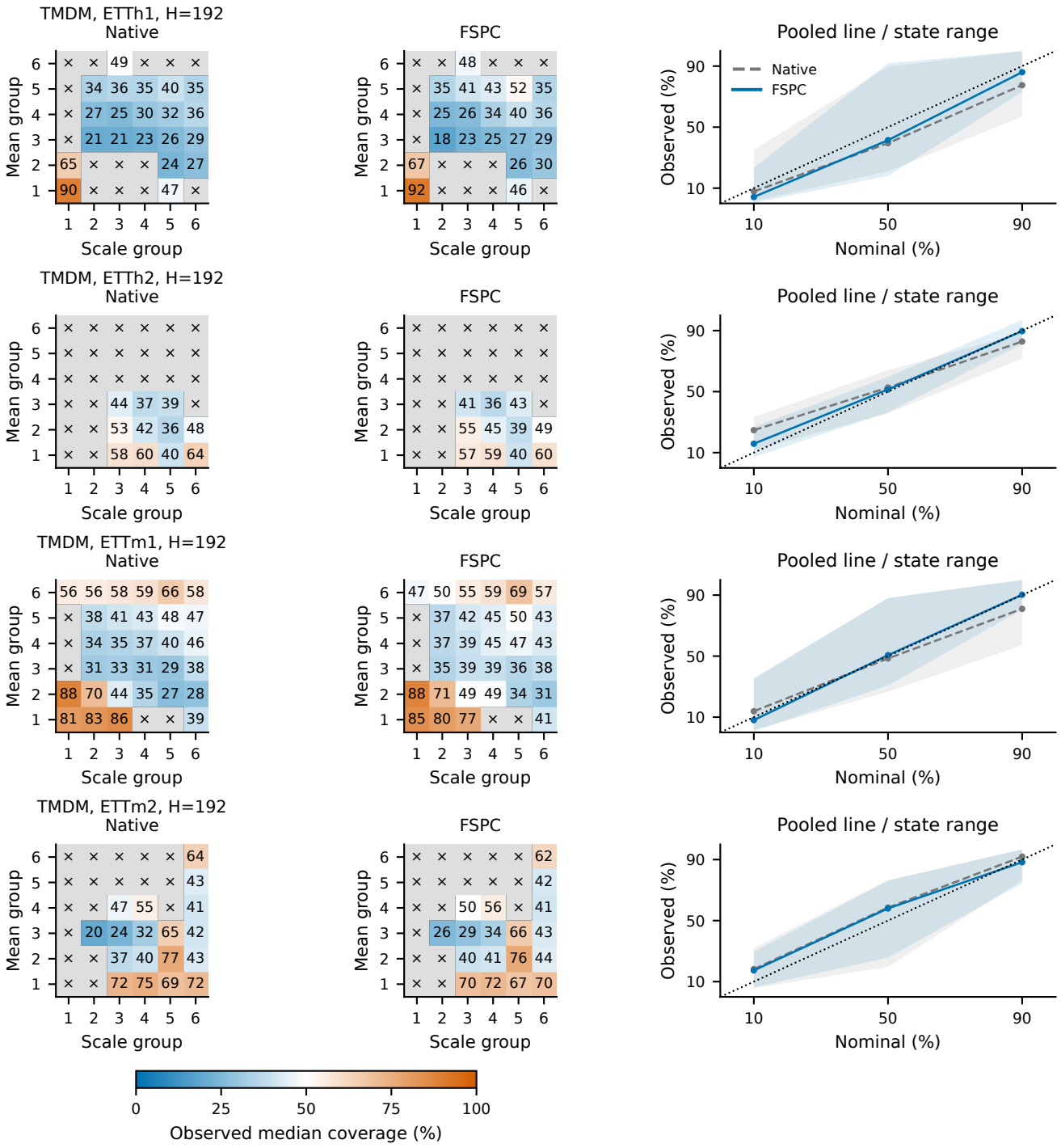


Fig. S7. Test-set extension of Figure 2: TMDM, ETTh1, $H = 192$; TMDM, ETTh2, $H = 192$; TMDM, ETTm1, $H = 192$; TMDM, ETTm2, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

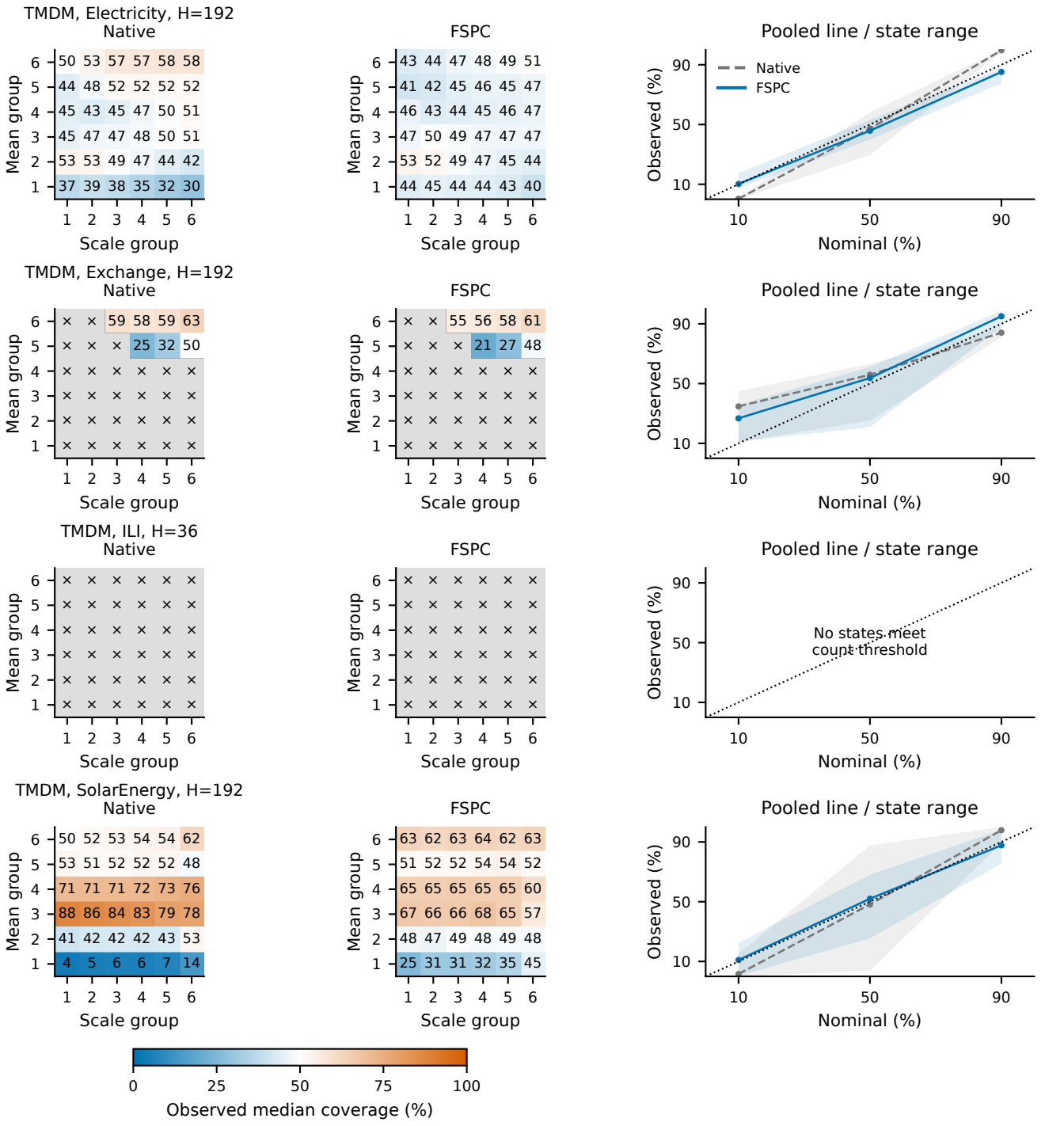


Fig. S8. Test-set extension of Figure 2: TMDM, Electricity, $H = 192$; TMDM, Exchange, $H = 192$; TMDM, ILI, $H = 36$; TMDM, SolarEnergy, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

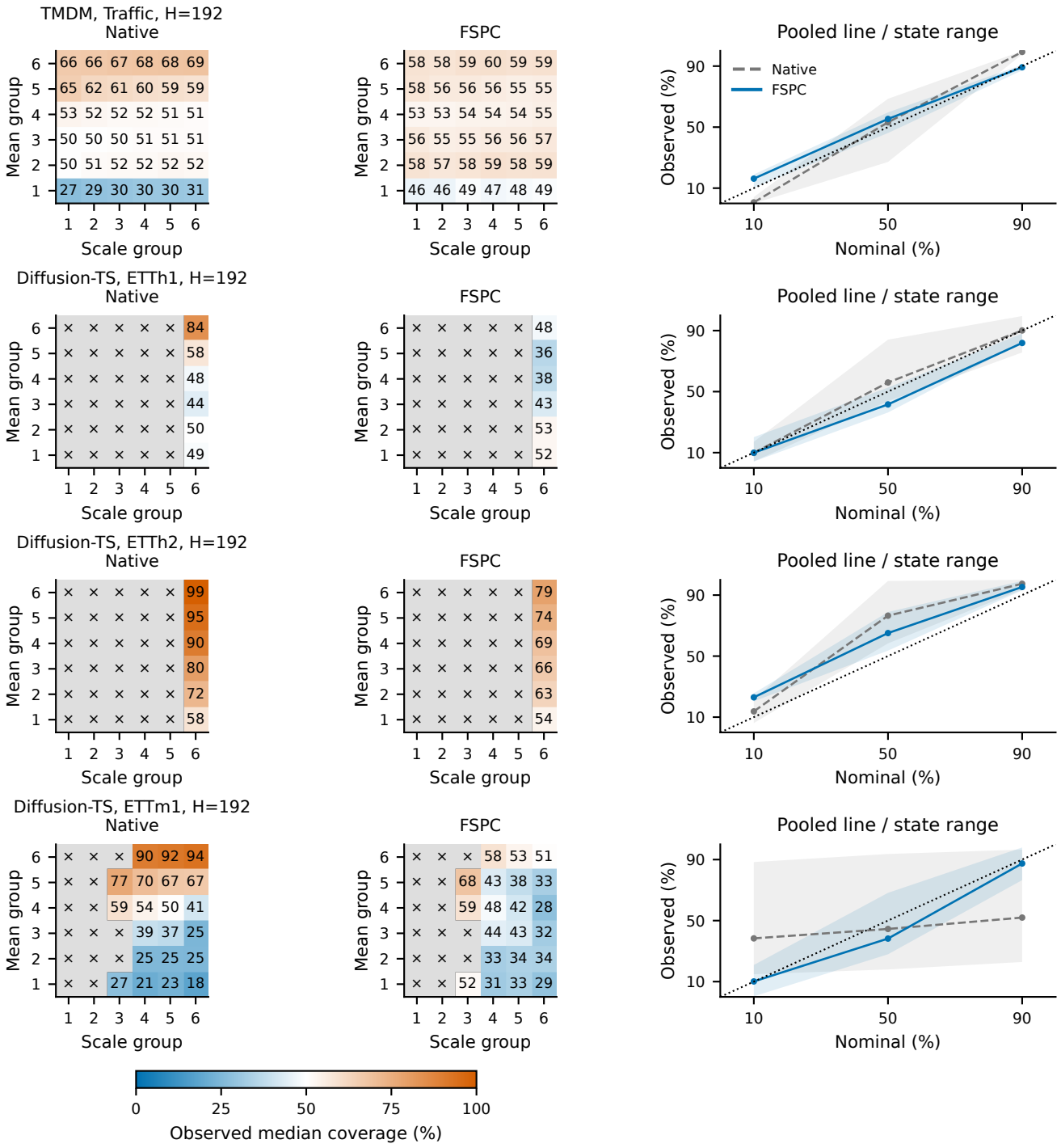


Fig. S9. Test-set extension of Figure 2: TMDM, Traffic, $H = 192$; Diffusion-TS, ETTh1, $H = 192$; Diffusion-TS, ETTh2, $H = 192$; Diffusion-TS, ETTm1, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

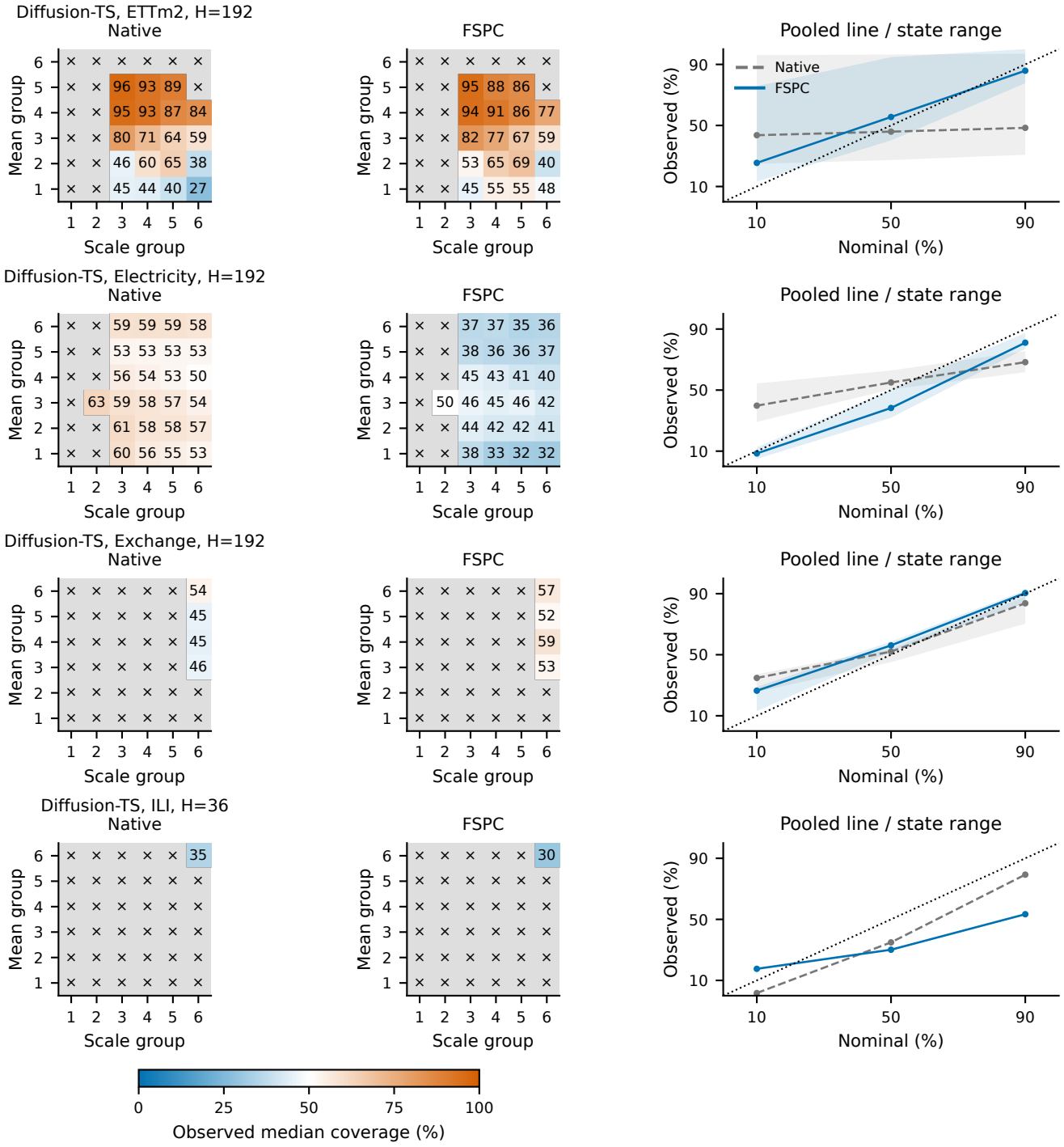


Fig. S10. Test-set extension of Figure 2: Diffusion-TS, ETTm2, $H = 192$; Diffusion-TS, Electricity, $H = 192$; Diffusion-TS, Exchange, $H = 192$; Diffusion-TS, ILI, $H = 36$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

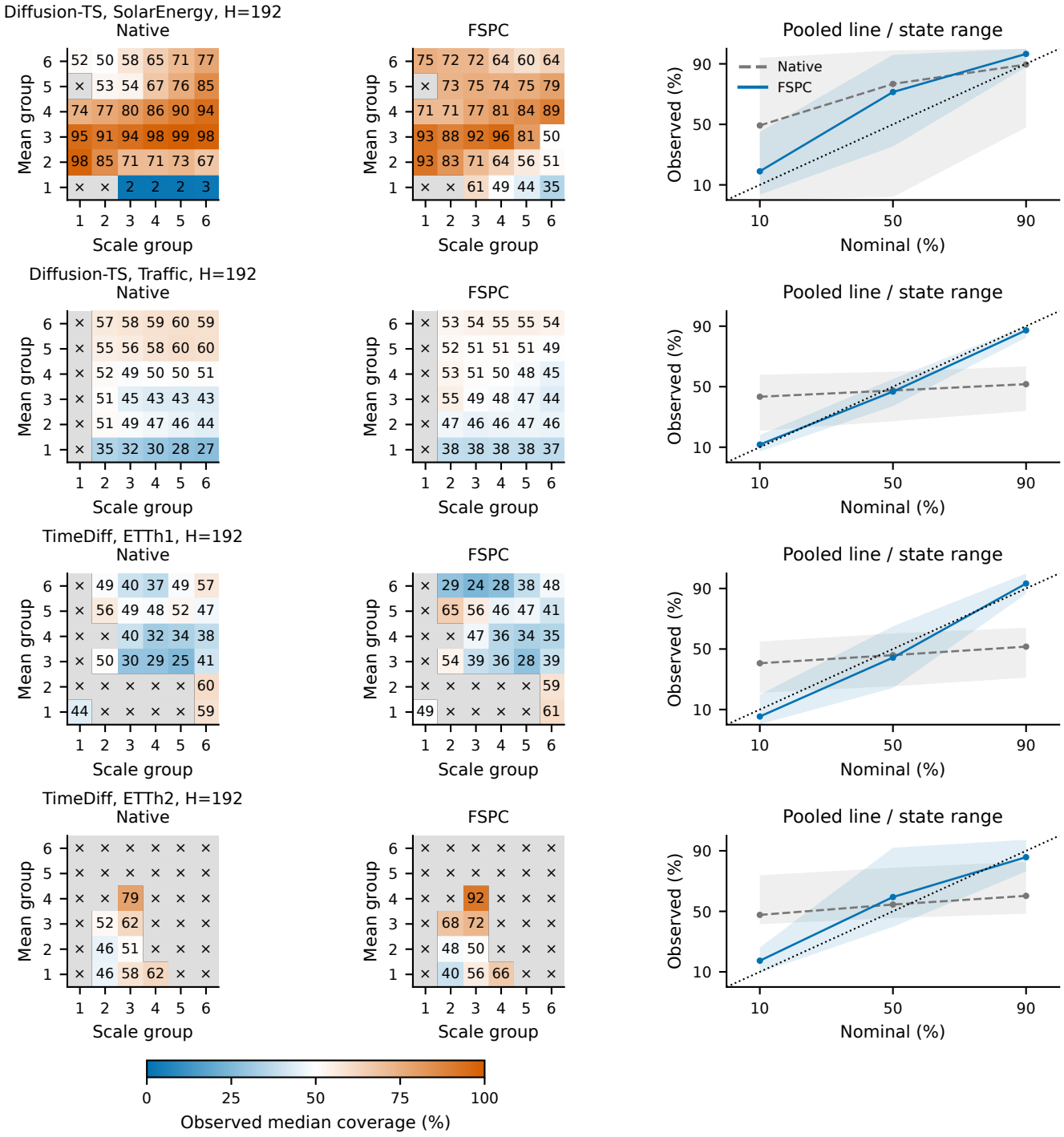


Fig. S11. Test-set extension of Figure 2: Diffusion-TS, SolarEnergy, $H = 192$; Diffusion-TS, Traffic, $H = 192$; TimeDiff, ETTh1, $H = 192$; TimeDiff, ETTh2, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

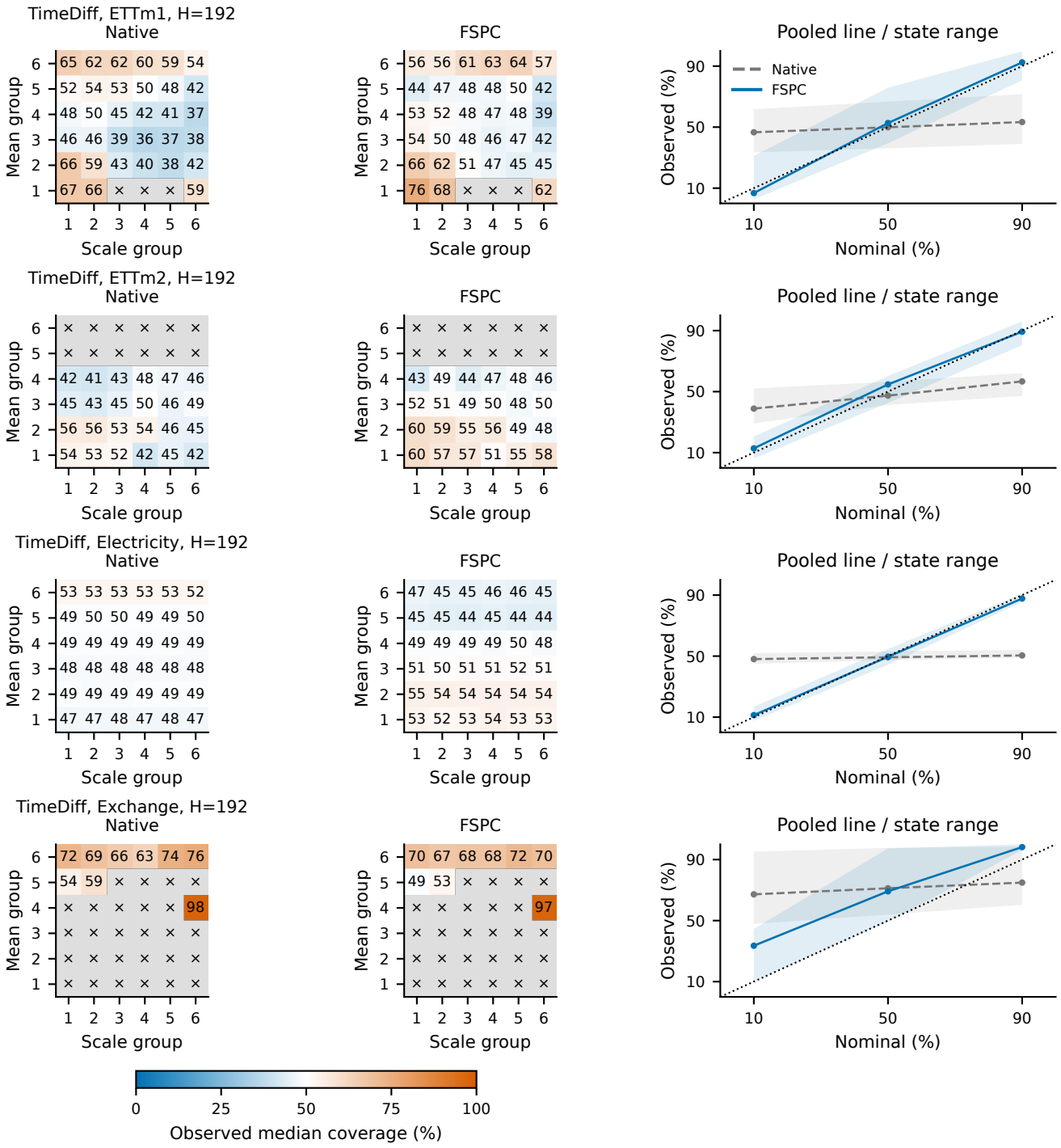


Fig. S12. Test-set extension of Figure 2: TimeDiff, ETTm1, $H = 192$; TimeDiff, ETTm2, $H = 192$; TimeDiff, Electricity, $H = 192$; TimeDiff, Exchange, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

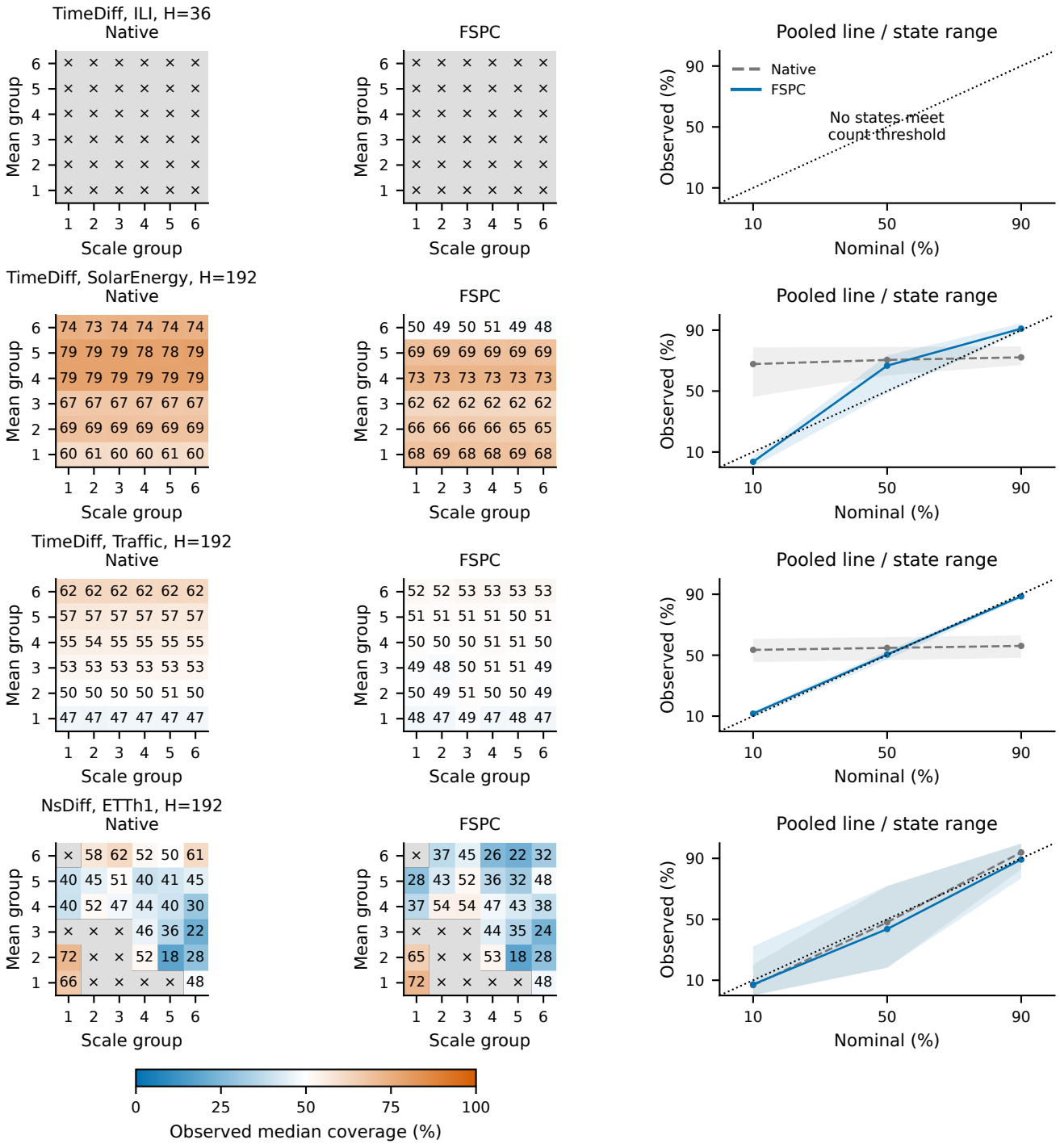


Fig. S13. Test-set extension of Figure 2: TimeDiff, ILI, $H = 36$; TimeDiff, SolarEnergy, $H = 192$; TimeDiff, Traffic, $H = 192$; NsDiff, ETTh1, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

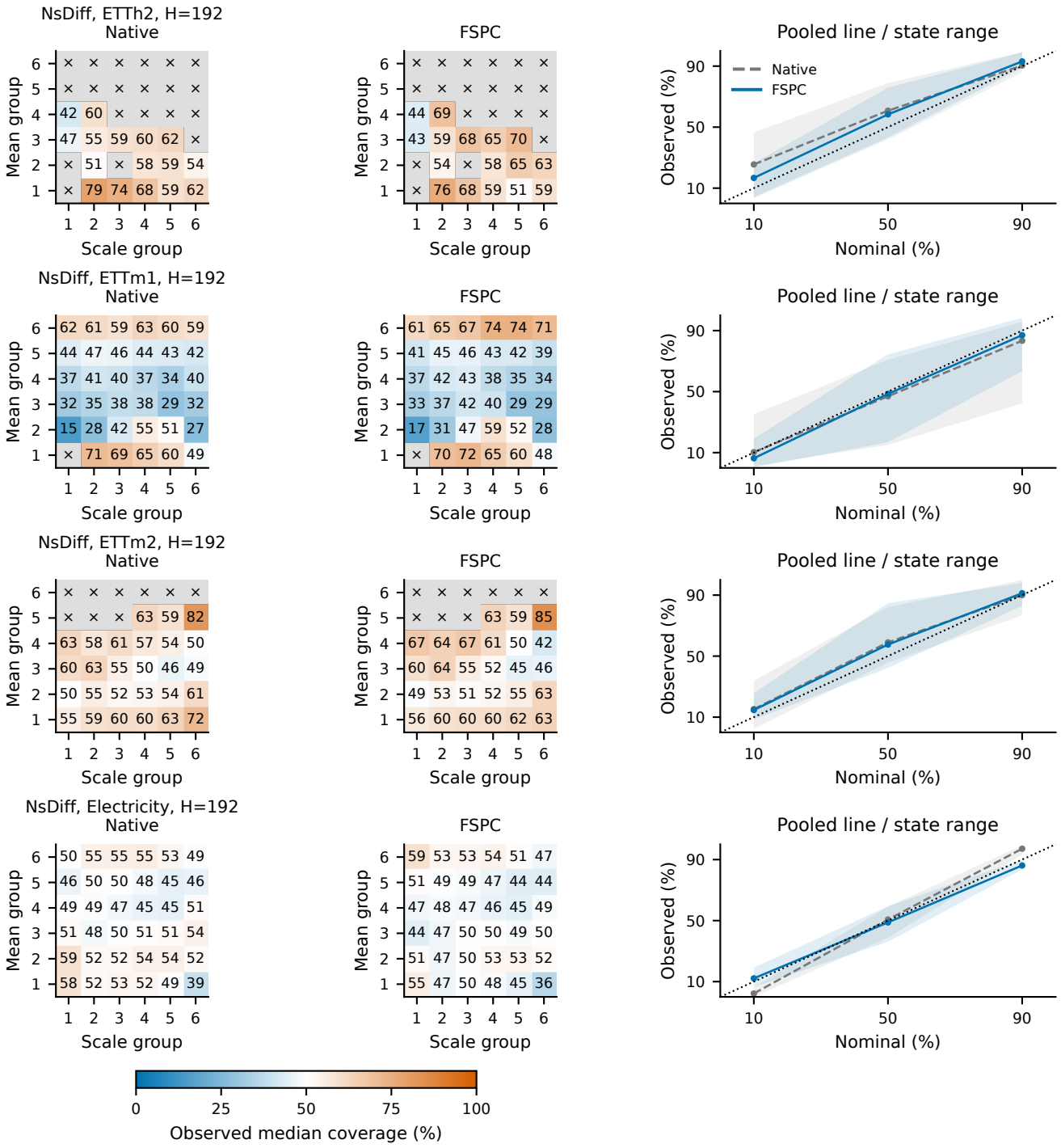


Fig. S14. Test-set extension of Figure 2: NsDiff, ETTh2, $H = 192$; NsDiff, ETTm1, $H = 192$; NsDiff, ETTm2, $H = 192$; NsDiff, Electricity, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

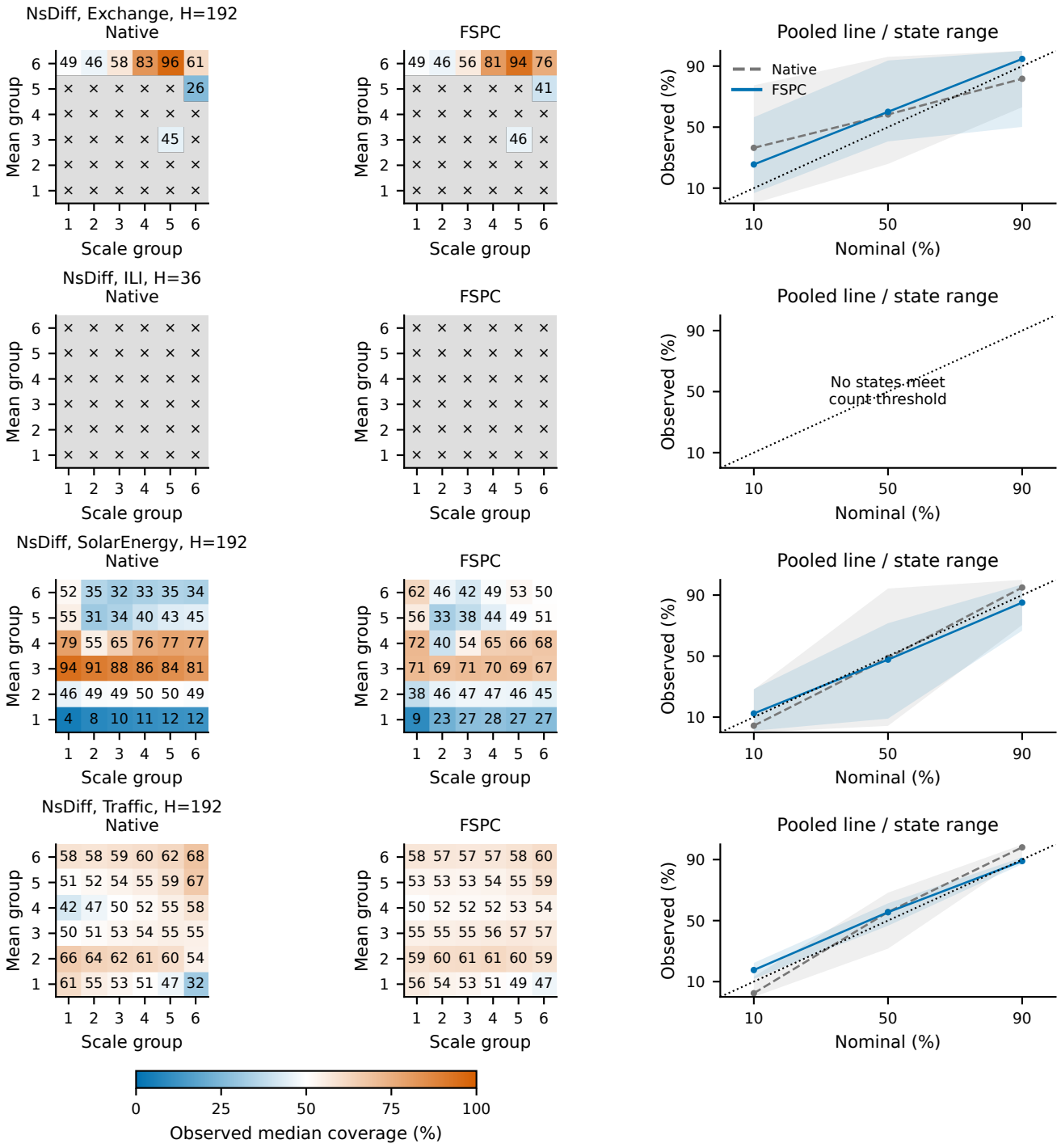


Fig. S15. Test-set extension of Figure 2: NsDiff, Exchange, $H = 192$; NsDiff, ILI, $H = 36$; NsDiff, SolarEnergy, $H = 192$; NsDiff, Traffic, $H = 192$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

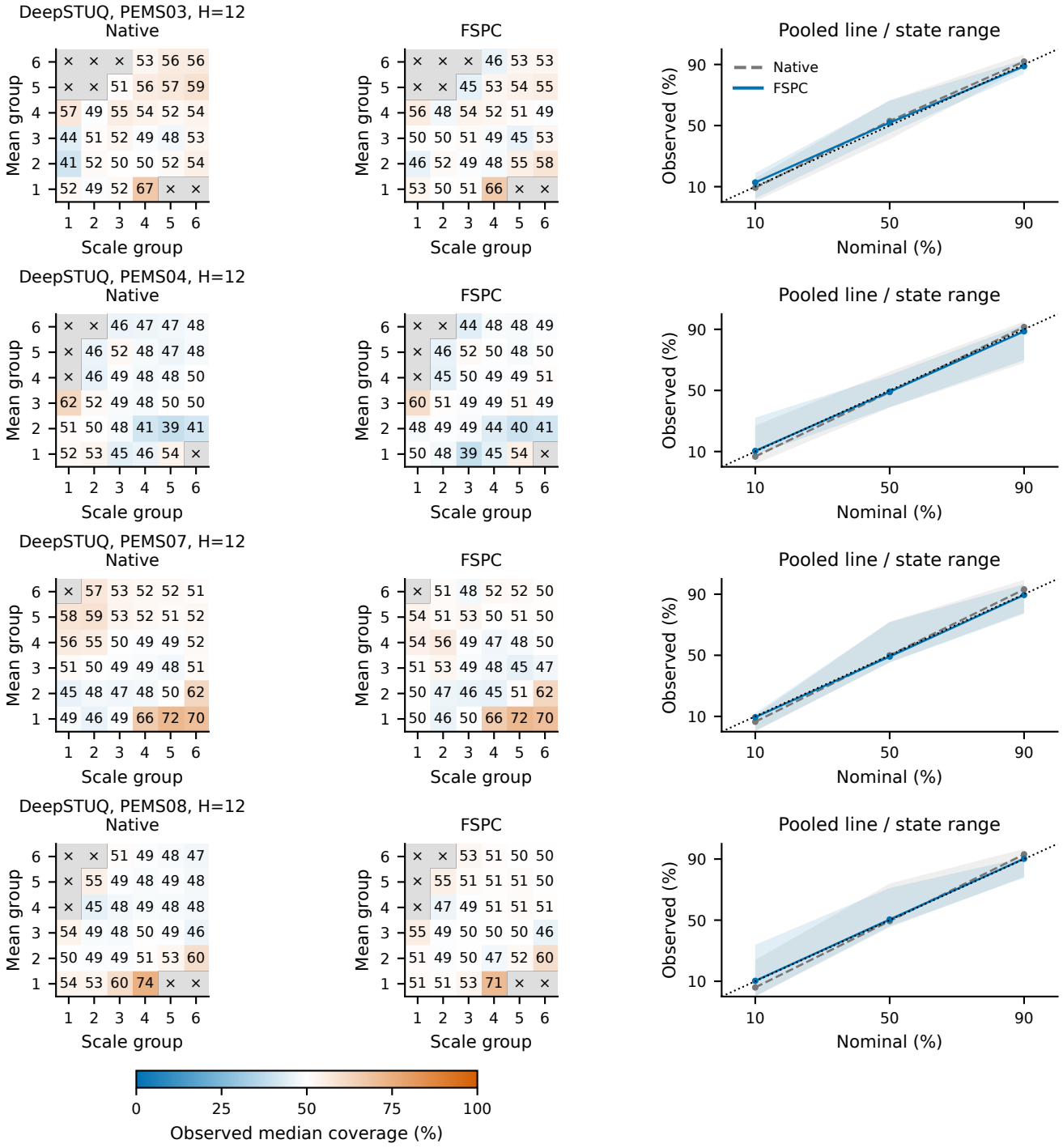


Fig. S16. Test-set extension of Figure 2: DeepSTUQ, PEMS03, $H = 12$; DeepSTUQ, PEMS04, $H = 12$; DeepSTUQ, PEMS07, $H = 12$; DeepSTUQ, PEMS08, $H = 12$. Left and center: Native and FSPC median coverage in identical training-defined 6×6 mean–scale groups. Annotations average three training runs and round to the nearest percentage point; gray cells lack 128 coordinates in at least one run. Full-precision values accompany the figure data. Right: pooled quantile coverage and the minimum–maximum range across eligible states at nominal probabilities 10%, 50%, and 90%. Bands describe state heterogeneity, not confidence intervals. The nominal median is 50%.

Residual-source ablations and full-method examples

The source-filtering score ratios isolate residual-source validity before mean control. Selected-coordinate plots compare Native, full FSPC and w/o Mean/Filter using frozen calibration parameters.

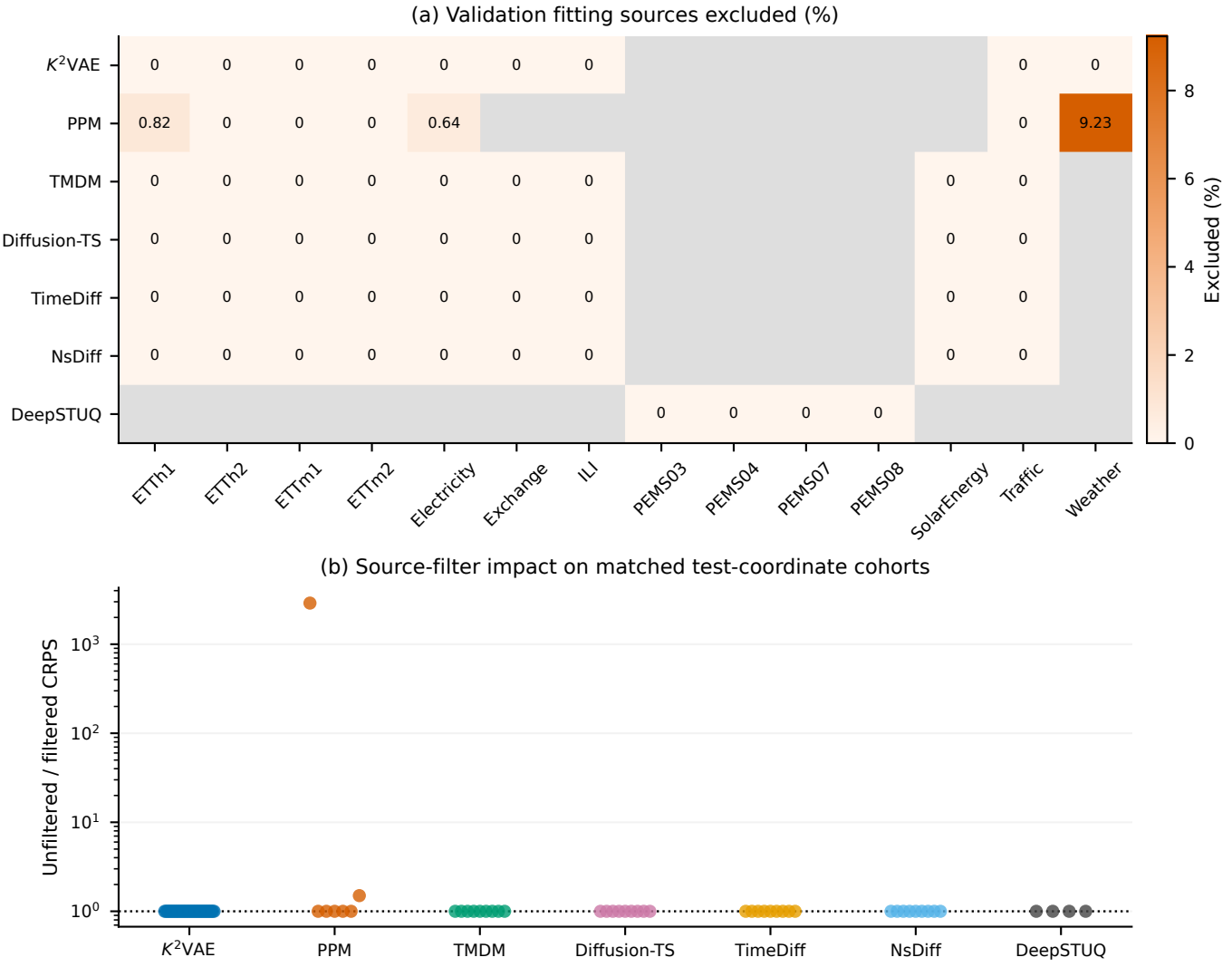


Fig. S17. Residual-source ablation: validation source exclusions and their scoring effect. Top: mean excluded-source percentage for all backbone/dataset pairs at the gallery horizon. Bottom: unfiltered/filtered w/o Mean Control CRPS ratios of three-run configuration means on matched retained test cohorts. This comparison isolates source filtering before mean control; FSPC performance is reported separately in the preceding main-method figures.

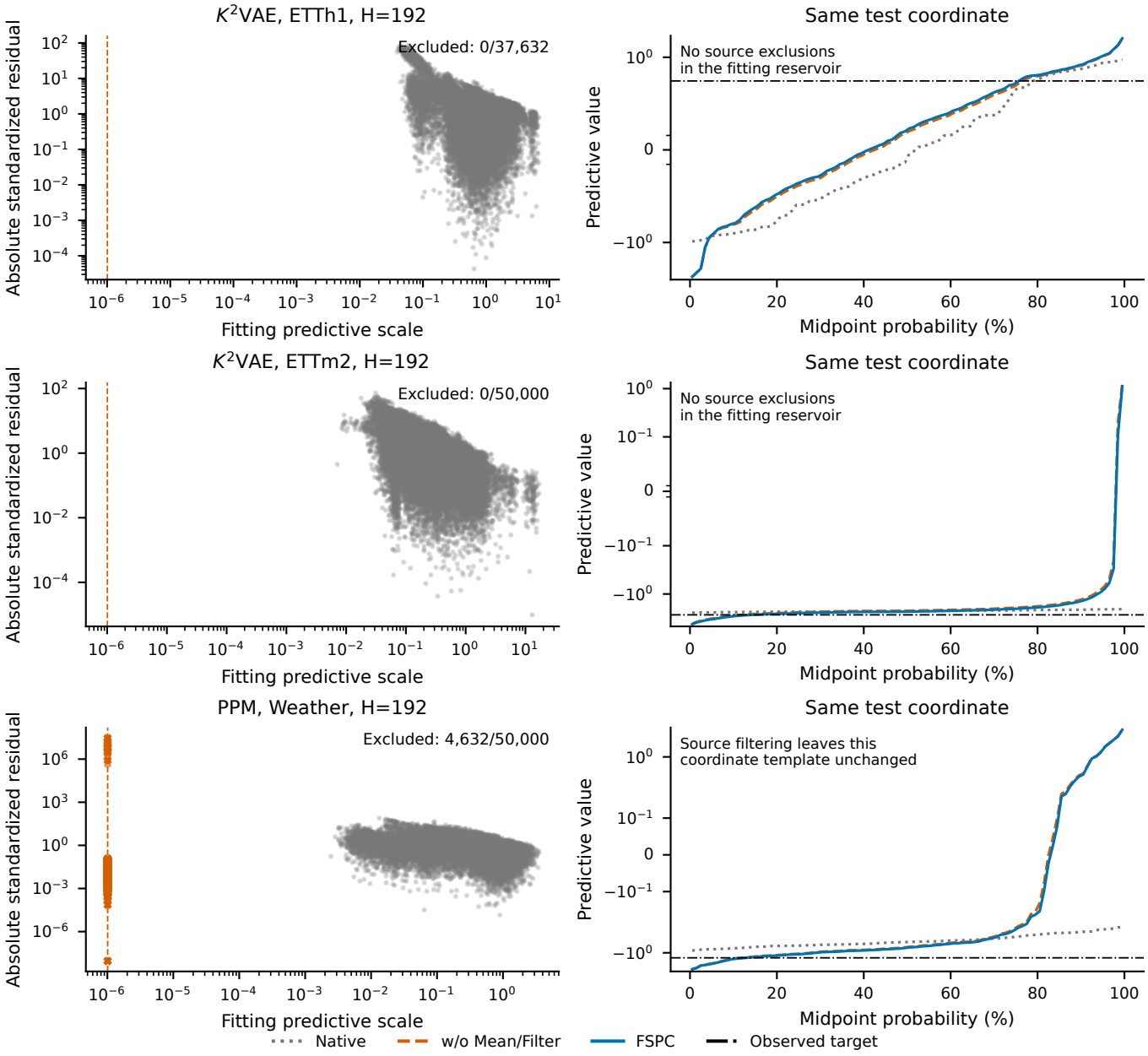


Fig. S18. Measured extension of Figure 4: K^2VAE, ETTh1, $H = 192$; K^2VAE, ETTm2, $H = 192$; PPM, Weather, $H = 192$. Left: fitting scale versus standardized residual magnitude; orange marks excluded near-floor sources. Right: Native, full FSPC and w/o Mean/Filter at the identical test coordinate; black marks the target. FSPC uses the frozen affine-plus-state mean and validation-selected blend. Dataset selection uses validation residuals; coordinate selection uses forecasts without test targets. Backbones without source exclusions retain identical residual-source pools; mean control can still change the full FSPC prediction. The vertical prediction axes use symmetric logarithmic scaling and each backbone's scoring units.

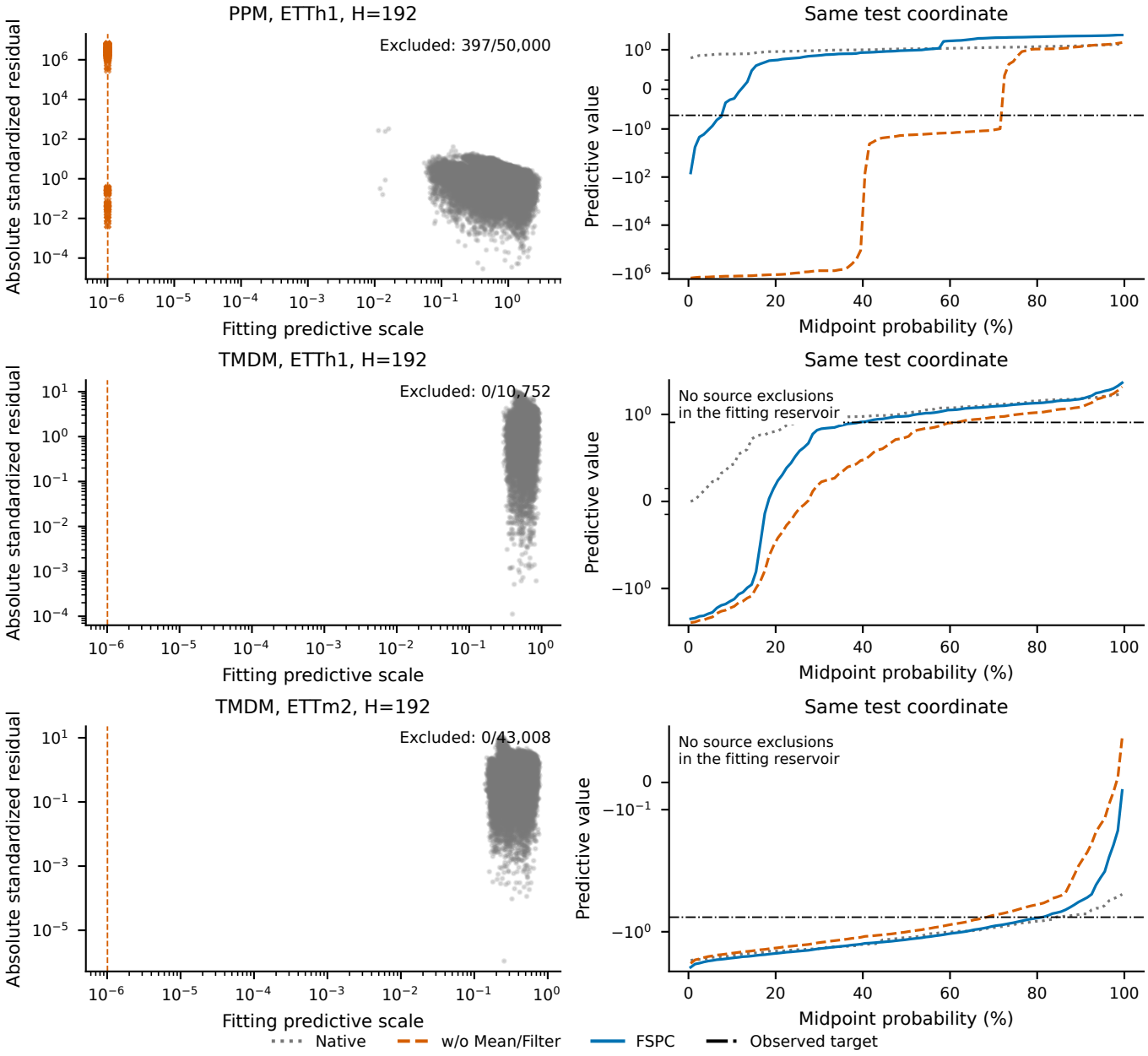


Fig. S19. Measured extension of Figure 4: PPM, ETTh1, $H = 192$; TMDM, ETTh1, $H = 192$; TMDM, ETTm2, $H = 192$. Left: fitting scale versus standardized residual magnitude; orange marks excluded near-floor sources. Right: Native, full FSPC and w/o Mean/Filter at the identical test coordinate; black marks the target. FSPC uses the frozen affine-plus-state mean and validation-selected blend. Dataset selection uses validation residuals; coordinate selection uses forecasts without test targets. Backbones without source exclusions retain identical residual-source pools; mean control can still change the full FSPC prediction. The vertical prediction axes use symmetric logarithmic scaling and each backbone's scoring units.

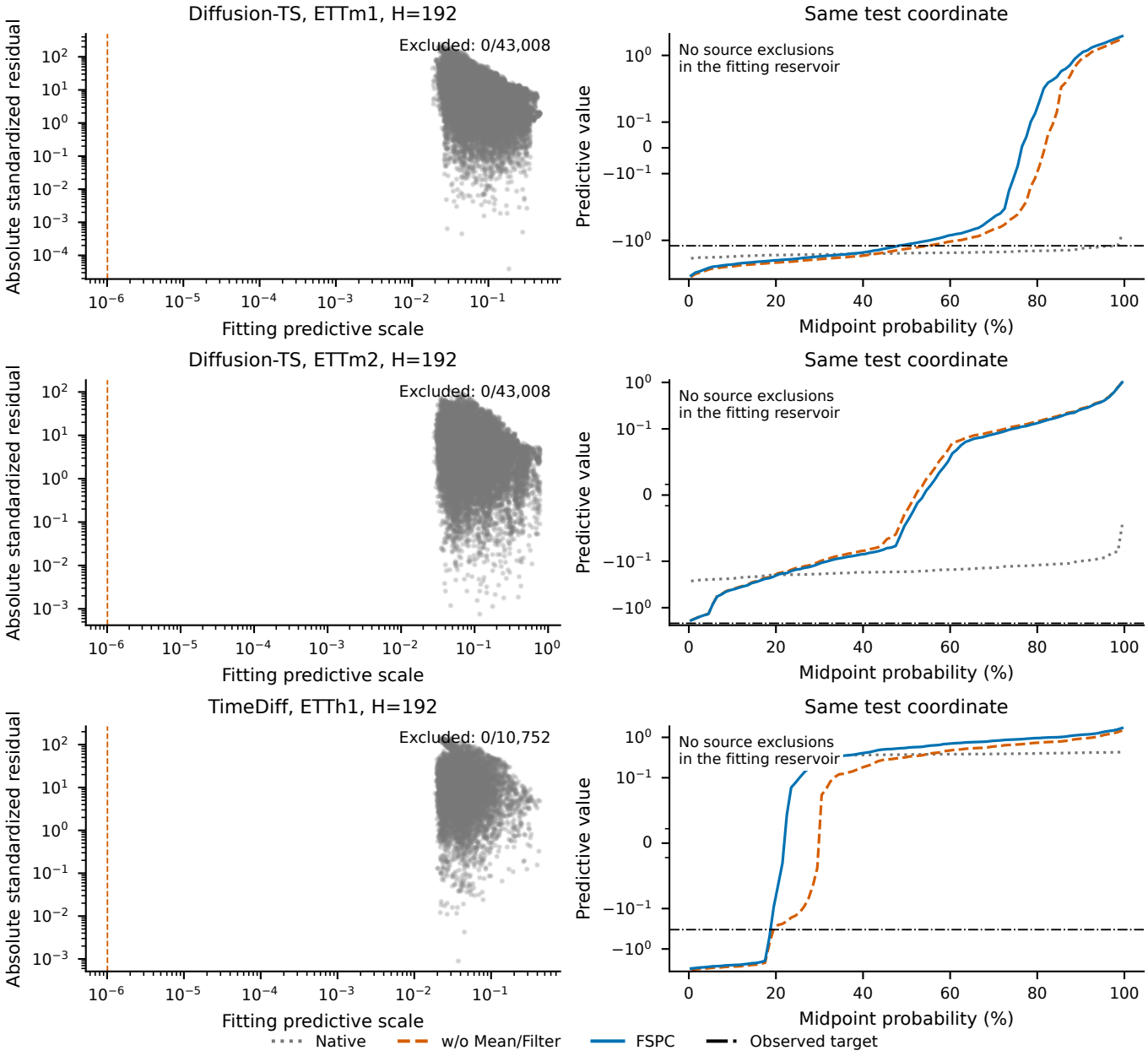


Fig. S20. Measured extension of Figure 4: Diffusion-TS, ETTm1, $H = 192$; Diffusion-TS, ETTm2, $H = 192$; TimeDiff, ETTh1, $H = 192$. Left: fitting scale versus standardized residual magnitude; orange marks excluded near-floor sources. Right: Native, full FSPC and w/o Mean/Filter at the identical test coordinate; black marks the target. FSPC uses the frozen affine-plus-state mean and validation-selected blend. Dataset selection uses validation residuals; coordinate selection uses forecasts without test targets. Backbones without source exclusions retain identical residual-source pools; mean control can still change the full FSPC prediction. The vertical prediction axes use symmetric logarithmic scaling and each backbone's scoring units.

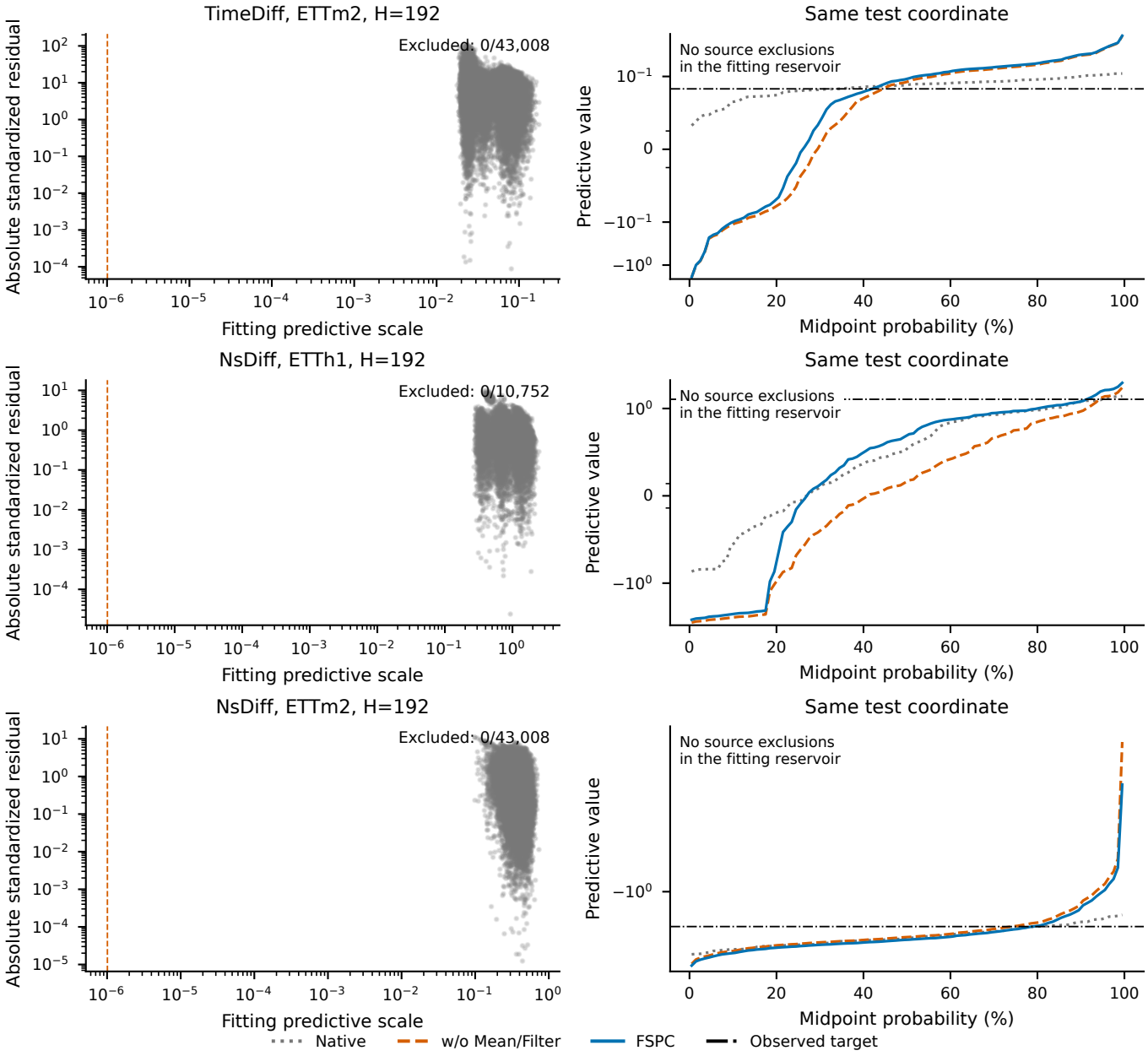


Fig. S21. Measured extension of Figure 4: TimeDiff, ETTm2, $H = 192$; NsDiff, ETTh1, $H = 192$; NsDiff, ETTm2, $H = 192$. Left: fitting scale versus standardized residual magnitude; orange marks excluded near-floor sources. Right: Native, full FSPC and w/o Mean/Filter at the identical test coordinate; black marks the target. FSPC uses the frozen affine-plus-state mean and validation-selected blend. Dataset selection uses validation residuals; coordinate selection uses forecasts without test targets. Backbones without source exclusions retain identical residual-source pools; mean control can still change the full FSPC prediction. The vertical prediction axes use symmetric logarithmic scaling and each backbone's scoring units.

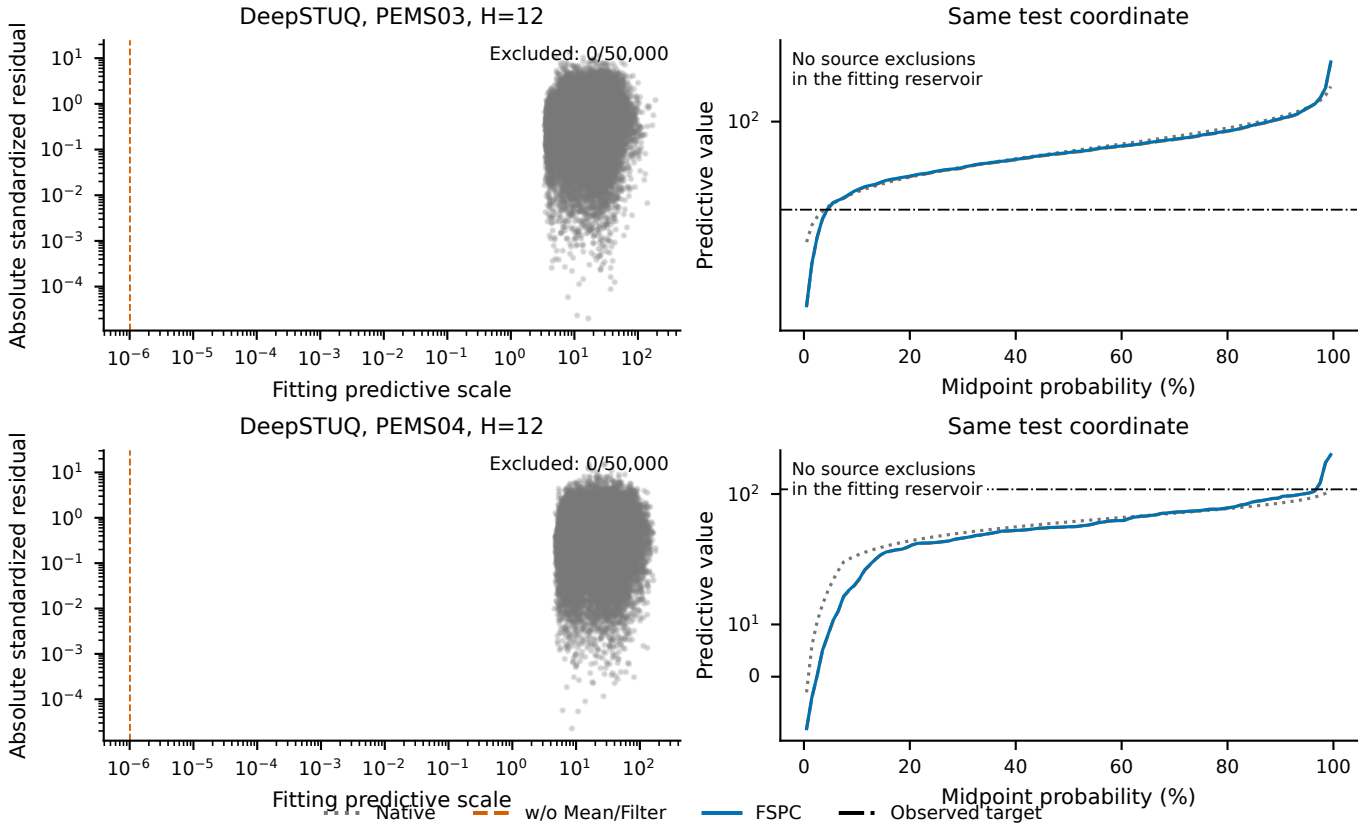


Fig. S22. Measured extension of Figure 4: DeepSTUQ, PEMS03, $H = 12$; DeepSTUQ, PEMS04, $H = 12$. Left: fitting scale versus standardized residual magnitude; orange marks excluded near-floor sources. Right: Native, full FSPC and w/o Mean/Filter at the identical test coordinate; black marks the target. FSPC uses the frozen affine-plus-state mean and validation-selected blend. Dataset selection uses validation residuals; coordinate selection uses forecasts without test targets. Backbones without source exclusions retain identical residual-source pools; mean control can still change the full FSPC prediction. The vertical prediction axes use symmetric logarithmic scaling and each backbone's scoring units.

Population transfer and empirical quantiles

Main-paper Equation (8) describes the continuum reconstruction before template centering and mean adjustment. Fix the fitted partition and templates, and let F_g^{fit} and F_g^{test} denote the population CDFs of standardized residuals conditional on state g in the fitting and test segments. If F_g^{fit} is continuous, its population quantile $q_g(\alpha)$ satisfies $F_g^{\mathrm{fit}}(q_g(\alpha)) = \alpha$ for $0 < \alpha < 1$. Using this population quantile gives nominal state coverage when $F_g^{\mathrm{test}} = F_g^{\mathrm{fit}}$.

The implemented template Q_g instead estimates quantiles from a finite residual pool. Its continuum coverage error decomposes as

$$\begin{aligned} F_g^{\mathrm{test}}(Q_g(\alpha)) - \alpha = &\left[F_g^{\mathrm{test}}(Q_g(\alpha)) - F_g^{\mathrm{fit}}(Q_g(\alpha))\right] \\ &+ \left[F_g^{\mathrm{fit}}(Q_g(\alpha)) - \alpha\right]. \end{aligned} \tag{1}$$

The first term measures within-state distribution shift. The second measures how the empirical template quantile departs from the fitting population's nominal quantile. Even with exact distribution transfer and an unfiltered, state-matched pool, this second term generally remains nonzero because the pool is finite. Quantile interpolation is part of this empirical approximation. Source filtering and global fallback can further change the residual population represented by a template, so the second term can also include a mismatch with the fitting state's unfiltered distribution.

The $S = 100$ midpoint output grid introduces a separate discretization of the continuum template. Centering and mean adjustment subsequently change its thresholds. Thus exact nominal coverage characterizes the population reconstruction under the stated transfer condition; the empirical, finite-grid FSPC forecasts are assessed by the reported coverage and scoring measurements.

EXPERIMENTAL PROTOCOL

Datasets, training and forecast sampling

Long-horizon datasets were ETTh1, ETTh2, ETTm1, ETTm2, Electricity, Exchange, Weather, Traffic, ILI and SolarEnergy, following each backbone's dataset and horizon settings. DeepSTUQ used PEMS03, PEMS04, PEMS07 and PEMS08. Main-paper Table II lists the 83 configurations and 249 runs.

Training used seeds 1, 2 and 3, with checkpoint selection based on each backbone's validation criterion. K^2VAE used at most 20 epochs, patience five and 100 training batches per epoch; PPM used at most 30 epochs and patience five. The diffusion backbones followed their native training settings. NsDiff on ETTh1 used shared pretrained mean and variance estimators to initialize independently seeded diffusion training. On ETTm1, ETTm2 and Exchange, NsDiff used float64 attention when float32 operations overflowed, and nonfinite validation forecasts counted as non-improvement for early stopping.

DeepSTUQ used its combined spatio-temporal architecture with up to 100 pretraining epochs, 20 adaptive weight averaging epochs and ten averaged snapshots. Twelve five-minute observations predicted the next twelve, using chronological 60/20/20 training, validation and test splits and training-segment normalization. Its predictive Gaussian was estimated from ten dropout passes and validation-fitted temperature calibration, then represented by 100 deterministic midpoint quantiles. Other backbones also supplied 100 members. Diffusion-TS used an unclamped denoised output because its targets were standard-score normalized; Native and every calibration method shared this sampling setting.

Calibration inputs and diagnostic populations

FSPC uses 6×6 states. Training forecasts supply reference biases; validation forecasts fit templates and the affine-plus-state mean. The main method and External calibration methods above specify chronological blend selection, training-based candidate fitting when validation blocks are insufficient, and final refitting. Reservoirs retain at most 50,000 coordinates from at most 100 native batches or windows. Parameters are frozen before testing.

Conditional PIT uses four groups each for horizon, scale and channel; both PIT maps have 41 monotone knots. Empirical residual methods share source filtering and storage limits. EMOS, Learned PIT and QRF fit the same validation reservoir and return 100 midpoint quantiles; their inputs and optimization settings are specified above.

The 77 eligible test configurations comprise 33 for K^2VAE, seven for PPM, nine for Diffusion-TS, eight each for TMDM, TimeDiff and NsDiff, and four for DeepSTUQ. All have eligible states in all three runs. Diagnostic groups use Native training forecasts for every correction. Diffusion-TS retains clamped training diagnostic boundaries and unclamped validation calibration boundaries.

Threshold-event scoring

Threshold-event evaluation tested whether the distributions supplied useful exceedance probabilities. For a channel-specific threshold τ_c,

$$\widehat{p}_k = \frac{1}{S} \sum_{s=1}^{S} \mathbb{I}\{\widehat{y}_k^{(s)} > \tau_c\}. \tag{2}$$

With $e_k = \mathbb{I}\{y_k > \tau_c\}$, the Brier score over all N test coordinates was

$$\mathrm{BS} = \frac{1}{N} \sum_{k} (\widehat{p}_k - e_k)^2. \tag{3}$$

The thresholds were each channel's 0.8, 0.9 and 0.95 validation-target quantiles, estimated from a uniform reservoir of at most 4096 horizon-coordinate rows and shared by all methods. Writing BS_r for the score at threshold quantile level r, we reported

$$\overline{\mathrm{BS}} = \frac{1}{3} \sum_{r \in \{0.8, 0.9, 0.95\}} \mathrm{BS}_r \,. \tag{4}$$

Lower values indicate more accurate event probabilities. Ten probability bins also compared predicted probabilities with observed event frequencies.

Fitting and deployment timing

Measurements use an Intel Xeon Gold 6426Y server, NumPy 1.26.4 and PyTorch 2.4.1. Four CPU workers each use a disjoint pair of cores, with BLAS and PyTorch limited to two threads. Complete fitting includes residual templates, training-reference biases, affine and state means, chronological blend selection and final refitting, given cached Native forecasts. Three fitting repetitions per task reproduce the historical selection CRPS and reference-MSE constraint. Online operations use one warm-up and three randomized paired timing repetitions with frozen parameters. Each task contributes its median duration; speedups are formed within tasks before configuration and three-run aggregation. Native generation, file access and metric accumulation are excluded from online timings. Direct queries evaluate all three historical thresholds, either including Native summary computation or reusing cached summaries.